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UPPER SIXTH — FURTHER MATHEMATICS

CHAPTER 3

PLANE TRANSFORMATIONS

SUGGESTED TIME: 18 PERIODS

IN THIS CHAPTER

This chapter studies the composition of plane transformations — reflections, translations, rotations, glide reflections and homotheties — leading to the classification of plane isometries and similarities. Students learn to identify, construct and combine these transformations geometrically, analytically and with complex numbers, in preparation for the GCE Advanced Level examination.

- By the end of this chapter I can determine the nature and characteristic elements of the composite of two orthogonal symmetries (parallel or intersecting axes).
- By the end of this chapter I can compose two translations and a translation with a reflection, and recognise a glide reflection.
- By the end of this chapter I can compose two rotations of the same or different centres and identify the resulting transformation.
- By the end of this chapter I can define a plane isometry and classify any isometry as a translation, rotation, reflection or glide reflection.
- By the end of this chapter I can construct the image of a point or figure under a homothety and state its basic and advanced properties.
- By the end of this chapter I can distinguish direct from indirect homothety and use the sign of the ratio correctly.

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THIS SIGNED COURSE CONTAINS

COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. Composing Reflections and Translations
2. Composing Rotations; Introduction to Isometries
3. Classification of Plane Isometries and Glide Reflections
4. Homothety: Definition, Construction and Basic Properties
5. Analytical and Complex Representation of Homothety
6. Composition of Homotheties and Advanced Properties
7. Plane Similarities: Concept and Characteristic Elements
8. Analytical Study of Similarities and Applications
9. Integration activity
10. Methods and worked examples
11. Exercises
12. Solutions to the exercises
13. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can determine the nature and characteristic elements of the composite of two orthogonal symmetries (parallel or intersecting axes).
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- By the end of this chapter I can construct the image of a point or figure under a homothety and state its basic and advanced properties.
- By the end of this chapter I can distinguish direct from indirect homothety and use the sign of the ratio correctly.
- By the end of this chapter I can represent a homothety analytically and by a complex number or vector expression.
- By the end of this chapter I can compose two homotheties with the same or distinct centres and find the centre and ratio of the result.

- By the end of this chapter I can define a plane similarity, give its analytical form, and solve problems on triangles, circles and spiral similarities.

PREREQUISITES

- Translations, reflections (orthogonal symmetries) and rotations as studied in Lower Sixth: definitions and basic properties
- Vectors in the plane: addition, scalar multiplication, coordinates and collinearity
- Complex numbers: modulus, argument, and the geometric meaning of $z' = az + b$
- Cartesian coordinates, equations of lines and circles in the plane
- Congruent and similar triangles, Thales' theorem and ratios of lengths

1 COMPOSING REFLECTIONS AND TRANSLATIONS

When a tailor in Bamenda folds a wax-print fabric along two parallel lines before cutting, he is unknowingly composing two reflections — and the result is exactly a translation. When a mechanic rotates a wheel nut with a spanner, or when a photographer in Douala enlarges a passport photo without distorting it, plane transformations are silently at work. What single transformation is hidden behind the composition of two symmetries, two rotations, or two enlargements? This chapter gives you the algebraic and geometric tools to answer such questions — and they appear every year in the GCE Advanced Level Further Mathematics paper.

In this first section we master the two simplest building blocks — **reflections** and **translations** — and we discover the beautiful fact that composing two reflections produces either a translation or a rotation, while composing a reflection with a translation produces a new creature: the **glide reflection**.

1.1 Recall: Orthogonal Symmetry (Reflection) Across a Line

DEFINITION **Orthogonal symmetry (reflection) across a line**

Let d be a line in the plane. The **orthogonal symmetry** (or **reflection**) across d, denoted s_d, is the transformation of the plane defined by:

- if $M \in d$, then $s_d(M) = M$;
- if $M \notin d$, then $s_d(M) = M'$, where d is the perpendicular bisector of the segment $[MM']$.

The line d is called the **axis** of the reflection.

KEY POINTS **Essential facts about a reflection**

- The invariant (fixed) points of s_d are exactly the points of the axis d.
- s_d is an involution: $s_d \circ s_d = \mathrm{Id}$ (reflecting twice brings every point back).
- s_d preserves distances (it is an isometry) but reverses orientation: a clockwise triangle becomes anticlockwise.
- To construct the image of M: drop the perpendicular from M to d, meeting d at H, then mark M' on the same line with $HM' = HM$ on the other side.

1.2 Composition of Two Reflections with Parallel Axes

Let d_1 and d_2 be two **parallel** lines. What is the nature of $s_{d_2} \circ s_{d_1}$? Take a point M, reflect it in d_1 to get M_1, then reflect M_1 in d_2 to get M'. Whatever the position of M, the segment $[MM']$ is perpendicular to both axes, and its length is always twice the distance between the axes. This suggests a translation.

PROPERTY **Two reflections with parallel axes give a translation — proof examinable**

Let $d_1 \parallel d_2$, and let $\vec{u}$ be the vector going from d_1 to d_2 along a common perpendicular (so $\|\vec{u}\|$ is the distance between the axes). Then

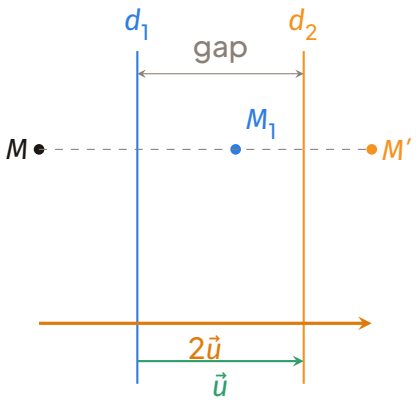
$$s_{d_2} \circ s_{d_1} = t_{2\vec{u}},$$

the translation of vector $2\vec{u}$.

Proof. Let M be any point, $M_1 = s_{d_1}(M)$ and $M' = s_{d_2}(M_1)$. Let H_1 and H_2 be the orthogonal projections of M onto d_1 and d_2 respectively; since $d_1 \parallel d_2$, the points M, M_1, M', H_1, H_2 all lie on one line perpendicular to the axes. Using directed distances along this line:

$$\overline{MM'} = \overline{MM_1} + \overline{M_1M'} = 2\overline{H_1M_1} + 2\overline{M_1H_2} = 2\overline{H_1H_2}.$$

But $\vec{H_1H_2} = \vec{u}$, independent of M. Hence $\vec{MM'} = 2\vec{u}$ for every point M, which is exactly the definition of the translation $t_{2\vec{u}}$. ■



$M_1 = s_{d_1}(M)$, $M' = s_{d_2}(M_1)$: the composite is the translation of vector $2\vec{u}$, twice the perpendicular gap between the axes.

WARNING **Order matters!**

$s_{d_2} \circ s_{d_1} = t_{2\vec{u}}$ where $\vec{u}$ goes from d_1 to d_2. Reversing the order gives the opposite translation:

$$s_{d_1} \circ s_{d_2} = t_{-2\vec{u}}.$$

Composition of transformations is not commutative in general. Always read $s_{d_2} \circ s_{d_1}$ as: "first s_{d_1}, then s_{d_2}".

1.3 Composition of Two Reflections with Intersecting Axes

Now suppose d_1 and d_2 meet at a point O, making an oriented angle $\theta = (d_1, \hat{d}_2)$. Reflecting a point M successively in d_1 then d_2 keeps its distance to O unchanged (each reflection does), but swings it around O. The total swing is exactly 2θ.

PROPERTY **Two reflections with intersecting axes give a rotation — proof examinable**

If $d_1 \cap d_2 = \{O\}$ and θ is the oriented angle from d_1 to d_2, then

$$s_{d_2} \circ s_{d_1} = r(O, 2\theta),$$

the rotation of centre O and angle 2θ.

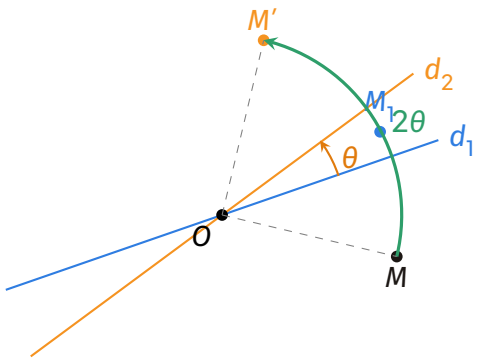
Proof. Let $M \neq O$, $M_1 = s_{d_1}(M)$, $M' = s_{d_2}(M_1)$. Since reflections preserve distances and fix O, we have $OM = OM_1 = OM'$, so M' lies on the circle of centre O through M. For the angle, use oriented angles of lines. Reflection in a line d sends any ray $[OM)$ to a ray symmetric with respect to d, so:

$$(\vec{OM}, \hat{\vec{OM'}}) = (\vec{OM}, \hat{d_1}) + (d_1, \hat{d_2}) + (d_2, \hat{\vec{OM'}}).$$

By symmetry, $(d_2, \hat{\vec{OM'}}) = (\vec{OM_1}, \hat{d_2})$ and $(\vec{OM}, \hat{d_1}) = (d_1, \hat{\vec{OM_1}})$. Hence

$$(\vec{OM}, \hat{\vec{OM'}}) = (d_1, \hat{\vec{OM_1}}) + (\vec{OM_1}, \hat{d_2}) + (d_1, \hat{d_2}) = 2(d_1, \hat{d_2}) = 2\theta.$$

Also O is fixed, and the map preserves distances, so $s_{d_2} \circ s_{d_1}$ is the rotation of centre O and angle 2θ. ■



Axes d_1, d_2 meeting at O with angle θ: the composite $s_{d_2} \circ s_{d_1}$ is the rotation of centre O and angle 2θ.

WARNING **The factor 2 — the classic trap**

A very common mistake at the GCE is to forget the factor 2:

- parallel axes: the translation vector is twice the gap between the axes, not the gap itself;
- intersecting axes: the rotation angle is twice the angle between the axes, not the angle itself.

Also remember: reversing the order of composition reverses the translation vector (resp. the rotation angle).

1.4 Composition of Two Translations

PROPERTY **Composite of two translations**

Let $t_{\vec{u}}$ and $t_{\vec{v}}$ be translations. Then

$$t_{\vec{v}} \circ t_{\vec{u}} = t_{\vec{u}+\vec{v}}.$$

In particular, translations commute: $t_{\vec{v}} \circ t_{\vec{u}} = t_{\vec{u}} \circ t_{\vec{v}}$.

Proof. For any point M, write $M_1 = t_{\vec{u}}(M)$ and $M' = t_{\vec{v}}(M_1)$. Then

$$\vec{MM'} = \vec{MM_1} + \vec{M_1M'} = \vec{u} + \vec{v},$$

so the composite is $t_{\vec{u}+\vec{v}}$. Since vector addition is commutative, $\vec{u} + \vec{v} = \vec{v} + \vec{u}$, and the two orders coincide. ■

EXEMPLE **Composing translations**

A delivery rider in Yaoundé moves a parcel by $\vec{u} = (3, -1)$ km, then by $\vec{v} = (-2, 4)$ km. The total displacement is $\vec{u} + \vec{v} = (1, 3)$ km: a single translation $t_{(1,3)}$ achieves the same result, in either order.

1.5 Composition of a Translation and a Reflection: the Glide Reflection

What happens if we compose a reflection s_d with a translation $t_{\vec{v}}$? Split $\vec{v} = \vec{v}_{\parallel} + \vec{v}_{\perp}$ into components parallel and perpendicular to d. The perpendicular part can be absorbed by shifting the axis d (using the theorem on parallel axes above), and only the parallel part survives. The result is a new transformation.

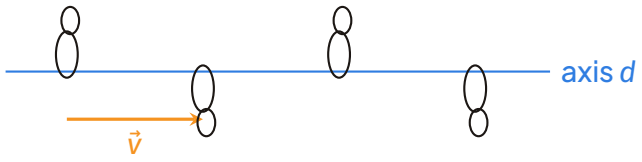
DEFINITION **Glide reflection**

A **glide reflection** is the composite of a reflection across a line d and a translation of vector $\vec{v}$ parallel to d (with $\vec{v} \neq \vec{0}$):

$$g = t_{\vec{v}} \circ s_d = s_d \circ t_{\vec{v}}.$$

The line d is the **axis** and $\vec{v}$ the **vector** of the glide reflection. (The two orders coincide here because $\vec{v} \parallel d$; indeed, reflecting a vector parallel to the axis leaves it unchanged.)

Think of footprints in the sand along a straight path: each step is the mirror image of the previous one across the line of the path, shifted forward by one pace. That is exactly a glide reflection.



Footprints along a line: each print is obtained from the previous one by reflection across d followed by a translation of vector $\vec{v}$ parallel to d — a glide reflection.

KEY POINTS **Recognition criteria for a glide reflection**

A plane transformation g is a glide reflection (with $\vec{v} \neq \vec{0}$) if and only if:

- g preserves distances (it is an isometry);
- g reverses orientation (it is an indirect isometry);
- g has no fixed point.

Its axis is the unique invariant line, and $g \circ g = t_{2\vec{v}}$.

METHOD **Identifying a composite $s_{d_2} \circ s_{d_1}$**

1. Draw the axes. Are they parallel or intersecting?
2. Parallel axes: the composite is a translation; measure the perpendicular gap from d_1 to d_2 and double it (mind the direction: $d_1 \to d_2$).
3. Intersecting axes: the composite is a rotation; the centre is the intersection point, the angle is twice the oriented angle from d_1 to d_2.
4. Check with one test point: pick a convenient point M, construct $M_1 = s_{d_1}(M)$ then $M' = s_{d_2}(M_1)$, and verify that $M\vec{M}'$ (or the angle $M\hat{O}M'$) matches your answer.

WORKED EXAMPLE 1 **Parallel axes**

The lines $d_1 : x = 1$ and $d_2 : x = 4$ are given in an orthonormal frame. Determine the nature and characteristic elements of $f = s_{d_2} \circ s_{d_1}$, and find the image of $A(2, 3)$.

SOLUTION

Solution. The axes are parallel (both vertical). The vector from d_1 to d_2 along a common perpendicular is $\vec{u} = (3, 0)$. By the theorem, $f = t_{2\vec{u}} = t_{(6,0)}$. Check with $A(2, 3)$: $s_{d_1}(A) = A_1(0, 3)$ (since A is 1 unit right of d_1), then $s_{d_2}(A_1) = A'(8, 3)$ (since A_1 is 4 units left of d_2). Indeed $\vec{AA'} = (6, 0) = 2\vec{u}$. ■

WORKED EXAMPLE 2 **Intersecting axes**

d_1 and d_2 pass through the origin, with d_1 the x-axis and d_2 making an angle of $30°$ with d_1. Determine $g = s_{d_2} \circ s_{d_1}$ and the image of $B(2, 0)$.

SOLUTION

Solution. The axes intersect at O with oriented angle $\theta = 30°$. Hence $g = r(O, 2\theta) = r(O, 60°)$. The image of $B(2, 0)$ is

$$B' = \left(2\cos 60°,\ 2\sin 60°\right) = \left(1,\ \sqrt{3}\right).$$

Check: $s_{d_1}(B) = B$ (since $B \in d_1$), and reflecting B across d_2 (line at $30°$) sends the ray $[OB)$ to the ray at angle $2 \times 30° = 60°$, giving $B'(1, \sqrt{3})$. ■

EXERCISE 1 **Practice**

1. $d_1 : y = 0$ and $d_2 : y = 2$. Give the nature and vector of $s_{d_2} \circ s_{d_1}$, then of $s_{d_1} \circ s_{d_2}$.
2. Two lines through O make an angle of $45°$. Show that $s_{d_2} \circ s_{d_1}$ is a quarter turn, and deduce that $(s_{d_2} \circ s_{d_1})^2$ is a half-turn.
3. Let $g = t_{\vec{v}} \circ s_d$ with $d : y = 0$ and $\vec{v} = (3, 0)$. Find the image of $P(1, 2)$ and show that g has no fixed point.

SOLUTION **Exercise 1**

1. $\vec{u} = (0, 2)$ from d_1 to d_2, so $s_{d_2} \circ s_{d_1} = t_{(0,4)}$ and $s_{d_1} \circ s_{d_2} = t_{(0,-4)}$.
2. $2\theta = 90°$: a quarter turn about O. Its square has angle $180°$: a half-turn about O.
3. $s_d(P) = (1, -2)$, then $g(P) = (4, -2)$. A fixed point would need $(x + 3, -y) = (x, y)$, i.e. $3 = 0$: impossible. So g has no fixed point — consistent with a glide reflection.

DID YOU KNOW? **Why this matters**

The theorems of this section are the key to the classification of isometries (Section 3): every plane isometry is a translation, a rotation, a reflection or a glide reflection, because every isometry can be written as a composite of at most three reflections. Glide reflections also describe the symmetry of footprints, of woven raffia patterns, and of many crystal structures studied in materials science.

2 COMPOSING ROTATIONS; INTRODUCTION TO ISOMETRIES

In the previous section we composed reflections and translations, and we discovered a powerful principle: composing two reflections with parallel axes gives a translation, while composing two reflections with intersecting axes gives a rotation. We now turn this principle around and use it as a tool. Since every rotation can be written as a composite of two reflections, we can study the composition of two rotations by cleverly decomposing them into reflections whose axes we choose ourselves. This beautiful technique, sometimes called the reflection-decomposition trick, will allow us to determine completely the nature of any composite of rotations — and it will lead us naturally to the general notion of a **plane isometry**.

2.1 Recall: Rotations and their Decomposition into Reflections

DEFINITION **Rotation**

Let O be a point of the plane and θ a real number (an angle, measured in degrees or radians). The **rotation** of centre O and angle θ, denoted $r(O, \theta)$, is the transformation of the plane that fixes O and sends every point $M \neq O$ to the point M' such that

$$OM' = OM \qquad \text{and} \qquad \left(\vec{OM}, \vec{OM'}\right) = \theta \pmod{360^\circ}.$$

The key structural fact, established in the previous section, is the following.

PROPERTY **A rotation is a composite of two reflections**

Let (d_1) and (d_2) be two lines intersecting at O, with $\big((d_1),(d_2)\big) = \frac{\theta}{2} \pmod{180^\circ}$. Then

$$r(O,\theta) = s_{(d_2)} \circ s_{(d_1)}.$$

Moreover, for a given rotation $r(O,\theta)$, one of the two axes may be chosen freely through O; the second axis is then forced: it is the image of the first by the rotation $r\left(O, \frac{\theta}{2}\right)$.

WARNING **The order matters**

In $s_{(d_2)} \circ s_{(d_1)}$ we reflect first in (d_1), then in (d_2), and the resulting rotation goes from (d_1) towards (d_2). Swapping the axes reverses the angle: $s_{(d_1)} \circ s_{(d_2)} = r(O,-\theta)$.

The freedom to choose one axis arbitrarily is exactly what makes the composition of rotations tractable, as we now see.

2.2 Composition of Two Rotations with the Same Centre

Let us start with the simplest situation. Take a point M in the plane and rotate it about O through an angle α, obtaining M_1; then rotate M_1 about the same centre O through an angle β, obtaining M'. Geometrically, the point has simply turned through α and then through β about the same pivot: the total effect is a turn through $\alpha + \beta$.

PROPERTY **Same centre**

For any point O and any angles α, β,

$$r(O,\beta) \circ r(O,\alpha) = r(O,\alpha+\beta).$$

In particular, two rotations with the same centre commute: $r(O,\beta) \circ r(O,\alpha) = r(O,\alpha) \circ r(O,\beta)$.

Proof (examinable). Write $r(O,\alpha) = s_{(d_2)} \circ s_{(d_1)}$ where $(d_1),(d_2)$ pass through O with $\big((d_1),(d_2)\big) = \frac{\alpha}{2}$, and write $r(O,\beta) = s_{(d_3)} \circ s_{(d_2)}$, choosing the same line (d_2) as first axis of the second rotation (this is allowed by the freedom of choice above). Then

$$r(O,\beta) \circ r(O,\alpha) = s_{(d_3)} \circ s_{(d_2)} \circ s_{(d_2)} \circ s_{(d_1)} = s_{(d_3)} \circ s_{(d_1)},$$

because $s_{(d_2)} \circ s_{(d_2)} = \mathrm{Id}$. The composite $s_{(d_3)} \circ s_{(d_1)}$ is a rotation about O of angle $2\left(\frac{\alpha}{2} + \frac{\beta}{2}\right) = \alpha + \beta$. ■

EXEMPLE **Same centre**

$r(O, 70^\circ) \circ r(O, 50^\circ) = r(O, 120^\circ)$, and $r(O, -30^\circ) \circ r(O, 90^\circ) = r(O, 60^\circ)$. Also $r(O, 180^\circ) \circ r(O, 180^\circ) = r(O, 360^\circ) = \mathrm{Id}$, which confirms that a half-turn is its own inverse.

2.3 Composition of Two Rotations with Different Centres

Now suppose the centres differ: $r_1 = r(O_1, \alpha)$ and $r_2 = r(O_2, \beta)$ with $O_1 \neq O_2$. What is $r_2 \circ r_1$? The pivot changes midway, so the answer is no longer obvious. The reflection-decomposition trick settles the question completely: we decompose each rotation into two reflections, and we choose the common axis to be the line (O_1O_2), so that the two middle reflections cancel.

Concretely:

- Write $r_1 = s_{(O_1O_2)} \circ s_{(d_1)}$, where (d_1) is the line through O_1 such that $\big((d_1), (O_1O_2)\big) = \frac{\alpha}{2}$.
- Write $r_2 = s_{(d_2)} \circ s_{(O_1O_2)}$, where (d_2) is the line through O_2 such that $\big((O_1O_2), (d_2)\big) = \frac{\beta}{2}$.

Then

$$r_2 \circ r_1 = s_{(d_2)} \circ s_{(O_1O_2)} \circ s_{(O_1O_2)} \circ s_{(d_1)} = s_{(d_2)} \circ s_{(d_1)}.$$

Two cases arise, according to whether (d_1) and (d_2) intersect or are parallel.

PROPERTY **Composite of two rotations with distinct centres — proof examinable**

Let $r_1 = r(O_1, \alpha)$ and $r_2 = r(O_2, \beta)$ with $O_1 \neq O_2$.

- If $\alpha + \beta \not\equiv 0 \pmod{360^\circ}$, the lines (d_1) and (d_2) constructed above meet at a point Ω, and

  $$r_2 \circ r_1 = r(\Omega, \alpha + \beta).$$

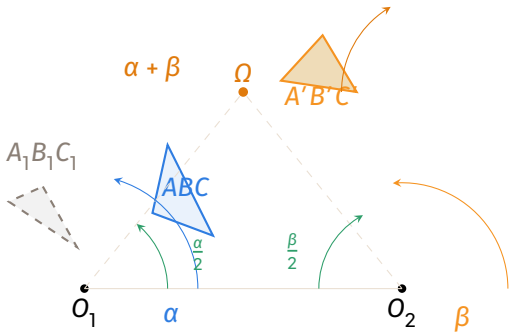
  The centre Ω is characterised by the half-angle conditions: the line (ΩO_1) makes angle $-\frac{\alpha}{2}$ with (O_1O_2) at O_1, and the line (ΩO_2) makes angle $+\frac{\beta}{2}$ with (O_2O_1) at O_2 (on the appropriate sides).
- If $\alpha + \beta \equiv 0 \pmod{360^\circ}$, then $(d_1) \parallel (d_2)$ and $r_2 \circ r_1$ is a **translation**.

Justification of the two cases. The angle between (d_1) and (d_2) equals $\frac{\alpha}{2} + \frac{\beta}{2} = \frac{\alpha+\beta}{2}$ (as directed angles of lines, modulo 180°). If $\alpha + \beta \not\equiv 0 \pmod{360^\circ}$, this angle is not a multiple of 180°, so (d_1) and (d_2) intersect at a point Ω, and $s_{(d_2)} \circ s_{(d_1)}$ is the rotation of centre Ω and angle $2 \cdot \frac{\alpha+\beta}{2} = \alpha + \beta$. If $\alpha + \beta \equiv 0 \pmod{360^\circ}$, then $\frac{\alpha+\beta}{2}$ is a multiple of 180°, so (d_1) and (d_2) are parallel (and distinct, since $O_1 \neq O_2$), and the composite of two reflections with parallel distinct axes is a translation. ■

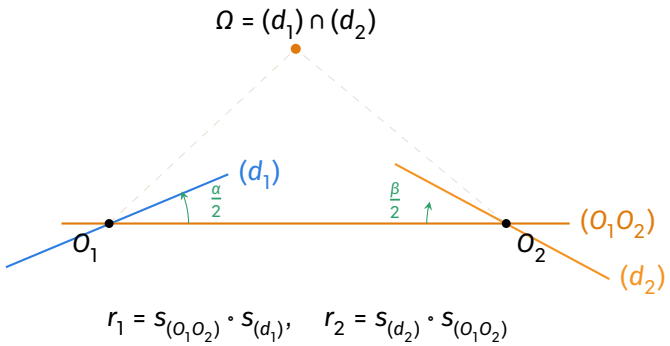
METHOD **Finding the centre of $r(O_2, \beta) \circ r(O_1, \alpha)$**

1. Check that $\alpha + \beta \not\equiv 0 \pmod{360^\circ}$; otherwise the composite is a translation.
2. Draw the line (O_1O_2).
3. Through O_1, draw the ray obtained by rotating the ray $[O_1O_2)$ by $-\frac{\alpha}{2}$.
4. Through O_2, draw the ray obtained by rotating the ray $[O_2O_1)$ by $+\frac{\beta}{2}$.
5. The two rays meet at Ω, the centre of the composite rotation; the angle of the composite is $\alpha + \beta$.

Equivalently, the triangle $O_1O_2\Omega$ has base angles $\frac{\alpha}{2}$ at O_1 and $\frac{\beta}{2}$ at O_2 (taken with the appropriate orientation).



The triangle ABC is rotated about O_1 through α (giving $A_1B_1C_1$), then about O_2 through β (giving $A'B'C'$). The net effect is a single rotation about Ω through $\alpha + \beta$. The base angles of triangle $O_1O_2\Omega$ are $\frac{\alpha}{2}$ and $\frac{\beta}{2}$.



The reflection-axes construction: both rotations share the middle axis (O_1O_2), which cancels; the composite is $s_{(d_2)} \circ s_{(d_1)}$, a rotation about the intersection Ω of the two outer axes.

WORKED EXAMPLE 3 **Composite of two rotations with distinct centres**

Let O_1 and O_2 be two distinct points. Determine the nature and characteristic elements of $f = r(O_2, 60^\circ) \circ r(O_1, 90^\circ)$.

SOLUTION

Solution. The sum of the angles is $90^\circ + 60^\circ = 150^\circ \not\equiv 0 \pmod{360^\circ}$, so f is a rotation of angle 150°. To find its centre Ω:

- Through O_1, draw the ray obtained by rotating $[O_1O_2)$ by $-\frac{90^\circ}{2} = -45^\circ$.
- Through O_2, draw the ray obtained by rotating $[O_2O_1)$ by $+\frac{60^\circ}{2} = +30^\circ$.

These rays intersect at Ω. Hence $f = r(\Omega, 150^\circ)$, where Ω is the point such that the triangle $O_1O_2\Omega$ has angles 45° at O_1 and 30° at O_2 (and therefore 105° at Ω).

2.4 The Special Case: Angles Summing to a Multiple of 360°

When $\alpha + \beta \equiv 0 \pmod{360^\circ}$, that is $\beta = -\alpha$, the two axes (d_1) and (d_2) are parallel and distinct, and

$$r(O_2, -\alpha) \circ r(O_1, \alpha) = s_{(d_2)} \circ s_{(d_1)} = t_{\vec{u}},$$

a translation. Recall that the composite of two reflections with parallel axes is the translation by the vector $2\vec{H_1H_2}$, where H_1, H_2 are the feet of a common perpendicular on (d_1) and (d_2). Hence the translation vector is perpendicular to (d_1) and (d_2), with magnitude twice the distance between the two axes.

METHOD **Finding the translation vector when $\alpha + \beta \equiv 0 \pmod{360^\circ}$**

The most reliable exam technique is to compute the image of one well-chosen point.

Since $f = r(O_2, -\alpha) \circ r(O_1, \alpha)$ is a translation $t_{\vec{u}}$, we have $\vec{u} = M\,\vec{f}(M)$ for any point M. The clever choice is the point M whose first image is the centre of the second rotation:

1. Choose M such that $r(O_1, \alpha)(M) = O_2$, i.e. M is the image of O_2 under $r(O_1, -\alpha)$.
2. Then $f(M) = r(O_2, -\alpha)(O_2) = O_2$, because the centre of a rotation is fixed.
3. Conclude: $\vec{u} = M\vec{O}_2$.

The geometry of the triangle O_1MO_2 (with $O_1M = O_1O_2$ and angle α at O_1) then gives the magnitude and direction of $\vec{u}$ explicitly.

WORKED EXAMPLE 4 **A composite that is a translation**

Let $O_1 \neq O_2$ and let $f = r(O_2, -90^\circ) \circ r(O_1, 90^\circ)$. Show that f is a translation and determine its vector in terms of $O_1\vec{O}_2$.

SOLUTION

Solution. The sum of the angles is $90^\circ + (-90^\circ) = 0$, so f is a translation $t_{\vec{u}}$. To find $\vec{u}$, let M be the point such that $r(O_1, 90^\circ)(M) = O_2$, i.e. M is obtained from O_2 by rotating about O_1 through -90°. Then

$$f(M) = r(O_2, -90^\circ)\big(r(O_1, 90^\circ)(M)\big) = r(O_2, -90^\circ)(O_2) = O_2,$$

since O_2, being the centre of the second rotation, is fixed by it. Hence

$$\vec{u} = M\vec{O}_2.$$

Now M satisfies $O_1M = O_1O_2$ and $\left(O_1\vec{M}, O_1\vec{O}_2\right) = 90^\circ$, so the triangle O_1MO_2 is right-angled isosceles at O_1. By the Pythagorean theorem, $MO_2 = O_1O_2\sqrt{2}$, and the vector $M\vec{O}_2$ makes an angle of -45° with $O_1\vec{O}_2$. Thus f is the translation by the vector $M\vec{O}_2$, of magnitude $O_1O_2\sqrt{2}$, directed along the hypotenuse of the right isosceles triangle built on $[O_1O_2]$.

WARNING **The composite of two rotations is not always a rotation!**

This is the most frequent trap at the GCE. Before announcing "rotation of angle $\alpha + \beta$", you must check whether $\alpha + \beta \equiv 0 \pmod{360^\circ}$. If it is, the composite is a translation (or the identity, when the centres coincide and the angles cancel). Writing "$r(O_2, -60^\circ) \circ r(O_1, 60^\circ) = r(\Omega, 0^\circ)$" is a serious error: a rotation of angle 0 is the identity, but here the composite is a genuine translation by a non-zero vector.

2.5 Composition of a Rotation and a Translation

The same reflection-decomposition technique handles mixed composites. A translation $t_{\vec{u}}$ is the composite of two reflections with parallel axes perpendicular to $\vec{u}$; a rotation is the composite of two reflections with axes through its centre. Composing $t_{\vec{u}}$ with $r(O,\theta)$ (with $\theta \neq 0$) and choosing the shared axis to be the line through O perpendicular to $\vec{u}$, the middle reflections cancel and one obtains a composite of two reflections with intersecting axes: a **rotation of angle** θ about a new centre.

PROPERTY **Rotation composed with translation — admitted**

If $\theta \not\equiv 0 \pmod{360^\circ}$, then both $t_{\vec{u}} \circ r(O,\theta)$ and $r(O,\theta) \circ t_{\vec{u}}$ are rotations of angle θ, with (generally different) new centres that can be constructed by the shared-axis method. If $\theta \equiv 0$, the composite is of course a translation.

EXEMPLE **Quarter-turn followed by a translation**

Let $r = r(O, 90^\circ)$ and $t = t_{\vec{u}}$. Then $t \circ r$ is a rotation of angle 90° whose centre Ω is obtained as follows: through O draw the line (d) perpendicular to $\vec{u}$; write $r = s_{(d)} \circ s_{(d_1)}$ and $t = s_{(d_2)} \circ s_{(d)}$ with $(d_2) \parallel (d)$; then $t \circ r = s_{(d_2)} \circ s_{(d_1)}$, a quarter-turn about $\Omega = (d_1) \cap (d_2)$.

2.6 Introduction to Plane Isometric Transformations

Let us step back. Translations, rotations and reflections all share one fundamental feature: they do not distort distances. A triangle carried by any of them lands on a congruent triangle. This common feature deserves a name.

DEFINITION **Plane isometry**

A **plane isometry** (or **isometric transformation**) is a transformation f of the plane such that for all points M and N, with images $M' = f(M)$ and $N' = f(N)$,

$$M'N' = MN.$$

In words: an isometry is a distance-preserving transformation of the plane.

PROPERTY **Conservation properties of isometries — proof examinable for collinearity and midpoints**

Every plane isometry preserves:

- collinearity: if A, B, C are collinear, so are their images (indeed $B \in [AC] \iff AB + BC =$

AC, an equality of distances preserved by f);

- midpoints: the image of the midpoint of $[AB]$ is the midpoint of $[A'B']$ (it is the unique point of $[A'B']$ equidistant from A' and B');
- angles (in absolute value): by the SSS congruence case, the triangle $A'B'C'$ is congruent to ABC, so corresponding angles are equal in magnitude;
- areas: congruent triangles have equal areas, and every polygonal region decomposes into triangles.

Consequently, the image of a line is a line, of a segment a congruent segment, of a circle a circle of the same radius.

Translations, rotations, reflections and the identity are isometries; so is any composite of them, since a composite of distance-preserving maps still preserves distances. A natural question arises: are there other isometries? The fixed-point analysis below answers this completely.

2.7 Fixed-Point Analysis of an Isometry

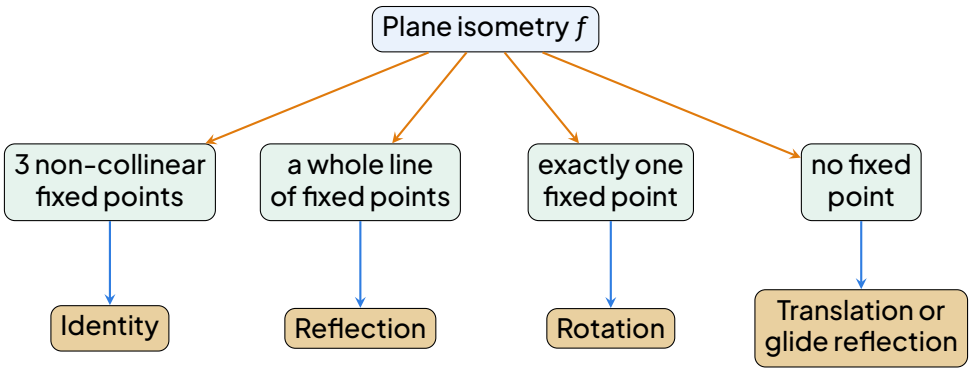
A **fixed point** of f is a point M such that $f(M) = M$. The number and arrangement of fixed points almost completely determine an isometry.

PROPERTY **Fixed points determine the isometry**

Let f be a plane isometry.

1. If f fixes three non-collinear points, then $f = \mathrm{Id}$ (the identity).
2. If f fixes exactly the points of one line (d) (and no other point), then f is the reflection $s_{(d)}$.
3. If f fixes exactly one point O, then f is a rotation of centre O (with $\theta \neq 0$).
4. If f has no fixed point, then f is a translation or a glide reflection.

Justification of the key step (examinable). Suppose f fixes three non-collinear points A, B, C, and let M be any point, $M' = f(M)$. Since f preserves distances, $M'A = MA$, $M'B = MB$, $M'C = MC$. A point of the plane is uniquely determined by its distances to three non-collinear points: the circles of centre A and radius MA and of centre B and radius MB meet in at most two points, symmetric about (AB), and the third distance to C (which lies off (AB)) selects exactly one of them. Hence $M' = M$ for every M: $f = \mathrm{Id}$. It follows that an isometry is entirely determined by the images of three non-collinear points, and that the isometries fixing exactly a line, exactly one point, or no point are precisely the reflections, the rotations, and the translations or glide reflections respectively. ■



Classification of plane isometries according to their fixed points.

2.8 Direct and Indirect Isometries

There is one last fundamental distinction. Draw the letter "R" on a sheet of paper. Apply a translation or a rotation: the letter still reads "R". Apply a reflection: you see a mirror image, reversed. Isometries fall into two families according to whether they preserve or reverse orientation.

DEFINITION **Direct and indirect isometries**

An isometry is called **direct** (or positive) if it preserves the orientation of angles: the oriented angle $(\vec{A'B'}, \vec{A'C'})$ equals $(\vec{AB}, \vec{AC})$. It is called **indirect** (or opposite, negative) if it reverses orientation: oriented angles are changed into their opposites.

KEY POINTS **The two families**

- Direct isometries: the identity, translations, rotations — and every composite of an even number of reflections.
- Indirect isometries: reflections, glide reflections — every composite of an odd number of reflections.
- The composite of two direct or of two indirect isometries is direct; the composite of a direct and an indirect isometry is indirect.

EXEMPLE **Orientation check**

The composite $r(O_2, \beta) \circ r(O_1, \alpha)$ of two rotations is a composite of four reflections — an even number — so it is direct, consistent with it being a rotation or a translation. A reflection

followed by a rotation is indirect: it must be a reflection or a glide reflection.

METHOD **GCE exam technique: identifying a composite transformation**

When asked to determine $g \circ f$:

1. State the nature first (identity, translation, rotation, reflection, glide reflection), using the angle-sum test for rotations and the parity of the number of reflections for orientation.
2. Give all characteristic elements: centre and angle for a rotation; vector for a translation; axis for a reflection; axis and vector for a glide reflection.
3. Justify with a fixed-point argument or by computing the image of one or two well-chosen points.

An answer giving only the nature (e.g. “it is a rotation”) without the centre and angle earns only part of the marks.

EXERCISE 2 **Practice**

Let O_1, O_2 be distinct points, $f = r(O_2, 120^\circ) \circ r(O_1, -60^\circ)$ and $g = r(O_2, 60^\circ) \circ r(O_1, -60^\circ)$.

1. Determine the nature and characteristic elements of f.
2. Show that g is a translation, and describe a method to construct its vector.
3. Is f a direct or an indirect isometry? Justify.

SOLUTION **Exercise 1**

1. The angle sum is $-60^\circ + 120^\circ = 60^\circ \not\equiv 0 \pmod{360^\circ}$, so f is a rotation of angle 60°. Its centre Ω is the intersection of the ray from O_1 making angle $-\frac{-60^\circ}{2} = +30^\circ$ with $[O_1O_2)$ and the ray from O_2 making angle $+\frac{120^\circ}{2} = +60^\circ$ with $[O_2O_1)$; the triangle $O_1O_2\Omega$ has base angles 30° and 60°, hence a right angle at Ω.
2. The angle sum is $-60^\circ + 60^\circ = 0$, so g is a translation. Its vector is $\vec{u} = M\,\vec{g(M)}$ for any convenient point M; choosing M with $r(O_1, -60^\circ)(M) = O_2$ gives $g(M) = O_2$, so $\vec{u} = \vec{MO_2}$.
3. f is a composite of two rotations, i.e. of four reflections (an even number); equivalently it is a rotation. Hence f is a direct isometry.

DID YOU KNOW? **Why this matters beyond the classroom**

The classification of plane isometries is the starting point of crystallography: the decorative friezes on traditional Cameroonian woven textiles and carved door frames repeat motifs by translations, rotations, reflections and glide reflections, and mathematicians have proved that exactly seven types of frieze patterns and seventeen types of wallpaper patterns exist — no more, no less. Every repeating pattern you see, from embroidered garments to modern fabric prints in Douala markets, belongs to one of these seventeen types!

In the next section we complete this picture by classifying all plane isometries, including the glide reflection, the subtle composite of a reflection and a translation parallel to its axis.

3 CLASSIFICATION OF PLANE ISOMETRIES AND GLIDE REFLECTIONS

In the previous section we composed rotations and discovered the family of **plane isometries** — transformations of the plane that preserve distances. We met translations, rotations, reflections and a curious newcomer, the **glide reflection**. The natural question is now: are there any others? Could some clever composition produce a fifth, exotic type of isometry? The remarkable answer, which we prove in this section, is **no**: the four types we already know exhaust all possibilities. This is the classification theorem of plane isometries, one of the most beautiful results of the Upper Sixth syllabus and a favourite of GCE examiners.

3.1 Orientation: the key that separates isometries into two families

Before classifying, we need a tool to distinguish isometries that "slide and turn" the plane from those that "flip it over". That tool is **orientation**.

DEFINITION **Direct and indirect isometries**

Let f be a plane isometry.

- f is a **direct isometry** (or positive isometry) if it preserves the orientation of angles: for any angle $(\vec{u}, \vec{v})$, the image angle $(f(\vec{u}), f(\vec{v}))$ has the same measure (same sign) as $(\vec{u}, \vec{v})$.
- f is an **indirect isometry** (or opposite isometry) if it reverses the orientation of angles: every oriented angle is sent to its opposite.

Concretely, take a triangle ABC labelled anticlockwise. Under a direct isometry, the image $A'B'C'$ is still labelled anticlockwise; under an indirect isometry, $A'B'C'$ is labelled clockwise — as if the triangle had been turned face down, like a page flipped over.

PROPERTY **Orientation of the known isometries**

- Direct: the identity, every translation, every rotation (and any composite of an even number of reflections).
- Indirect: every reflection, every glide reflection (and any composite of an odd number of reflections).

This follows because a single reflection reverses orientation, and orientation behaviour is multiplicative under composition: reversing twice restores the original orientation.

3.2 The four types of plane isometries

PROPERTY **Classification theorem — proof examinable in outline**

Every plane isometry f is exactly one of the following:

1. a translation (the identity being the translation by $\vec{0}$);
2. a rotation of angle $\theta \not\equiv 0 \pmod{2\pi}$;
3. a reflection across a line;
4. a glide reflection (reflection across a line Δ followed by translation by a non-zero vector parallel to Δ).

The first two are direct; the last two are indirect.

Why is this true? Proof sketch. The argument rests on two lemmas.

Lemma 1: an isometry is determined by three non-collinear points (see the property box below). Lemma 2: every isometry is a composite of at most three reflections. Indeed, let f be an isometry and A, B, C three non-collinear points with images A', B', C'. We move A to A' by the reflection s_1 across the perpendicular bisector of $[AA']$ (if $A = A'$, skip). The point $s_1(B)$ is at distance $AB = A'B'$ from A', so a second reflection s_2 across the perpendicular bisector of $[s_1(B)\,B']$ — which passes through A' — fixes A' and sends $s_1(B)$ to B'. Finally $s_2 \circ s_1(C)$ lies at the correct distances from both A' and B', so it equals either C' or its mirror image across line $(A'B')$; one more reflection s_3 across $(A'B')$ finishes the job if needed. Then $f = s_3 \circ s_2 \circ s_1$ (with possibly fewer factors).

Now count the reflections:

- **One reflection:** a reflection (indirect).
- **Two reflections:** parallel axes give a translation; intersecting axes give a rotation (Lessons 28 and 31). Both are direct.
- **Three reflections:** one shows (by regrouping two of them) that the composite reduces to a reflection or to a reflection followed by a translation parallel to its axis — a glide reflection

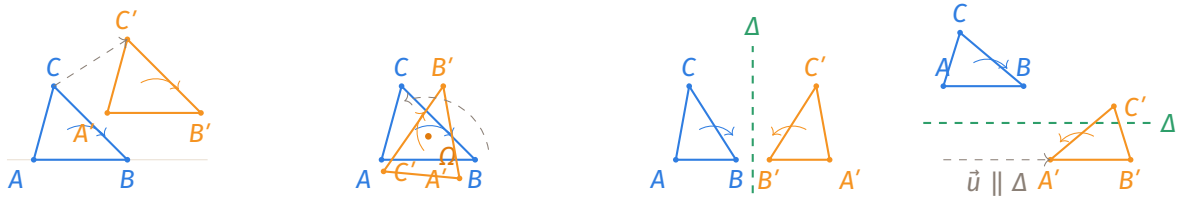
(indirect).

Hence the four types, and no others. ■

PROPERTY **Determination by three points**

An isometry is completely determined by the images of three non-collinear points: if two isometries f and g agree on three non-collinear points A, B, C, then $f = g$.

Justification. If M is any point, its distances to A, B, C equal the distances from $f(M)$ to A', B', C'. But a point of the plane is uniquely located by its distances to three non-collinear points (the circles centred at A', B', C' with those radii meet in exactly one point). Hence $f(M)$ is forced, so $f = g$.



The same triangle ABC (blue, anticlockwise) and its images (orange) under the four types of isometry. Direct isometries keep the anticlockwise arrow; indirect ones reverse it.

3.3 Analytical (complex) form of isometries

Identifying the plane with $\mathbb{C}$, every isometry has a remarkably simple algebraic description.

PROPERTY **Complex form of isometries**

Every plane isometry has one of the following forms, where $a, b \in \mathbb{C}$ and $|a| = 1$:

- Direct isometry: $z' = az + b$. If $a = 1$: translation by the vector of affix b. If $a \neq 1$: rotation of angle $\arg a$ about the unique fixed point $\omega = \dfrac{b}{1 - a}$.
- Indirect isometry: $z' = a\bar{z} + b$. This is a reflection if it has fixed points (equivalently $a\bar{b} + b = 0$), otherwise a glide reflection.

Why $|a| = 1$? Distances are preserved, so $|z_1' - z_2'| = |a|\,|z_1 - z_2| = |z_1 - z_2|$ for all z_1, z_2, forcing $|a| = 1$. The conjugate $\bar{z}$ appears precisely in the indirect case because conjugation reverses the sign of every angle.

EXEMPLE **Reading the nature from the formula**

$f : z' = iz + 2 - i$. Here $a = i$, $|a| = 1$, $a \neq 1$: a rotation of angle $\arg i = \frac{\pi}{2}$, with centre $\omega = \frac{2-i}{1-i} = \frac{(2-i)(1+i)}{2} = \frac{3+i}{2}$.

3.4 Recognition strategy: how to classify a given isometry

METHOD **Classifying an isometry f**

1. Look for fixed points. Solve $f(M) = M$.
   - Exactly one fixed point ⇒ rotation.
   - A whole line of fixed points ⇒ reflection.
   - Every point fixed ⇒ identity.
   - No fixed point ⇒ translation or glide reflection: go to step 2.
2. Test orientation on one triangle ABC: compare the senses of ABC and $A'B'C'$.
   - Same sense ⇒ direct ⇒ translation.
   - Opposite sense ⇒ indirect ⇒ glide reflection.
3. Conclude and give characteristic elements: vector of the translation; centre and angle of the rotation; axis of the reflection; axis and glide vector of the glide reflection.

WARNING **A glide reflection is not a translation**

A glide reflection has no fixed point, exactly like a (non-identity) translation. Fixed points alone cannot separate them! The orientation test is decisive: a translation preserves the sense of every triangle, a glide reflection reverses it. Alternatively, apply f twice: for a glide reflection, $f \circ f$ is a translation by $2\vec{u}$ where $\vec{u}$ is the glide vector, while the midpoint of $[M\, f(M)]$ always lies on the axis — never true for a translation.

3.5 Composition table of isometries

Since orientation is multiplicative (reversing twice restores), we obtain:

PROPERTY **Parity rule for composition**

Let f, g be isometries. Then $f \circ g$ is

- direct if f and g have the same type (both direct or both indirect);
- indirect if f and g have different types.

In symbols: direct $\circ$ direct = direct; indirect $\circ$ indirect = direct; direct $\circ$ indirect = indirect $\circ$ direct = indirect.

$f \circ g$	g direct	g indirect
f direct	direct	indirect
f indirect	indirect	direct

Green cells: same parity gives a direct isometry.

Composition table of direct and indirect isometries (parity rule).

Beware: the table gives the orientation class, not the exact type. For instance, indirect $\circ$ indirect is direct, but it may be a translation or a rotation — fixed points decide.

WORKED EXAMPLE 5 **Identifying reflection $\circ$ rotation**

Let r be the rotation of centre O and angle $\frac{\pi}{2}$, and s the reflection across a line Δ through O. Determine the nature of $f = s \circ r$.

SOLUTION

Orientation: s is indirect, r is direct, so f is indirect: a reflection or a glide reflection. Fixed points: $f(O) = s(r(O)) = s(O) = O$, so O is fixed. An indirect isometry with a fixed point is a reflection (a glide reflection has none). Axis: take $A \notin \Delta$, $A \neq O$. Then $r(A) = A'$ with $A\hat{O}A' = 90^\circ$, and $f(A) = s(A') = A''$. The axis of f is the perpendicular bisector of $[AA'']$, which passes through O: it is the line through O making angle $\frac{\pi}{4}$ with Δ. Hence f is the reflection across the line through O inclined at $\frac{\pi}{4}$ to Δ.

WORKED EXAMPLE 6 **Glide reflection $\circ$ translation**

Let g be the glide reflection with axis $\Delta : y = 0$ and glide vector $\vec{u}$ of affix 2 (i.e. $g : z' = \bar{z}+2$), and t the translation by vector of affix i. Determine the nature of $h = g \circ t$.

SOLUTION

Complex form: $t(z) = z + i$, so $h(z) = g(z + i) = \overline{z + i} + 2 = \bar{z} + 2 - i$. This has the form $z' = a\bar{z} + b$ with $|a| = 1$: h is indirect. Fixed points? $z = \bar{z} + 2 - i$. Writing $z = x + iy$: $x + iy = x - iy + 2 - i$ gives $0 = 2$ on the real parts — impossible. No fixed point. An indirect isometry with no fixed point is a glide reflection. Its axis is found from midpoints: the midpoint of $[M\, h(M)]$ has affix $\frac{z + \bar{z} + 2 - i}{2} = x + 1 - \frac{i}{2}$, which describes the line $y = -\frac{1}{2}$. Squaring: $h \circ h(z) = \overline{\bar{z} + 2 - i} + 2 - i = z + 2 + i + 2 - i = z + 4$, a translation by affix 4, so the glide vector has affix $\frac{4}{2} = 2$. Conclusion: h is the glide reflection of axis $y = -\frac{1}{2}$ and glide vector of affix 2 (i.e. horizontal translation by 2).

METHOD **GCE exam technique — "deduce the nature of $f \circ g$"**

1. Use the parity rule to decide direct/indirect immediately.
2. Compute the images of two or three well-chosen points (the centre of a rotation, a point of the axis, the origin) instead of heavy algebra.
3. Count fixed points among those images, then conclude with the classification table.
4. Finish by giving the characteristic elements (centre, angle, axis, vector).

WARNING **Frequent traps at the GCE**

- Saying "no fixed point ⇒ translation": false, a glide reflection also has none.
- Forgetting that the identity counts as a translation (by $\vec{0}$) and as a rotation (of angle 0).
- Claiming rotation ∘ rotation is always a rotation: if the angles sum to $0 \pmod{2\pi}$, it is a translation.
- Giving the orientation class when the question asks for the precise nature with characteristic elements.

EXERCISE 3 **Practice**

1. Classify $f : z' = -i\bar{z} + 1 + i$ (nature and characteristic elements).
2. r_1 and r_2 are rotations of angles $\frac{\pi}{3}$ and $-\frac{\pi}{3}$ about distinct centres. What is the nature of

$r_1 \circ r_2$?

3. s_1, s_2 are reflections across perpendicular lines meeting at O. Give the nature of $s_2 \circ s_1$.

SOLUTION **Exercise 1**

1. $|-i| = 1$ with a conjugate: indirect. Fixed points: $z = -i\bar{z} + 1 + i$; with $z = x + iy$, $x + iy = -i(x - iy) + 1 + i = -y + 1 + i(1 - x)$, so $x = 1 - y$ and $y = 1 - x$: the whole line $x + y = 1$ is fixed. Hence f is the reflection across the line $x + y = 1$. 2. Angles sum to 0: $r_1 \circ r_2$ is a translation. 3. Two reflections across intersecting axes: a rotation about O of angle $2 \times 90^\circ = 180^\circ$, i.e. the half-turn (central symmetry) about O.

KEY POINTS **Key points of the section**

- Every plane isometry is a translation, rotation, reflection or glide reflection — and is a composite of at most three reflections.
- Direct isometries (translation, rotation): $z' = az + b$, $|a| = 1$, preserve orientation. Indirect (reflection, glide reflection): $z' = a\bar{z} + b$, $|a| = 1$, reverse it.
- An isometry is determined by the images of three non-collinear points.
- To classify: fixed points first, then orientation, then characteristic elements.
- Parity rule: same type $\circ$ same type = direct; mixed = indirect.

DID YOU KNOW? **Why four and only four?**

The classification of plane isometries is a model of mathematical economy: a single hypothesis (preservation of distances) forces every transformation of the plane into one of four rigid "moves". The same spirit drives modern crystallography, where the symmetries of tilings and crystals — think of the repetitive patterns on Ndop or Bamileke traditional fabrics — are classified into exactly 17 "wallpaper groups", all built from these same four isometries.

4 HOMOTHETY: DEFINITION, CONSTRUCTION AND BASIC PROPERTIES

In the previous section we completed the classification of plane isometries: every transformation that preserves distances is a translation, a rotation, a reflection or a glide reflection. We now leave the world of isometries. Photocopiers in a Yaoundé business centre offer "enlarge to 141%" or "reduce to 71%"; the map of Cameroon hanging in your classroom is a reduced copy of the country; the shadow cast by a lamp grows as you move your hand towards the bulb. In each case the shape is preserved but the size changes, and all lengths are multiplied by the same number. The mathematical model of this situation is the **homothety**, the simplest plane transformation that is not an isometry. It is the building block of all plane similarities, which we shall study later in this chapter.

4.1 Definition of a Homothety

EXPERIMENT **From a lamp shadow to a formula**

Place a point source of light O and a small object at a point M. The shadow of M on a screen is a point M', and O, M, M' are always aligned. If the screen is twice as far from O as the object, then $OM' = 2\,OM$ and M' lies on the same side of O as M: in vector language $\vec{OM'} = 2\vec{OM}$. If we could place the screen on the other side of O, we would obtain $\vec{OM'} = -2\vec{OM}$. The whole geometry of the situation is captured by one centre O and one number k.

DEFINITION **Homothety of centre O and ratio k**

Let O be a fixed point of the plane and k a non-zero real number. The **homothety** of **centre** O and **ratio** k, denoted $h(O, k)$, is the plane transformation h which maps every point M to the point $M' = h(M)$ defined by

$$\vec{OM'} = k\,\vec{OM}.$$

The centre O is the unique fixed point when $k \neq 1$.

Two special cases are already familiar:

- If $k = 1$, then $\vec{OM'} = \vec{OM}$, so $M' = M$ for every M: $h(O, 1)$ is the **identity**.
- If $k = -1$, then $\vec{OM'} = -\vec{OM}$, so O is the midpoint of $[MM']$: $h(O, -1)$ is the **central symmetry** about O.

WARNING **The ratio must be non-zero**

The ratio $k = 0$ is excluded: it would send every point of the plane onto O, which is not a

transformation (it is not bijective). Whenever you write $h(O, k)$ in an examination, keep in mind $k \in \mathbb{R}^*$.

4.2 Construction of the Image of a Point

METHOD **Constructing $M' = h(M)$ given O, k and M**

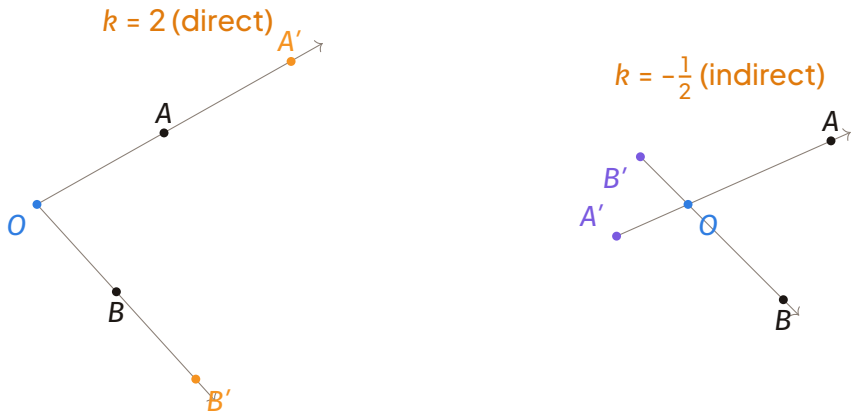
1. Draw the line (OM) and the ray $[OM)$ starting at O through M.
2. Compute the distance $OM' = |k| \cdot OM$.
3. If $k > 0$, place M' on the ray $[OM)$ (same side as M). If $k < 0$, place M' on the opposite ray.
4. Check: O, M, M' must be collinear, with $\vec{OM'} = k\vec{OM}$.

DEFINITION **Direct and indirect homothety**

A homothety $h(O, k)$ is called **direct** when $k > 0$: each point and its image lie on the same side of O. It is called **indirect** when $k < 0$: each point and its image lie on opposite sides of O. An indirect homothety is the composite of the direct homothety $h(O, |k|)$ with the central symmetry about O:

$$h(O, k) = h(O, -1) \circ h(O, |k|) \qquad (k < 0),$$

since $k\vec{OM} = -\left(|k|\vec{OM}\right)$.



Images under $h(O, 2)$ (same side of O) and under $h(O, -\frac{1}{2})$ (opposite side, halved distances).

WARNING **The classic trap with $k < 0$**

For a negative ratio the image lies on the opposite ray. Many candidates construct M' on the same side as M and lose all the construction marks. Always write the sign analysis first: "$k = -\frac{1}{2} < 0$, so $M' \in$ ray opposite to $[OM)$, with $OM' = \frac{1}{2}OM$."

4.3 Basic Properties: Distances, Areas, Collinearity, Angles

PROPERTY **Multiplication of distances and areas**

Let $h = h(O, k)$ and let $A' = h(A)$, $B' = h(B)$. Then

$$\vec{A'B'} = k\,\vec{AB}, \qquad A'B' = |k|\,AB,$$

and every area is multiplied by k^2: $\mathcal{A}' = k^2\,\mathcal{A}$.

Proof (examinable). By the Chasles relation,

$$\vec{A'B'} = \vec{A'O} + \vec{OB'} = -\vec{OA'} + \vec{OB'} = -k\vec{OA} + k\vec{OB} = k\left(\vec{OB} - \vec{OA}\right) = k\vec{AB}.$$

Taking norms gives $A'B' = |k|\,AB$. For areas, note that a triangle ABC has area $\mathcal{A} = \frac{1}{2}\,AB \cdot AC\,|\sin B\hat{A}C|$; since $A'B' = |k|AB, A'C' = |k|AC$ and the angle is preserved (see below), $\mathcal{A}' = \frac{1}{2}|k|^2 AB \cdot AC\,|\sin B\hat{A}C| = k^2\mathcal{A}$. Any polygonal region is a union of triangles, so the result extends. ■

PROPERTY **Conservation properties**

A homothety preserves:

- **collinearity**: if A, B, C are aligned, so are A', B', C';
- **barycentres** (hence midpoints): if G is the barycentre of $\{(A, \alpha), (B, \beta)\}$, then $G' = h(G)$ is the barycentre of $\{(A', \alpha), (B', \beta)\}$;
- **angles**: $A'\hat{B}'C' = A\hat{B}C$ (geometric angles; oriented angles are preserved because $\vec{A'B'} = k\vec{AB}$ rotates no vector);
- **parallelism** and **orthogonality**.

Indeed, if G is the midpoint of $[AB]$, then $\vec{OG} = \frac{1}{2}(\vec{OA} + \vec{OB})$, so $\vec{OG'} = k\vec{OG} = \frac{1}{2}(\vec{OA'} + \vec{OB'})$, which says exactly that G' is the midpoint of $[A'B']$.

4.4 Image of a Line and of a Circle

PROPERTY **Image of a line**

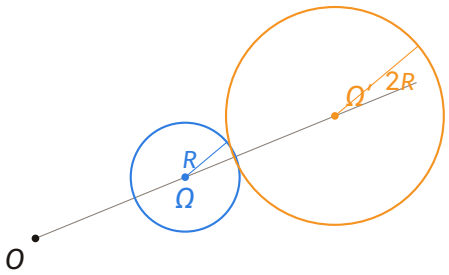
The image of a line (d) under $h(O,k)$ is a line (d') parallel to (d). If $O \in (d)$, then $(d') = (d)$: every line through the centre is globally invariant.

Proof. Take two distinct points A, B on (d). Any point $M \in (d)$ satisfies $\vec{AM} = t\vec{AB}$ for some $t \in \mathbb{R}$. Then $\vec{A'M'} = k\vec{AM} = t(k\vec{AB}) = t\vec{A'B'}$, so M' lies on the line $(A'B')$, which is parallel to $(AB) = (d)$ since $\vec{A'B'} = k\vec{AB}$. If $O \in (d)$, then for $M \in (d)$ the vector $\vec{OM'} = k\vec{OM}$ is still along (d), so $(d') = (d)$. ■

PROPERTY **Image of a circle**

The image under $h(O,k)$ of a circle C of centre Ω and radius R is the circle C' of centre $\Omega' = h(\Omega)$ and radius $R' = |k|R$. In particular O, Ω, Ω' are collinear, with $\vec{O\Omega'} = k\,\vec{O\Omega}$.

Proof. $M \in C \iff \Omega M = R$. Since homothety multiplies all distances by $|k|$, $\Omega' M' = |k|\,\Omega M = |k|R$, so M' describes the circle of centre Ω' and radius $|k|R$. ■



$h(O,2)$: $\vec{O\Omega'} = 2\vec{O\Omega}$, $R' = 2R$

Image of a circle under $h(O,2)$: aligned centres, doubled radius.

PROPERTY **Characteristic property**

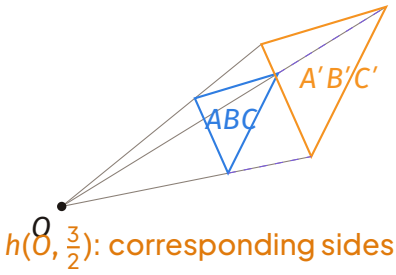
A plane transformation that maps every line to a parallel line is either a **translation** or a **homothety**. If moreover it has a fixed point O, it is the homothety of centre O (or the identity).

This converse is the standard tool for recognising a homothety in examination problems: show that every line is sent to a parallel line and exhibit a fixed point.

4.5 Image of a Figure: Enlargement and Reduction

DEFINITION **Enlargement, reduction, scale factor**

Under $h(O, k)$, a figure F is mapped to a figure F' of the same shape whose lengths are multiplied by $|k|$. We speak of an **enlargement** when $|k| > 1$, of a **reduction** when $|k| < 1$; $|k|$ is the **scale factor**. When $|k| = 1$ ($k = \pm 1$) the homothety is an isometry (identity or central symmetry).



$h(O, \frac{3}{2})$: corresponding sides are parallel

Triangle ABC and its image under $h(O, \frac{3}{2})$: $(A'B') \parallel (AB)$, etc., and all lengths are multiplied by $\frac{3}{2}$.

WORKED EXAMPLE 7 **Distances, area and image of a circle**

Let $h = h(O, -\frac{3}{2})$. Given $AB = 4$ cm, a triangle ABC of area 6 cm^2, and a circle $C(\Omega, R)$ with $R = 2$ cm and $O\Omega = 5$ cm: find $A'B'$, the area of $A'B'C'$, the radius of $C' = h(C)$ and the distance $O\Omega'$.

SOLUTION

Solution. Here $k = -\frac{3}{2}$, so $|k| = \frac{3}{2}$ and $k^2 = \frac{9}{4}$.

$$A'B' = |k|\,AB = \tfrac{3}{2} \times 4 = 6 \text{ cm},$$
$$A' = k^2 A = \tfrac{9}{4} \times 6 = 13.5 \text{ cm}^2,$$
$$R' = |k|R = \tfrac{3}{2} \times 2 = 3 \text{ cm},$$
$$O\Omega' = |k|\,O\Omega = \tfrac{3}{2} \times 5 = 7.5 \text{ cm}.$$

Because $k < 0$, Ω' lies on the ray opposite to $[O\Omega)$, and $A'B'C'$ is an enlargement (scale factor $\frac{3}{2} > 1$) of ABC.

EXEMPLE **Recognising a homothety**

Two circles $C(\Omega, 2)$ and $C'(\Omega', 6)$ are such that O, Ω, Ω' are aligned with $\vec{O\Omega'} = 3\vec{O\Omega}$. Since the radii satisfy $6 = 3 \times 2$, the homothety $h(O, 3)$ maps C onto C': the point O is a centre of

similitude of the two circles. (There is a second one, giving ratio -3; this idea returns in the study of similarities.)

WARNING **Areas are multiplied by k^2, not by k**

A very frequent error at the GCE: "the ratio is 2, so the area is doubled." Wrong — lengths are multiplied by $|k|$, areas by k^2 (and volumes by $|k|^3$ in space). Also remember k^2 is positive even for an indirect homothety: an area is never negative.

KEY POINTS **Key points**

- $h(O, k)$: $\vec{OM'} = k\vec{OM}$, with $k \neq 0$; O is the only fixed point if $k \neq 1$.
- $k > 0$: direct homothety (same side); $k < 0$: indirect homothety (opposite side), equal to $h(O, |k|)$ composed with the central symmetry about O.
- $h(O, 1) = \mathrm{id}$; $h(O, -1)$ = central symmetry about O.
- Distances are multiplied by $|k|$, areas by k^2; collinearity, midpoints, barycentres, angles and parallelism are preserved.
- Image of a line: a parallel line (lines through O are invariant). Image of a circle (Ω, R): circle $(\Omega', |k|R)$ with $\vec{O\Omega'} = k\vec{O\Omega}$.
- $|k| > 1$: enlargement; $|k| < 1$: reduction. A transformation sending every line to a parallel line is a translation or a homothety.

EXERCISE 4 **Construction and properties**

Let OAB be a triangle with $OA = 3$ cm, $OB = 4$ cm and $A\hat{O}B = 60^\circ$.

1. Construct $A' = h(A)$ and $B' = h(B)$ where $h = h(O, -2)$.
2. Compute $A'B'$ given that $AB = \sqrt{13}$ cm.
3. Show that $(A'B') \parallel (AB)$ and compute the area of $OA'B'$ knowing that the area of OAB is $3\sqrt{3}\,\mathrm{cm}^2$.

SOLUTION **Exercise 1**

1. Since $k = -2 < 0$, A' is on the ray opposite to $[OA)$ with $OA' = 2 \times 3 = 6$ cm; similarly B' is opposite to $[OB)$ with $OB' = 8$ cm.
2. $A'B' = |k|\,AB = 2\sqrt{13}$ cm.
3. $\vec{A'B'} = -2\vec{AB}$, so the vectors are collinear and $(A'B') \parallel (AB)$. The area is multiplied by $k^2 = 4$: $A' = 4 \times 3\sqrt{3} = 12\sqrt{3}$ cm^2.

DID YOU KNOW? **Homotheties around us**

The "zoom" of a camera lens, the scaling of a technical drawing at 1/50 used by builders in Douala, and the pantograph — a hinged parallelogram invented in the 17th century to copy drawings at a different scale — are all physical realisations of homotheties. In the next sections we shall describe homotheties analytically and with complex numbers, and then compose them, which will lead us to the general notion of plane similarity.

5 ANALYTICAL AND COMPLEX REPRESENTATION OF HOMOTHETY

In the previous sections we defined a homothety geometrically: $h(O, k)$ sends each point M to the point M' on the line (OM) such that $\vec{OM'} = k\,\vec{OM}$. Geometry alone, however, becomes heavy when we must compute the images of many points, compose several homotheties, or recognise a homothety hidden inside a formula. The remedy is to translate the vector relation into **coordinates** and into **complex numbers**. This is exactly the spirit of the GCE Advanced Level paper: a transformation is given by a formula such as $z' = 2z - 3 + i$, and you must identify it, find its centre and ratio, and compute images quickly. This section gives you all the tools.

5.1 Analytical Expression in Cartesian Coordinates

From the vector relation to coordinates.

Let the plane be equipped with an orthonormal frame $(O_0; \vec{i}, \vec{j})$. Suppose the centre of the homothety is $O(a, b)$ and the ratio is $k \in \mathbb{R}$, $k \neq 0$. Let $M(x, y)$ and $M'(x', y') = h(M)$. By definition,

$$\vec{OM'} = k\,\vec{OM}.$$

In coordinates, $\vec{OM} = (x - a,\ y - b)$ and $\vec{OM'} = (x' - a,\ y' - b)$. The vector equality therefore gives

$$x' - a = k(x - a) \qquad \text{and} \qquad y' - b = k(y - b).$$

Expanding and rearranging:

PROPERTY **Analytical expression of a homothety**

The homothety $h(O, k)$ with centre $O(a, b)$ and ratio k sends $M(x, y)$ to $M'(x', y')$ where

$$x' = kx + (1 - k)a, \qquad y' = ky + (1 - k)b.$$

Equivalently, in vector form with position vectors $\vec{p} = \vec{OM}$, $\vec{p}\,' = \vec{OM'}$ and $\vec{o} = \vec{OO}$:

$$\vec{p}\,' = k\,\vec{p} + (1 - k)\,\vec{o}.$$

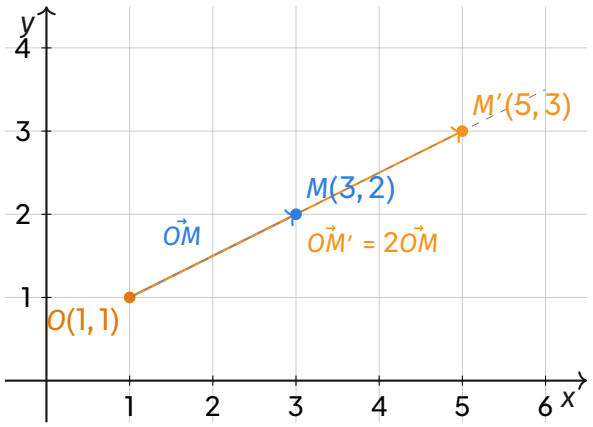
Notice the shape of the formula: the coordinates of M are multiplied by k, and a constant vector $(1 - k)\vec{o}$ is added. When $k = 1$ the constant vanishes and $M' = M$: the homothety of ratio 1 is the **identity**. When $k = -1$, we get $x' = -x + 2a$, $y' = -y + 2b$, which is exactly the **central symmetry** of centre O.

EXEMPLE **Computing an image with coordinates**

Let h be the homothety of centre $O(1, 1)$ and ratio $k = 2$. Find the image of $M(3, 2)$.
We apply the formula with $a = b = 1$ and $1 - k = -1$:

$$x' = 2(3) + (-1)(1) = 5, \qquad y' = 2(2) + (-1)(1) = 3.$$

Hence $M'(5, 3)$. Check: $\vec{OM} = (2, 1)$ and $\vec{OM'} = (4, 2) = 2\vec{OM}$. ✓



Homothety of centre $O(1, 1)$ and ratio $k = 2$: $M(3, 2) \mapsto M'(5, 3)$, with O, M, M' collinear.

5.2 Complex Representation of a Homothety

Why complex numbers?

Identify the plane with the complex plane: each point $M(x, y)$ has affix $z = x + iy$. The two real equations $x' = kx + (1 - k)a$ and $y' = ky + (1 - k)b$ then merge into one single complex equation.

Indeed, writing $\omega = a + ib$ for the affix of the centre,

$$z' = x' + iy' = k(x + iy) + (1 - k)(a + ib) = kz + (1 - k)\omega.$$

PROPERTY **Complex expression of a homothety — key formula**

The homothety $h(O, k)$, where O has affix ω and $k \in \mathbb{R}$, $k \neq 0$, has complex expression

$$z' = kz + (1 - k)\omega.$$

Conversely, every transformation of the form $z' = az + b$ with $a \in \mathbb{R}$, $a \neq 1$, is a homothety of ratio a whose centre has affix $\omega = \dfrac{b}{1 - a}$. (If $a = 1$ and $b \neq 0$, it is a translation.)

The proof of the converse is instructive and examinable. Suppose $z' = az + b$ with a real, $a \neq 1$. Look for a fixed point: $z = az + b$ gives $z(1 - a) = b$, hence the unique fixed point $\omega = \dfrac{b}{1 - a}$. Then

$$z' - \omega = az + b - \omega = az + (1 - a)\omega - \omega = a(z - \omega),$$

which says exactly $\vec{OM'} = a\,\vec{OM}$ in complex form: the transformation is the homothety of centre ω and ratio a. ■

KEY POINTS **The two faces of the formula**

- To write the expression of a known homothety: $z' = kz + (1 - k)\omega$.
- To identify a homothety from $z' = az + b$ (a real, $\neq 1$): ratio $= a$, centre $= \dfrac{b}{1 - a}$.

METHOD **Identifying a homothety from its complex expression**

Given $z' = az + b$:

1. Check that a is real. If not, it is not a pure homothety (see the warning below).
2. If $a = 1$: the map is a translation by the vector of affix b (or the identity if $b = 0$).
3. If $a \neq 1$: it is a homothety of ratio $k = a$.
4. Find the centre by solving the fixed-point equation $z = az + b$, i.e. $\omega = \dfrac{b}{1 - a}$.

WARNING **The coefficient of z must be real**

The formula $z' = kz + (1 - k)\omega$ is only valid when $k \in \mathbb{R}$. If the coefficient of z is a non-real

complex number, for example $z' = 2iz + 1$, the transformation also rotates the plane: it is a **similarity** (here a rotation composed with a homothety), not a pure homothety. Always check the nature of the coefficient before concluding.

5.3 Collinearity of O, M, M' and Fast Computation of Images

The complex form makes the fundamental geometric property transparent. From $z' - \omega = k(z - \omega)$, the vectors $\vec{OM}$ and $\vec{OM'}$ have affixes that are real multiples of each other; hence they are collinear.

PROPERTY **Fundamental property**

For a homothety $h(O, k)$, the points O, M and $M' = h(M)$ are always collinear, and $\vec{OM'} = k\,\vec{OM}$. In complex form: $z' - \omega = k(z - \omega)$, so

$$\arg\left(\frac{z' - \omega}{z - \omega}\right) = 0 \pmod{\pi},$$

since the quotient is the real number k.

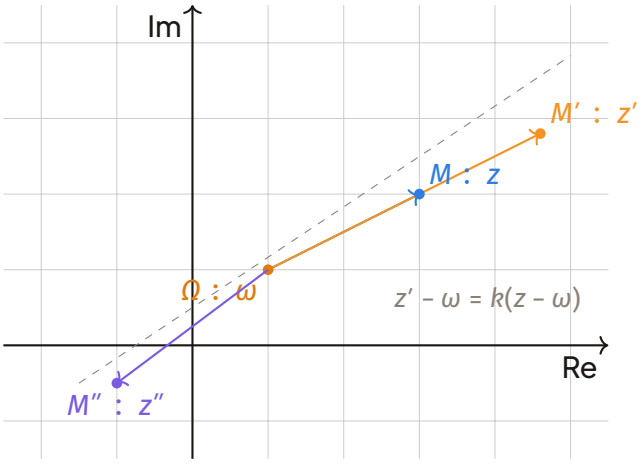
This gives a rapid computational strategy: to find the image of M of affix z, compute $z' = \omega + k(z - \omega)$ — subtract the centre, multiply by the ratio, add the centre back.

EXEMPLE **Fast image computation**

Let h have centre $\omega = 1 + i$ and ratio $k = -\frac{1}{2}$. Image of M with affix $z = 3 + 3i$:

$$z' = \omega + k(z - \omega) = (1 + i) - \tfrac{1}{2}\big((3 + 3i) - (1 + i)\big) = (1 + i) - \tfrac{1}{2}(2 + 2i) = 0.$$

So M' has affix 0 (the origin of the frame). Check: $\vec{OM}$ has affix $2 + 2i$, and $-\frac{1}{2}(2 + 2i) = -1 - i$, so M' has affix $\omega + (-1 - i) = 0$. ✓



Argand diagram: Ω, M, M' are collinear since $z' - \omega = k(z - \omega)$ with k real. Here $k = 1.8$ for M' and $k = -1$ for M''.

5.4 Recovering Centre and Ratio: a GCE-Style Worked Problem

WORKED EXAMPLE 8 **Identifying $z' = 2z - 3 + i$**

The transformation T of the complex plane is defined by $z' = 2z - 3 + i$. Show that T is a homothety and determine its centre and ratio.

SOLUTION

Solution.
Step 1: nature of the coefficient. The expression has the form $z' = az + b$ with $a = 2 \in \mathbb{R}$ and $b = -3 + i$. Since a is real and $a \neq 1$, T is a homothety of ratio $k = 2$.
Step 2: the centre is the unique fixed point. Set $z' = z$:

$$z = 2z - 3 + i \implies -z = -3 + i \implies z = 3 - i.$$

So the centre is Ω of affix $\omega = 3 - i$.
Step 3: verification (recommended in the exam). Compute

$$z' - \omega = 2z - 3 + i - (3 - i) = 2z - 6 + 2i = 2(z - 3 + i) = 2(z - \omega).$$

Thus $z' - \omega = 2(z - \omega)$, i.e. $\Omega\vec{M}' = 2\,\Omega\vec{M}$, confirming that $T = h(\Omega, 2)$ with $\omega = 3 - i$. ■
Remark. Using the formula $\omega = \dfrac{b}{1 - a}$ directly: $\omega = \dfrac{-3 + i}{1 - 2} = \dfrac{-3 + i}{-1} = 3 - i$. Same answer.

METHOD **Exam tip (GCE): fixed point first**

The centre of a homothety is its unique fixed point. Rather than memorising $\omega = \dfrac{b}{1-a}$, simply set $z' = z$ in the given expression and solve. This is faster, works even when you have forgotten the formula, and the equation $z' - \omega = k(z - \omega)$ then gives the ratio for free.

5.5 Direct and Indirect Homotheties

The sign of the ratio k determines the relative position of M and M' with respect to the centre O.

PROPERTY **Direct and indirect homothety**

- If $k > 0$ ($k \neq 1$): the homothety is **direct**. The points M and M' lie on the same ray starting at O. The orientation of any triangle is preserved.
- If $k < 0$: the homothety is **indirect**. The points M and M' lie on opposite sides of O. The orientation of any triangle is reversed.
- The special case $k = -1$ is the central symmetry (half-turn), which is indirect.

EXEMPLE **Direct vs indirect**

- $h(O, 3)$: direct. M and M' are on the same side of O, $OM' = 3\,OM$.
- $h(O, -\frac{1}{2})$: indirect. M' is on the opposite side of O from M, and $OM' = \frac{1}{2}OM$.

5.6 Composition of Two Homotheties (Complex Approach)

The complex representation makes composition almost trivial. Let $h_1 = h(\omega_1, k_1)$ and $h_2 = h(\omega_2, k_2)$ with $k_1, k_2 \in \mathbb{R}^*$. Their complex forms are

$$z_1 = k_1 z + (1 - k_1)\omega_1, \qquad z' = k_2 z_1 + (1 - k_2)\omega_2.$$

Substituting the first into the second:

$$z' = k_2\big(k_1 z + (1 - k_1)\omega_1\big) + (1 - k_2)\omega_2 = (k_1 k_2)z + \big(k_2(1 - k_1)\omega_1 + (1 - k_2)\omega_2\big).$$

PROPERTY **Composition of two homotheties**

Let $h_1 = h(\omega_1, k_1)$ and $h_2 = h(\omega_2, k_2)$ with $k_1, k_2 \in \mathbb{R}^*$. The composition $h_2 \circ h_1$ is:

- A homothety of ratio $k = k_1k_2$ if $k_1k_2 \neq 1$. Its centre ω satisfies

$$\omega = \frac{k_2(1-k_1)\omega_1 + (1-k_2)\omega_2}{1-k_1k_2}.$$

- A translation by the vector of affix $k_2(1-k_1)\omega_1 + (1-k_2)\omega_2$ if $k_1k_2 = 1$ (and this vector is non-zero).
- The identity if $k_1k_2 = 1$ and the translation vector is zero.

EXEMPLE **Composition: homothety then homothety**

Let $h_1 = h(0, 2)$ (centre 0, ratio 2) and $h_2 = h(1+i, 3)$ (centre $1+i$, ratio 3). Then $h_2 \circ h_1$ has ratio $k = 2 \times 3 = 6 \neq 1$, so it is a homothety. Centre $\omega = \dfrac{3(1-2)\cdot 0 + (1-3)(1+i)}{1-6} = \dfrac{-2(1+i)}{-5} = \dfrac{2}{5}(1+i)$. Check: $h_1(z) = 2z$, $h_2(z_1) = 3z_1 + (1-3)(1+i) = 3z_1 - 2 - 2i$. Composition: $z' = 3(2z) - 2 - 2i = 6z - 2 - 2i = 6z + (1-6)\cdot\frac{2}{5}(1+i)$. ✓

EXEMPLE **Composition giving a translation**

Let $h_1 = h(0, 2)$ and $h_2 = h(1, \frac{1}{2})$. Then $k_1k_2 = 1$. $h_1(z) = 2z$, $h_2(z_1) = \frac{1}{2}z_1 + (1-\frac{1}{2})\cdot 1 = \frac{1}{2}z_1 + \frac{1}{2}$. Composition: $z' = \frac{1}{2}(2z) + \frac{1}{2} = z + \frac{1}{2}$. This is a translation by vector $\frac{1}{2}$ (affix 0.5).

5.7 Links with the Other Plane Transformations

The complex formula $z' = kz + (1-k)\omega$ unifies several transformations you already know:

Value of k	Transformation	Complex form
$k = 1$	Identity	$z' = z$
$k = -1$	Central symmetry, centre ω	$z' = -z + 2\omega$
$k > 0$, $k \neq 1$	Direct homothety	$z' = kz + (1-k)\omega$
$k < 0$, $k \neq -1$	Indirect homothety	$z' = kz + (1-k)\omega$

What about **translations**? A translation $z' = z + b$ has coefficient 1 in front of z: it is the case $k = 1$ with a non-zero constant term. One may regard a translation as the limit of homotheties whose centre ω goes to infinity while $k \to 1$ in such a way that $(1-k)\omega \to b$. More concretely, a translation is the companion of homotheties inside the family of maps $z' = az + b$: the homotheties are exactly those with a fixed point, the translations those without one. This point of view will be essential when we classify plane similarities in the next sections.

WARNING **Common mistakes to avoid**

- Confusing b with the centre: in $z' = 2z - 3 + i$, the centre is not $-3 + i$; you must divide by $1 - a$ (or solve $z' = z$).
- Forgetting to check that the coefficient of z is real before declaring "homothety".
- Writing $z' = kz + (1 - k)\omega$ with ω the affix of the origin of the frame instead of the centre of the homothety.
- Sign errors when $k < 0$: then $1 - k > 1$, and M' lies on the opposite side of O from M.
- In composition, forgetting that the product of ratios $k_1 k_2$ determines the nature of the result (homothety if $\neq 1$, translation if $= 1$).

EXERCISE 5 **Practice**

1. Write the complex expression of the homothety of centre $\omega = -2 + i$ and ratio $k = 3$, then find the image of the point A of affix $1 - i$.
2. The map S is given by $z' = -\frac{1}{2}z + 6 - 3i$. Show that S is a homothety; give its centre and ratio.
3. Given $z' = 5z + 8 + 4i$, find the fixed point and deduce the centre and ratio of the homothety.
4. Is the map $z' = (1 + i)z + 2$ a homothety? Justify.
5. Let $h_1 = h(2i, -2)$ and $h_2 = h(1, 3)$. Find the complex expression of $h_2 \circ h_1$ and identify the resulting transformation (give its centre and ratio, or its translation vector).
6. A homothety h maps $A(1 + 2i)$ to $A'(4 + 5i)$ and $B(3 + i)$ to $B'(7 + 4i)$. Determine the centre and ratio of h using complex numbers.

SOLUTION **Exercise 1**

1. $z' = 3z + (1 - 3)(-2 + i) = 3z + 4 - 2i$. Image of A: $z'_A = 3(1 - i) + 4 - 2i = 7 - 5i$.
2. Coefficient $a = -\frac{1}{2} \in \mathbb{R}$, $a \neq 1$: homothety of ratio $k = -\frac{1}{2}$. Fixed point: $z = -\frac{1}{2}z + 6 - 3i \Rightarrow \frac{3}{2}z = 6 - 3i \Rightarrow \omega = 4 - 2i$. Check: $z' - \omega = -\frac{1}{2}z + 6 - 3i - 4 + 2i = -\frac{1}{2}z + 2 - i = -\frac{1}{2}(z - 4 + 2i) = -\frac{1}{2}(z - \omega)$. ✓
3. Fixed point: $z = 5z + 8 + 4i \Rightarrow -4z = 8 + 4i \Rightarrow \omega = -2 - i$; ratio $k = 5$.
4. No: the coefficient $1 + i$ is not real, so this is a similarity (rotation of angle $\frac{\pi}{4}$ composed with a homothety of ratio $\sqrt{2}$), not a pure homothety.

5. $h_1(z) = -2z + (1 - (-2))(2i) = -2z + 6i$. $h_2(z_1) = 3z_1 + (1 - 3)(1) = 3z_1 - 2$. Composition: $z' = 3(-2z + 6i) - 2 = -6z + 18i - 2$. Ratio $k = -6 \neq 1$, so it is a homothety. Centre $\omega = \dfrac{-2 + 18i}{1 - (-6)} = \dfrac{-2 + 18i}{7} = -\frac{2}{7} + \frac{18}{7}i$.

6. Let h have centre ω and ratio k. Then $z'_A - \omega = k(z_A - \omega)$ and $z'_B - \omega = k(z_B - \omega)$. Subtracting: $z'_A - z'_B = k(z_A - z_B)$. $z_A = 1 + 2i$, $z'_A = 4 + 5i$, $z_B = 3 + i$, $z'_B = 7 + 4i$. $z'_A - z'_B = -3 + i$, $z_A - z_B = -2 + i$. $k = \dfrac{-3 + i}{-2 + i} = \dfrac{(-3 + i)(-2 - i)}{(-2 + i)(-2 - i)} = \dfrac{6 + 3i - 2i - i^2}{4 + 1} = \dfrac{7 + i}{5} = 1.4 + 0.2i$. Since k is not real, no such homothety exists. (The data would correspond to a spiral similarity.)

KEY POINTS **Key points of the section**

- Analytical form: $x' = kx + (1 - k)a$, $y' = ky + (1 - k)b$; vector form: $\vec{p}\,' = k\vec{p} + (1 - k)\vec{o}$.
- Complex form: $z' = kz + (1 - k)\omega$, with k real and ω the affix of the centre.
- To identify: $z' = az + b$, $a \in \mathbb{R} \setminus \{1\} \Rightarrow$ homothety, ratio a, centre found by solving $z' = z$.
- O, M, M' are collinear because $z' - \omega = k(z - \omega)$ with k real.
- $k > 0$: direct homothety (preserves orientation); $k < 0$: indirect homothety (reverses orientation).
- $k = 1$: identity; $k = -1$: central symmetry; coefficient 1 with $b \neq 0$: translation.
- Composition $h_2 \circ h_1$: ratio k_1k_2. If $k_1k_2 \neq 1$, homothety with centre $\omega = \frac{k_2(1-k_1)\omega_1 + (1-k_2)\omega_2}{1-k_1k_2}$. If $k_1k_2 = 1$, translation (or identity).

6 COMPOSITION OF HOMOTHETIES AND ADVANCED PROPERTIES

In the previous section we learnt to describe a homothety analytically: $h(O, k)$ sends the point $M(z)$ to $M'(z')$ with $z' - \omega = k(z - \omega)$, where ω is the affix of the centre. This compact formula is a powerful tool: composing two homotheties now amounts to composing two affine maps of the complex plane, and the nature of the composite can be read directly from the algebra. That is exactly the programme of this section: we compose homotheties (same centre, then distinct centres), we handle the surprising special case where the product of the ratios equals 1, we mix homotheties with translations, reflections and rotations, and we finish with two jewels of geometry — the centres of similitude of two circles and the Euler line of a triangle.

6.1 Composition of Two Homotheties with the Same Centre

Observation. Enlarge a photograph from centre O by a factor 2, then enlarge the result from the same centre by a factor 3. A point M goes to M_1 with $\vec{OM_1} = 2\vec{OM}$, then to M' with $\vec{OM'} = 3\vec{OM_1} = 6\vec{OM}$. The combined effect is simply the homothety $h(O, 6)$: the ratios multiply.

PROPERTY **Composite of two homotheties with the same centre**

Let O be a point of the plane and k_1, k_2 two non-zero real numbers. Then

$$h(O, k_2) \circ h(O, k_1) = h(O, k_1 k_2).$$

In particular the composition is **commutative**: $h(O, k_2) \circ h(O, k_1) = h(O, k_1) \circ h(O, k_2)$. Moreover $h(O, k)^{-1} = h\left(O, \frac{1}{k}\right)$, so the homotheties of centre O form a **group** under composition.

Proof (examinable). For any point M, write $M_1 = h(O, k_1)(M)$ and $M' = h(O, k_2)(M_1)$. Then

$$\vec{OM'} = k_2\, \vec{OM_1} = k_2\left(k_1\, \vec{OM}\right) = k_1 k_2\, \vec{OM},$$

so $M' = h(O, k_1 k_2)(M)$ for every M. ■

EXEMPLE **Same centre**

$h\left(O, \frac{1}{2}\right) \circ h(O, -4) = h(O, -2)$: an indirect homothety of ratio -2. Note that $h(O, -1) \circ h(O, -1) = h(O, 1) = \mathrm{id}$: the central symmetry about O is its own inverse.

6.2 Composition of Two Homotheties with Distinct Centres

Observation. Now take two different centres O_1 and O_2. Apply $h_1 = h(O_1, k_1)$ then $h_2 = h(O_2, k_2)$. Using complex affixes with ω_1, ω_2 the affixes of the centres:

$$z \xrightarrow{h_1} z_1 = k_1 z + (1 - k_1)\omega_1 \xrightarrow{h_2} z' = k_2 z_1 + (1 - k_2)\omega_2.$$

Substituting,

$$z' = k_1 k_2\, z + \underbrace{k_2(1 - k_1)\omega_1 + (1 - k_2)\omega_2}_{\text{a fixed complex number } b}.$$

So the composite has the form $z' = k_1 k_2\, z + b$. Two cases arise, according to whether $k_1 k_2 = 1$ or not.

PROPERTY **Composite of two homotheties with distinct centres — proof examinable**

Let $h_1 = h(O_1, k_1)$ and $h_2 = h(O_2, k_2)$ with $O_1 \neq O_2$, and let $f = h_2 \circ h_1$.

- If $k_1 k_2 \neq 1$, then f is a **homothety** of ratio $k_1 k_2$ whose centre Ω lies on the line $(O_1 O_2)$.

- If $k_1k_2 = 1$, then f is a **translation** of vector $(1 - k_2)\,\vec{O_1O_2}$ (equivalently $(k_1 - 1)\,\vec{O_2O_1}$).

Proof. From the computation above, $z' = k_1k_2\,z + b$ with $b = k_2(1 - k_1)\omega_1 + (1 - k_2)\omega_2$.

Case $k_1k_2 \neq 1$. An affine map $z' = kz + b$ with $k \neq 1$ is the homothety of ratio k and centre of affix $\omega = \dfrac{b}{1 - k}$ (since $\omega = k\omega + b$). Here

$$\omega = \frac{k_2(1 - k_1)\omega_1 + (1 - k_2)\omega_2}{1 - k_1k_2},$$

a barycentre of ω_1 and ω_2 with real coefficients summing to $1 - k_1k_2 \neq 0$; hence $\Omega \in (O_1O_2)$.

Case $k_1k_2 = 1$. Then $k_2 = 1/k_1$ and $z' = z + b$: a translation. Its vector is obtained from b: with $k_2(1 - k_1) = k_2 - 1$,

$$b = (k_2 - 1)\omega_1 + (1 - k_2)\omega_2 = (1 - k_2)(\omega_2 - \omega_1),$$

so the translation vector is $(1 - k_2)\,\vec{O_1O_2}$. ■

METHOD **Finding the centre of the composite $h_2 \circ h_1$ when $k_1k_2 \neq 1$**

1. Compute the ratio: $k = k_1k_2$.
2. Choose a convenient point — the best choice is O_1 itself. Compute $P = h_1(O_1) = O_1$, then $P' = h_2(O_1)$: since h_1 fixes O_1, we get $f(O_1) = h_2(O_1)$.
3. The centre Ω is the fixed point of f; it lies on the line through a point and its image, and also on (O_1O_2). Concretely, Ω is the intersection of (O_1O_2) with the line through O_1 and $f(O_1)$ — or solve $\vec{\Omega f(O_1)} = k\,\vec{\Omega O_1}$.
4. Check with a second point if time allows.

WORKED EXAMPLE 9 **Composite with distinct centres**

Let $O_1(0, 0)$, $O_2(4, 0)$, $h_1 = h(O_1, 2)$ and $h_2 = h(O_2, \frac{1}{3})$. Determine the nature and the characteristic elements of $f = h_2 \circ h_1$.

SOLUTION

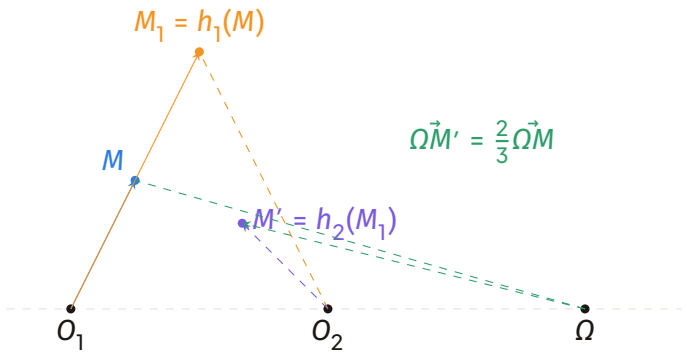
Solution. The product of the ratios is $k = 2 \times \frac{1}{3} = \frac{2}{3} \neq 1$, so f is a homothety of ratio $\frac{2}{3}$ with centre Ω on the x-axis (the line (O_1O_2)).

Image of O_1: $h_1(O_1) = O_1$, then $h_2(O_1) = P'$ with $\vec{O_2P'} = \frac{1}{3}\vec{O_2O_1}$, i.e. $P' = O_2 + \frac{1}{3}(O_1 - O_2) = \left(4 - \frac{4}{3}, 0\right) = \left(\frac{8}{3}, 0\right)$.

The centre $\Omega(x, 0)$ satisfies $\vec{\Omega P'} = \frac{2}{3}\vec{\Omega O_1}$:

$$\frac{8}{3} - x = \frac{2}{3}(0 - x) \implies \frac{8}{3} - x = -\frac{2}{3}x \implies \frac{8}{3} = \frac{1}{3}x \implies x = 8.$$

Hence $f = h\left(\Omega(8,0), \frac{2}{3}\right)$. Check with O_2: $h_1(O_2) = (8,0)$, $h_2(8,0) = O_2 + \frac{1}{3}(8-4,0) = \left(\frac{16}{3}, 0\right)$; and $f(O_2)$ should be $\Omega + \frac{2}{3}(O_2 - \Omega) = \left(8 - \frac{8}{3}, 0\right) = \left(\frac{16}{3}, 0\right)$. ✓



$f = h(O_2, \frac{1}{3}) \circ h(O_1, 2) = h(\Omega, \frac{2}{3})$: the centre Ω lies on the line (O_1O_2), and Ω, M, M' are collinear.

WARNING **Composition is not commutative in general**

When the centres differ, $h_2 \circ h_1 \neq h_1 \circ h_2$ in general: both are homotheties of the same ratio k_1k_2 but with different centres (both on (O_1O_2)). Commutativity holds only when $O_1 = O_2$. Always respect the order: in $h_2 \circ h_1$, it is h_1 that acts first.

6.3 The Special Case $k_1k_2 = 1$: a Translation

When $k_2 = \dfrac{1}{k_1}$, the composite $h(O_2, 1/k_1) \circ h(O_1, k_1)$ is a translation. Intuitively: the first homothety scales all distances by $|k_1|$, the second scales them back by $1/|k_1|$, so distances are preserved; and every line is mapped to a parallel line, so no point can be fixed — the only isometry with these features and no fixed point is a translation.

PROPERTY **Vector of the translation — exam tip**

If $k_1k_2 = 1$ with $O_1 \neq O_2$, then

$$h\left(O_2, \frac{1}{k_1}\right) \circ h(O_1, k_1) = t_{\vec{u}}, \qquad \vec{u} = \left(1 - \frac{1}{k_1}\right)\vec{O_1O_2} = (1 - k_2)\,\vec{O_1O_2}.$$

In particular $\vec{u}$ is parallel to (O_1O_2), and $\vec{u} = \vec{0}$ only if $k_1 = k_2 = 1$.

METHOD **GCE exam strategy when $k_1k_2 = 1$**

1. Do not look for a centre — there is none. State immediately that the composite is a translation.
2. To find the vector, compute the image of one easy point (usually O_1): $\vec{u} = O_1\vec{f(O_1)}$.
3. Alternatively, prove directly that for every M, the vector $M\vec{M''}$ is constant (independent of M): this is the definition of a translation and earns full marks.

WORKED EXAMPLE 10 **The case $k_1k_2 = 1$**

Let A and B be two distinct points, $h_1 = h(A, 3)$ and $h_2 = h\left(B, \frac{1}{3}\right)$. Determine $f = h_2 \circ h_1$.

SOLUTION

Solution. Since $3 \times \frac{1}{3} = 1$, f is a translation. Compute the image of A: $h_1(A) = A$, then $h_2(A) = A'$ with $B\vec{A'} = \frac{1}{3}\vec{BA}$. Hence

$$\vec{u} = \vec{AA'} = \vec{AB} + \vec{BA'} = \vec{AB} + \frac{1}{3}\vec{BA} = \frac{2}{3}\vec{AB}.$$

Therefore $f = t_{\vec{u}}$ with $\vec{u} = \frac{2}{3}\vec{AB}$, agreeing with the formula $(1 - \frac{1}{3})\vec{AB}$. Direct check for any M: $M\vec{M''} = \vec{MA} + \vec{AA'} + \vec{A'M''}$; since $\vec{A'M''} = \vec{AM_1} \cdot \frac{1}{3} - \ldots$ one verifies $M\vec{M''} = \frac{2}{3}\vec{AB}$, constant, as required.

6.4 Composing a Homothety with a Translation, a Reflection or a Rotation

Analytically, homotheties, translations and rotations all have the form $z' = az + b$ with $|a| = k$ (for a homothety $a = k \in \mathbb{R}$, for a rotation $a = e^{i\theta}$, for a translation $a = 1$); reflections have the form $z' = a\bar{z} + b$ with $|a| = 1$. Composing a homothety $h(O, k)$ with any of them multiplies all distances by $|k|$ while preserving angles (direct case) or reversing them (indirect case). The result is therefore never a plain isometry (unless $|k| = 1$) but a **similarity**.

KEY POINTS **Nature of the composite $h \circ g$ or $g \circ h$**

Let h be a homothety of ratio k ($|k| \neq 1$).

- $h \circ t_{\vec{u}}$ (or $t_{\vec{u}} \circ h$) is a **homothety** of ratio k (with a new centre). Indeed $z \mapsto kz + b$, $k \neq 1$, always has a fixed point.

- $h \circ r$ (or $r \circ h$), where r is a rotation of angle $\theta \not\equiv 0 \pmod{2\pi}$, is a **direct similarity**: $z \mapsto ke^{i\theta}z + b$ — a spiral similarity (rotation + homothety about the same fixed point).
- $h \circ s$ (or $s \circ h$), where s is a reflection, is an **indirect similarity**: $z \mapsto ke^{i\theta}\bar{z} + b$.

In all cases distances are multiplied by $|k|$ and the composite has exactly one fixed point (its centre).

EXEMPLE **Homothety and translation**

$h(O, 2) \circ t_{\vec{u}}$ sends z to $2(z + u) = 2z + 2u$: a homothety of ratio 2 whose centre Ω satisfies $\omega = 2\omega + 2u$, i.e. $\omega = -2u$. The centre has moved, the ratio has not.

6.5 Advanced Properties: Barycentres, Homothetic Polygons

PROPERTY **Homothety and barycentres**

Every homothety $h(O, k)$ preserves **barycentres**: if G is the barycentre of $\{(A_i, \alpha_i)\}$ with $\sum \alpha_i \neq 0$, then $h(G)$ is the barycentre of $\{(h(A_i), \alpha_i)\}$ with the same weights. In particular midpoints, centroids and centres of mass are mapped to their counterparts.

Proof. $\vec{OG'} = k\vec{OG} = \frac{k}{\sum \alpha_i} \sum \alpha_i \vec{OA_i} = \frac{1}{\sum \alpha_i} \sum \alpha_i \vec{OA'_i}$, so G' is the barycentre of the images with the same weights. ■

Consequences: a homothety preserves the directions of lines (a line maps to a parallel line), hence it preserves parallelism, orthogonality and angles; it maps a polygon to a **homothetic polygon** whose corresponding sides are parallel and proportional with factor $|k|$.

PROPERTY **Concurrency of lines joining corresponding vertices**

If $P' = h(P)$ is the image of a polygon P by $h(O, k)$, then all the lines joining corresponding vertices (AA'), (BB'), (CC'), … pass through the centre O: they are **concurrent**. Conversely, if two polygons have corresponding sides parallel and the lines joining corresponding vertices are concurrent at O, then one is the image of the other by a homothety of centre O.

This converse is a standard GCE tool: to prove two figures are homothetic, exhibit the concurrency point and check the constant ratio $\frac{OA'}{OA} = \frac{A'B'}{AB}$.

6.6 Application 1: Two Circles Are Always Homothetic

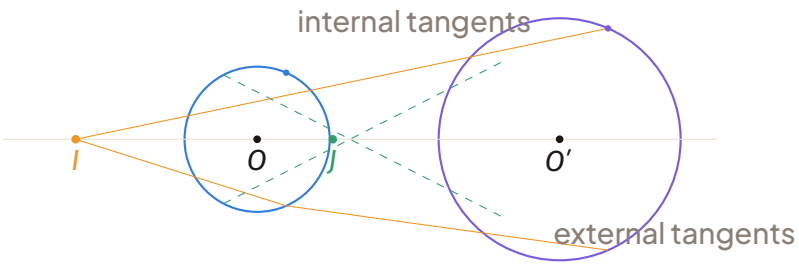
PROPERTY **Centres of similitude of two circles**

Let $C(O,R)$ and $C'(O',R')$ be two circles with $R \neq R'$. There are exactly two homotheties mapping C onto C':

- $h\left(I, \frac{R'}{R}\right)$, the **external centre of similitude** I, dividing $[OO']$ externally: $\vec{IO'} = \frac{R'}{R}\vec{IO}$ with $I \notin [OO']$;
- $h\left(J, -\frac{R'}{R}\right)$, the **internal centre of similitude** J, dividing $[OO']$ internally: $\vec{JO'} = -\frac{R'}{R}\vec{JO}$ with $J \in [OO']$.

Both I and J lie on the line of centres (OO'). The common external tangents meet at I; the common internal tangents (when they exist) meet at J. If $R = R'$ and $O \neq O'$, only the internal centre J (midpoint of $[OO']$, central symmetry) and a translation remain.

Why it works. Under $h(I, R'/R)$, $O \mapsto O'$ and radii scale by R'/R, so $C \mapsto C'$. A tangent from I to C touches at T; its image is a line through I tangent to C' at $T' = h(T)$ — hence the external tangents concur at I. The same argument with ratio $-R'/R$ gives J and the internal tangents.



Two circles with $R' \neq R$: the external centre of similitude I (external tangents) and the internal centre J (internal tangents), both on the line of centres (OO').

EXEMPLE **Locating the centres of similitude**

Circles $C(O,2)$ and $C'(O',6)$ with $OO' = 12$. Then $J \in [OO']$ with $OJ : JO' = R : R' = 1 : 3$, so $OJ = 3$; and I on line (OO') outside $[OO']$ with $IO' = 3\,IO$, giving I on the side of O with $OI = 6$. The two homotheties are $h(I, 3)$ and $h(J, -3)$.

6.7 Application 2: The Euler Line of a Triangle

Let ABC be a non-degenerate triangle, G its centroid, O its circumcentre, H its orthocentre, and C its circumcircle. Consider

$$h = h\left(G, -\tfrac{1}{2}\right).$$

1. **Step 1: h maps vertices to midpoints of the opposite sides.** Since $\vec{GA'} = -\frac{1}{2}\vec{GA}$ when A' is the

midpoint of $[BC]$ (a classical centroid relation, because $\vec{GA} + \vec{GB} + \vec{GC} = \vec{0}$ gives $\vec{GA} = -2\vec{GA'}$), we get $h(A) = A'$, $h(B) = B'$, $h(C) = C'$, the midpoints of the sides.

2. **Step 2: h maps the circumcircle to the nine-point circle.** The image of $C(O, R)$ is the circle of centre $O' = h(O)$ and radius $\frac{R}{2}$ passing through A', B', C': this is precisely the **nine-point circle** (circle through the three midpoints of the sides), whose centre $N = h(O)$ is the midpoint of $[OH]$.

3. **Step 3: the altitudes map to the perpendicular bisectors.** The altitude from A is the line through A perpendicular to (BC); its image is the line through A' parallel to it, i.e. perpendicular to (BC) through its midpoint: the perpendicular bisector of $[BC]$. Since the altitudes concur at H and a homothety preserves concurrency, their images concur at $h(H)$. But the perpendicular bisectors concur at O. Hence

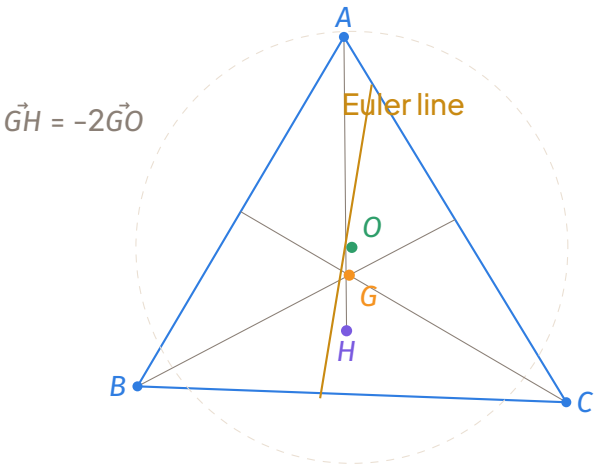
$$h(H) = O, \qquad \text{i.e.} \qquad \vec{GO} = -\frac{1}{2}\vec{GH} \iff \vec{OH} = 3\,\vec{OG}.$$

PROPERTY **Euler line**

In any non-equilateral triangle, the circumcentre O, the centroid G and the orthocentre H are collinear on the **Euler line**, with G between O and H and

$$\vec{GH} = -2\,\vec{GO}, \qquad OH = 3\,OG.$$

The homothety $h(G, -\frac{1}{2})$ maps $H \mapsto O$ and the circumcircle onto the nine-point circle; equivalently $h(G, -2)$ maps $O \mapsto H$.



The Euler line: $h(G, -\frac{1}{2})$ sends H to O and the circumcircle to the nine-point circle, forcing O, G, H to be collinear with $GH = 2\,GO$.

DID YOU KNOW? **Leonhard Euler**

The collinearity of O, G, H was published by Leonhard Euler in 1765. The nine-point circle — through the three midpoints of the sides, the three feet of the altitudes and the three midpoints of $[AH], [BH], [CH]$ — is a direct gift of the homothety $h(G, -\frac{1}{2})$ and of $h(H, \frac{1}{2})$. It is a beautiful example of how a single transformation can unify nine seemingly unrelated points.

WARNING **Common mistakes**

- Adding ratios instead of multiplying them: $h(O, k_2) \circ h(O, k_1)$ has ratio $k_1 k_2$, never $k_1 + k_2$.
- Forgetting the order: $h_2 \circ h_1$ means h_1 first; with distinct centres, reversing the order changes the centre.
- Looking for a centre when $k_1 k_2 = 1$: the composite is a translation — find its vector instead.
- Claiming the centre of the composite is the midpoint of $[O_1 O_2]$: it is a specific barycentre of O_1, O_2, generally not the midpoint.
- For the Euler line, writing $\vec{GH} = 2\vec{GO}$: the correct relation is $\vec{GH} = -2\vec{GO}$ (the homothety ratio $-\frac{1}{2}$ is negative).

EXERCISE 6 **Practice**

1. Let $O_1(1, 0)$, $O_2(5, 0)$, $h_1 = h(O_1, -2)$, $h_2 = h(O_2, 3)$. Determine the nature, ratio and centre of $h_2 \circ h_1$.
2. Let $A(0, 0)$, $B(6, 0)$. Show that $h(B, \frac{1}{2}) \circ h(A, 2)$ is a translation and give its vector.
3. Two circles have radii 3 and 5 and centres 8 cm apart. Locate their two centres of similitude on the line of centres.
4. In a triangle, $G(1, 1)$ and $O(4, 3)$ are the centroid and circumcentre. Find the orthocentre H.

SOLUTION **Exercise 1**

1. Ratio $k = -6 \neq 1$: a homothety. Image of O_1: $h_1(O_1) = O_1$, $h_2(O_1) = O_2 + 3(O_1 - O_2) = (5 - 12, 0) = (-7, 0)$. Centre $\Omega(x, 0)$: $-7 - x = -6(1 - x)$ gives $-7 - x = -6 + 6x$, so $x = -\frac{1}{7}$. Hence $h\left(\Omega\left(-\frac{1}{7}, 0\right), -6\right)$.
2. Ratios $2 \times \frac{1}{2} = 1$: translation. Vector $(1 - \frac{1}{2})\vec{AB} = \frac{1}{2}\vec{AB} = (3, 0)$.

3. Internal centre $J \in [OO']$ with $OJ : JO' = 3 : 5$, so $OJ = 3$ cm. External centre I outside $[OO']$ on the side of O with $IO' = \frac{5}{3}IO$: $IO' - IO = 8$ gives $\frac{2}{3}IO = 8$, $IO = 12$ cm.
4. $\vec{GH} = -2\vec{GO}$: $H = G - 2(O - G) = (1, 1) - 2(3, 2) = (-5, -3)$.

KEY POINTS **Key points of the section**

- $h(O, k_2) \circ h(O, k_1) = h(O, k_1 k_2)$ (same centre, commutative group).
- Distinct centres: $h_2 \circ h_1$ is a homothety of ratio $k_1 k_2$ with centre on (O_1O_2) if $k_1 k_2 \neq 1$; a translation of vector $(1 - k_2)\vec{O_1O_2}$ if $k_1 k_2 = 1$.
- Homothety composed with translation / rotation / reflection gives a similarity (homothety, spiral similarity, or indirect similarity respectively).
- Homotheties preserve barycentres and directions; lines joining corresponding vertices of homothetic polygons concur at the centre.
- Two circles of unequal radii admit exactly two centres of similitude I (ratio R'/R) and J (ratio $-R'/R$) on the line of centres.
- $h(G, -\frac{1}{2})$ sends circumcircle to nine-point circle and H to O: hence the Euler line with $\vec{GH} = -2\vec{GO}$.

7 PLANE SIMILARITIES: CONCEPT AND CHARACTERISTIC ELEMENTS

In the previous section we composed homotheties and discovered a striking phenomenon: the composite of two homotheties is either another homothety or a translation. In both cases, the resulting transformation shrinks or stretches every distance by the same factor. Photocopiers, zoom lenses, and the maps of Cameroon used by geographers in Yaoundé all share this feature: shapes are preserved, sizes are not. Transformations with this behaviour are called **similarities**, and they form the natural family unifying everything we have studied so far: isometries (ratio 1), homotheties, and their composites. This section gives the concept, the characteristic elements, and the first classification results.

7.1 Definition and First Examples

EXPERIMENT **From similar triangles to similar figures**

Draw a triangle ABC with $AB = 3$ cm, $BC = 4$ cm, $CA = 5$ cm. Now draw a second triangle

$A'B'C'$ with $A'B' = 4.5$ cm, $B'C' = 6$ cm, $C'A' = 7.5$ cm. Measure the angles of both triangles: they are equal in pairs, and every side of the second triangle is exactly 1.5 times the corresponding side of the first. The two triangles have the same shape but different sizes. A similarity is precisely the transformation that realises this passage from one figure to the other.

DEFINITION **Similarity of ratio k**

Let k be a strictly positive real number. A **similarity** of **ratio** k is a transformation s of the plane such that for all points M and N, with images $M' = s(M)$ and $N' = s(N)$,

$$M'N' = k \cdot MN.$$

A similarity of ratio $k = 1$ is an **isometry**.

EXEMPLE **Recognising similarities**

- Every translation, rotation, reflection and glide reflection is a similarity of ratio $k = 1$.
- Every homothety of ratio $\lambda \neq 0$ is a similarity of ratio $k = |\lambda|$.
- The composite of a homothety of ratio $|\lambda| = 2$ with a rotation is a similarity of ratio 2: the homothety multiplies distances by 2 and the rotation leaves them unchanged.

PROPERTY **Elementary properties of a similarity**

Let s be a similarity of ratio k. Then:

1. s is a bijection of the plane;
2. s preserves collinearity and midpoints: the image of a line is a line, and if I is the midpoint of $[MN]$ then $s(I)$ is the midpoint of $[s(M)s(N)]$;
3. s preserves angles (magnitude): for any three non-collinear points A, B, C, the angle $s(A)\widehat{s(B)}s(C)$ has the same measure as $A\hat{B}C$;
4. the image of a circle of radius R is a circle of radius kR;
5. areas are multiplied by k^2: if a domain D has area A, its image has area k^2A.

Justification (sketch). Points (1) and (2) follow from the fact that s multiplies all distances by k, exactly as for homotheties. For (3), triangles ABC and $A'B'C'$ have proportional sides with factor k, hence are similar in the classical sense, so their angles are equal. Point (4) is immediate since

a circle is a locus defined by a distance. Point (5) holds for a triangle (base and height both multiplied by k) and extends to any domain by decomposition into triangles. ■

WARNING **Shape is preserved, size is not**

A similarity never distorts a figure: a right-angled triangle stays right-angled, a square stays a square, a circle stays a circle (never an ellipse!). If a transformation sends a circle onto an ellipse, it is not a similarity. Conversely, do not confuse "same shape" with "same size": unless $k = 1$, lengths genuinely change.

7.2 Decomposition: Homothety times Isometry

The key structural result of the theory says that similarities are exactly the composites of the two families we already know.

PROPERTY **Decomposition theorem**

Every similarity s of ratio k can be written as

$$s = h \circ f = f' \circ h,$$

where h is a homothety of ratio k (with arbitrary centre) and f, f' are isometries. Conversely, every composite of a homothety of ratio k and an isometry is a similarity of ratio k.

Proof. Let s be a similarity of ratio k. Fix any point Ω and let h be the homothety of centre Ω and ratio $\frac{1}{k}$. Consider $f = h \circ s$. For all points M, N:

$$f(M)f(N) = \frac{1}{k}\, s(M)s(N) = \frac{1}{k} \cdot k \cdot MN = MN.$$

Hence f is an isometry, and $s = h^{-1} \circ f$ where h^{-1} is the homothety of centre Ω and ratio k. The equality $s = f' \circ h$ is obtained by choosing $f' = s \circ h^{-1}$. The converse is immediate: an isometry preserves distances and the homothety multiplies them by k, so the composite multiplies them by k. ■

KEY POINTS **The big picture**

Similarities = homotheties × isometries. Since isometries are translations, rotations, reflections and glide reflections, every similarity is built from these bricks together with one homothety. The ratio k comes entirely from the homothety part.

7.3 Direct and Indirect Similarities

DEFINITION **Direct and indirect similarities**

A similarity is called **direct** if it preserves the orientation of angles, and **indirect** if it reverses the orientation of angles. Equivalently, writing $s = h \circ f$ with h a homothety of positive ratio:

- s is direct when f is a direct isometry (translation or rotation);
- s is indirect when f is an indirect isometry (reflection or glide reflection).

Concretely: a direct similarity sends a triangle labelled anticlockwise A, B, C onto a triangle labelled anticlockwise A', B', C'; an indirect similarity flips the cyclic order, like the image seen in a mirror.

EXEMPLE **Classifying the familiar transformations**

- Direct: translations ($k = 1$), rotations ($k = 1$), homotheties of any ratio $\lambda \neq 0$ ($k = |\lambda|$), spiral similarities.
- Indirect: reflections and glide reflections ($k = 1$), and any composite of a reflection or glide reflection with a homothety ($k > 0$).

7.4 Spiral Similarities

Among direct similarities, the most important new objects are the **spiral similarities**, obtained by composing a rotation and a homothety about the same centre.

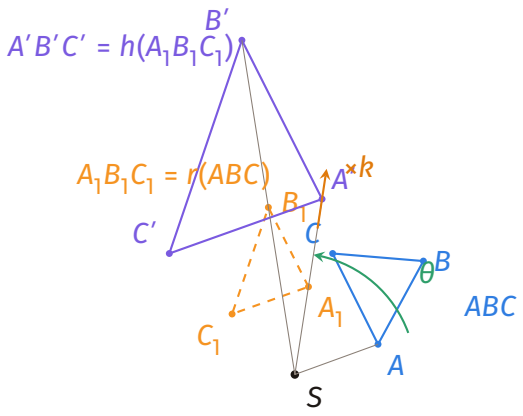
DEFINITION **Spiral similarity**

A **spiral similarity** is the composite $s = h \circ r$ (equivalently $r \circ h$) of a rotation r of angle θ and a homothety h of positive ratio k, both having the same centre S. It is completely characterised by three elements: its **centre** S, its **angle** θ and its **ratio** k. For every point $M \neq S$ with image $M' = s(M)$:

$$SM' = k \cdot SM \qquad \text{and} \qquad M\hat{S}M' = \theta.$$

WARNING **The ratio is always positive**

The ratio of a similarity is by definition a positive number. A homothety of negative ratio $-k$ about S is nothing other than the homothety of ratio k composed with the half-turn of centre S, i.e. a rotation of angle 180°. So the "minus sign" is absorbed into the rotation part: a spiral similarity of ratio k and angle θ with a negative homothety is simply a spiral similarity of ratio k and angle $\theta + 180^\circ$.



A spiral similarity of centre S, angle θ and ratio $k = 2$: the triangle ABC is first rotated about S (triangle $A_1B_1C_1$), then enlarged from S by the factor k (triangle $A'B'C'$). For each vertex, $SM' = k\,SM$ and $M\hat{S}M' = \theta$.

7.5 Determination of a Direct Similarity

PROPERTY **Two point–image pairs determine a direct similarity**

Let A, B be two distinct points and A', B' two distinct points. There exists a unique direct similarity s such that $s(A) = A'$ and $s(B) = B'$. Its ratio is necessarily

$$k = \frac{A'B'}{AB}.$$

Justification of uniqueness. If two direct similarities s_1, s_2 send A to A' and B to B', then $s_2^{-1} \circ s_1$ is a direct similarity fixing both A and B. Its ratio must be 1 (since AB is preserved), so it is a direct isometry fixing two points: a translation or rotation fixing two distinct points, hence the identity. Therefore $s_1 = s_2$. ■

METHOD **Finding the centre of a spiral similarity (geometric construction)**

Given $A \mapsto A'$ and $B \mapsto B'$ (with the segments not related by a mere translation or homothety), to construct the centre S of the spiral similarity:

1. Compute the ratio $k = \dfrac{A'B'}{AB}$ and read the angle $\theta = (AB, \hat{A'B'})$ between the two segments.
2. The centre S must see the segment $[AA']$ under the angle θ: it lies on an arc of circle through A and A' (locus of points M with $A\hat{M}A' = \theta$).
3. Likewise S sees $[BB']$ under the same angle θ: it lies on the corresponding arc through

B and B'.

4. The intersection of the two arcs (other than any trivial common point) is the centre S.

A useful shortcut: S also lies at the intersection of the circles through A, B and the intersection point of lines (AB) and $(A'B')$, and through A', B' and the same point.

METHOD **Finding the centre and parameters using complex numbers**

In an orthonormal frame identified with $\mathbb{C}$, a direct similarity has the form

$$z' = az + b \qquad \text{with } a = ke^{i\theta},\ k > 0.$$

Given $z_A \mapsto z_{A'}$ and $z_B \mapsto z_{B'}$:

1. Compute $a = \dfrac{z_{B'} - z_{A'}}{z_B - z_A}$. Then $k = |a|$ and $\theta = \arg(a)$.
2. Compute $b = z_{A'} - az_A$.
3. The centre (fixed point) is $z_S = \dfrac{b}{1-a}$ (provided $a \neq 1$, i.e. not a translation).

WORKED EXAMPLE 11 **Identifying and using a spiral similarity**

The plane is referred to an orthonormal frame. A direct similarity s sends $A(0,0)$ to $A'(2,1)$ and $B(4,0)$ to $B'(2,5)$.

1. Determine the ratio and the angle of s.
2. Give the complex expression of s and find its centre.

SOLUTION

Solution.

1. $AB = 4$ and $A'B' = \sqrt{0^2 + 4^2} = 4$, so $k = \dfrac{A'B'}{AB} = 1$. The vector $\vec{AB} = (4,0)$ points along the x-axis while $\vec{A'B'} = (0,4)$ points along the y-axis: the angle is $\theta = \dfrac{\pi}{2}$. Since $k = 1$, s is in fact a rotation of angle $\dfrac{\pi}{2}$.
2. A direct similarity has complex form $z' = az + b$ with $a = ke^{i\theta}$. Here $a = e^{i\pi/2} = i$. From $s(A) = A'$: $b = 2 + i$. Check with B: $z' = 4i + 2 + i = 2 + 5i$ ✓. The centre (fixed point) satisfies $z = iz + 2 + i$, so

$$z = \frac{2+i}{1-i} = \frac{(2+i)(1+i)}{(1-i)(1+i)} = \frac{1+3i}{2}.$$

The centre is $S\left(\frac{1}{2}, \frac{3}{2}\right)$.

WORKED EXAMPLE 12 **Spiral similarity with $k \neq 1$**

A direct similarity s maps $A(1,2)$ to $A'(3,0)$ and $B(3,2)$ to $B'(7,4)$. Find its ratio, angle, complex form and centre.

SOLUTION

Solution. In complex form: $z_A = 1+2i$, $z_{A'} = 3$, $z_B = 3+2i$, $z_{B'} = 7+4i$.

$$a = \frac{z_{B'} - z_{A'}}{z_B - z_A} = \frac{(7+4i)-3}{(3+2i)-(1+2i)} = \frac{4+4i}{2} = 2+2i = 2\sqrt{2}\,e^{i\pi/4}.$$

Hence $k = 2\sqrt{2}$, $\theta = \frac{\pi}{4}$. $b = z_{A'} - az_A = 3-(2+2i)(1+2i) = 3-(2+4i+2i-4) = 3-(-2+6i) = 5-6i$. Complex form: $z' = (2+2i)z + 5 - 6i$. Centre: $z_S = \frac{b}{1-a} = \frac{5-6i}{1-(2+2i)} = \frac{5-6i}{-1-2i} = \frac{(5-6i)(-1+2i)}{(-1-2i)(-1+2i)} = \frac{-5+10i+6i+12}{5} = \frac{7+16i}{5}$. Thus $S\left(\frac{7}{5}, \frac{16}{5}\right)$.

7.6 Analytical Representation of Similarities

PROPERTY **Analytical form of similarities**

In an orthonormal coordinate system (or complex plane), every similarity can be written in one of the two forms:

- Direct similarity: $z' = az + b$ with $a \in \mathbb{C}^*$, $b \in \mathbb{C}$. Ratio $k = |a|$, angle $\theta = \arg(a)$.
- Indirect similarity: $z' = a\bar{z} + b$ with $a \in \mathbb{C}^*$, $b \in \mathbb{C}$. Ratio $k = |a|$.

In vector notation (with $\vec{OM} = \vec{x}$):

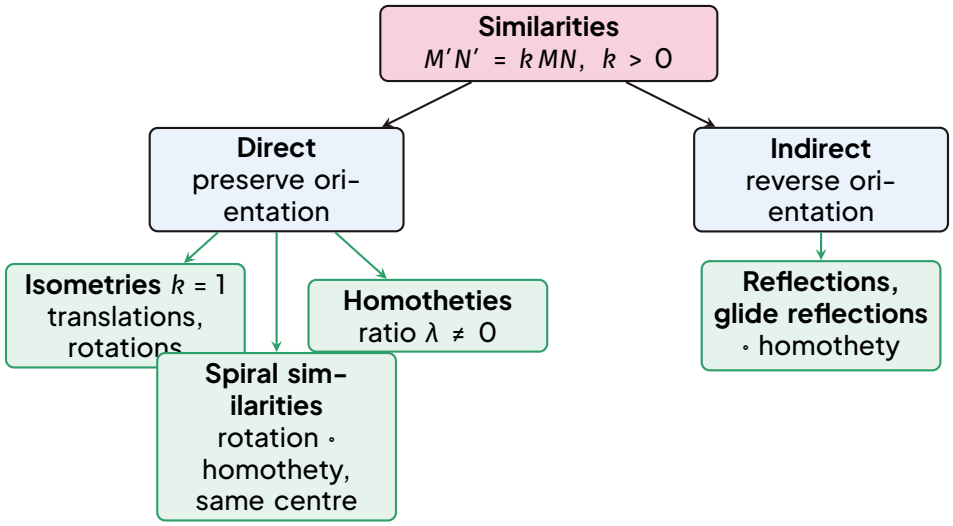
- Direct: $\vec{x}' = k\,R_\theta \vec{x} + \vec{b}$ where R_θ is the rotation matrix.
- Indirect: $\vec{x}' = k\,S_\alpha \vec{x} + \vec{b}$ where S_α is the reflection matrix across a line through the origin making angle $\alpha/2$ with the x-axis.

EXEMPLE **Special cases in analytical form**

- Translation: $z' = z + b$ ($a = 1$, $k = 1$, $\theta = 0$).

- Rotation about c: $z' = e^{i\theta}(z - c) + c = e^{i\theta}z + c(1 - e^{i\theta})$.
- Homothety of centre c, ratio λ: $z' = \lambda(z - c) + c = \lambda z + c(1 - \lambda)$. If $\lambda > 0$, direct ($k = \lambda$, $\theta = 0$); if $\lambda < 0$, direct ($k = |\lambda|$, $\theta = \pi$).
- Reflection across line through origin with angle α: $z' = e^{2i\alpha}\bar{z}$ ($k = 1$, indirect).
- Glide reflection: $z' = e^{2i\alpha}\bar{z} + b$ with b parallel to the reflection line.

7.7 The Family Tree of Plane Transformations



Family tree of plane transformations. Every similarity is direct or indirect; the direct family contains the isometries, the homotheties and their generic composites, the spiral similarities.

DID YOU KNOW? **Similarities everywhere**

When a cartographer at the National Institute of Cartography in Yaoundé produces a map of the Littoral region at scale $1 : 50\,000$, he is applying a similarity of ratio $k = \frac{1}{50000}$: every angle on the ground (road junctions, river bends) is exactly preserved on the map, which is why bearings read on the map are reliable. Spiral similarities also appear in nature: the growth of a snail shell is modelled by iterating a spiral similarity — each new chamber is the previous one rotated and enlarged by a fixed factor, producing the famous logarithmic spiral.

EXERCISE 7 **Recognising and determining similarities**

1. Classify each transformation as direct or indirect and give its ratio: (a) the reflection across the line $y = x$; (b) the homothety of centre O and ratio -3; (c) the composite of a rotation of angle $\frac{\pi}{3}$ with a homothety of ratio 2 about the same centre.
2. $A(1,1)$, $B(3,1)$ have images $A'(0,2)$, $B'(0,6)$ under a direct similarity s. Find the ratio, the angle, the complex form $z' = az + b$ and the centre of s.
3. A similarity of ratio $k = 3$ transforms a triangle of area $5\ \text{cm}^2$. What is the area of the image? What is the image of its inscribed circle of radius 1.2 cm?
4. A direct similarity s has ratio $k = \frac{1}{2}$ and angle $\theta = -\frac{\pi}{3}$. It maps $A(2,0)$ to A'. Find A' if the centre of s is the origin. What is the complex form of s?
5. The vertices of a triangle are $P(1,0)$, $Q(0,2)$, $R(-1,0)$. A similarity of ratio 2 maps this triangle to $P'Q'R'$ with $P'(3,1)$, $Q'(1,5)$. Determine whether the similarity is direct or indirect, find the image R', and give the transformation in complex form.

SOLUTION **Exercise n° 1**

1. (a) Indirect, $k = 1$. (b) Direct (a homothety preserves orientation), $k = |-3| = 3$; the negative sign corresponds to an extra half-turn. (c) Direct spiral similarity, $k = 2$, angle $\frac{\pi}{3}$.
2. $AB = 2$, $A'B' = 4$, so $k = 2$. $\vec{AB} = (2,0)$, $\vec{A'B'} = (0,4)$: angle $\frac{\pi}{2}$. Hence $a = 2i$; from $s(A) = A'$: $2i(1+i) + b = 2i$, giving $b = 2$. So $z' = 2iz + 2$. Fixed point: $z = \frac{2}{1-2i} = \frac{2+4i}{5}$; centre $S\left(\frac{2}{5}, \frac{4}{5}\right)$.
3. Area multiplied by $k^2 = 9$: $45\ \text{cm}^2$. The inscribed circle maps to a circle of radius $3 \times 1.2 = 3.6$ cm.
4. Centre at origin $\Rightarrow z' = az$ with $a = \frac{1}{2}e^{-i\pi/3} = \frac{1}{2}\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = \frac{1}{4} - i\frac{\sqrt{3}}{4}$. $z_A = 2$, so $z_{A'} = 2a = \frac{1}{2} - i\frac{\sqrt{3}}{2}$, i.e. $A'\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$.
5. $PQ = \sqrt{5}$, $P'Q' = \sqrt{(3-1)^2 + (1-5)^2} = \sqrt{20} = 2\sqrt{5}$, so $k = 2$ (matches). Vector $\vec{PQ} = (-1,2)$, $\vec{P'Q'} = (-2,4) = 2\vec{PQ}$. The angle is 0, so the similarity is a homothety (direct). $R' = P' + 2\vec{PR} = (3,1) + 2(-2,0) = (-1,1)$. Complex form: $a = 2$, $b = z_{P'} - 2z_P = (3+i) - 2 = 1 + i$. So $z' = 2z + 1 + i$.

KEY POINTS **Key points of the section**

- A similarity of ratio $k > 0$ multiplies all distances by k; it preserves angles and shapes, and multiplies areas by k^2.
- Every similarity is the composite of a homothety of ratio k and an isometry.
- Direct similarities preserve orientation (translations, rotations, homotheties, spiral similarities); indirect ones reverse it (reflections and glide reflections composed with homotheties).
- A spiral similarity is characterised by its centre, its angle and its ratio; the ratio is always positive.
- A direct similarity is uniquely determined by the images of two distinct points, with $k = \dfrac{A'B'}{AB}$.
- Analytical forms: direct $z' = az + b$ ($a = ke^{i\theta}$), indirect $z' = a\bar{z} + b$ ($k = |a|$).

8 ANALYTICAL STUDY OF SIMILARITIES AND APPLICATIONS

In the previous section we defined a **plane similarity** geometrically: a transformation that multiplies all distances by a fixed positive ratio k and preserves angles (in magnitude). We saw that a similarity is direct when it preserves orientation and indirect when it reverses it. This geometric description is beautiful, but for computation it is clumsy. The great power of the Argand diagram is that it turns every similarity into a one-line formula. In this section we give the complex (analytical) form of similarities, learn to read off their characteristic elements from the formula, study composition, and finish with GCE-style synthesis problems.

8.1 The Complex Form of a Similarity

Observation. Consider the map S of the plane into itself defined, in the Argand diagram, by

$$z' = az + b, \qquad a \in \mathbb{C},\ a \neq 0,\ b \in \mathbb{C}.$$

Take two points $M_1(z_1)$ and $M_2(z_2)$ with images $M_1'(z_1')$ and $M_2'(z_2')$. Then

$$|z_1' - z_2'| = |a(z_1 - z_2)| = |a|\,|z_1 - z_2|,$$

so $M_1'M_2' = |a|\,M_1M_2$: all distances are multiplied by the same positive constant $|a|$. Moreover,

$$\arg\left(\frac{z_2' - z_1'}{z_2 - z_1}\right) = \arg(a) \pmod{2\pi},$$

so every directed angle is increased by the same amount $\arg(a)$: angles are preserved in size and orientation. This is exactly a direct similarity of ratio $|a|$ and angle $\arg(a)$.

PROPERTY **Complex form of a similarity — proof examinable**

Every **direct plane similarity** has a unique complex expression

$$z' = az + b, \qquad a \neq 0,$$

with ratio $k = |a|$ and angle $\theta = \arg(a)$. Every **indirect similarity** has a unique complex expression

$$z' = a\bar{z} + b, \qquad a \neq 0,$$

with ratio $|a|$ (angles are preserved in magnitude but reversed in orientation). Conversely, every map of one of these two forms is a similarity.

Proof (direct case). Let S be a direct similarity of ratio k. Pick any point A with affix α and image $A'(\alpha')$. For any $M(z)$ with image $M'(z')$, the similarity preserves angles and multiplies distances by k, so

$$\frac{z' - \alpha'}{z - \alpha} \text{ has modulus } k \text{ and constant argument } \theta,$$

i.e. $z' - \alpha' = a(z - \alpha)$ with $a = ke^{i\theta}$. Setting $b = \alpha' - a\alpha$ gives $z' = az + b$. Uniqueness follows because $a = \dfrac{z_1' - z_2'}{z_1 - z_2}$ is forced by any two distinct points and their images. ■

PROPERTY **Reading off the geometric elements — direct case**

Given a direct similarity $S : z' = az + b$ with $a \neq 1$, the map has a unique fixed point Ω (the **centre**), whose affix is

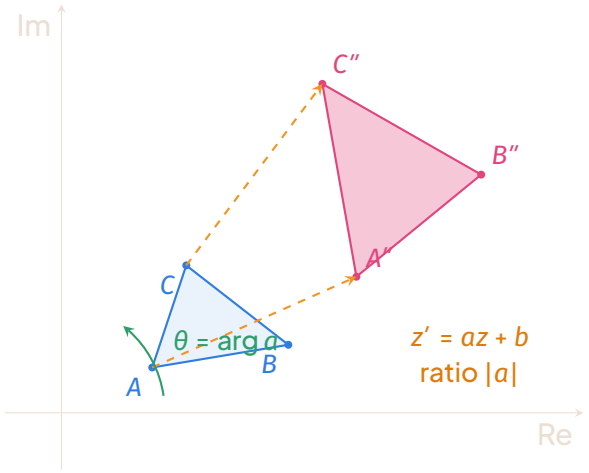
$$\omega = \frac{b}{1 - a}.$$

Then S is the composite (in either order) of the homothety $h(\Omega, |a|)$ and the rotation $r(\Omega, \arg a)$: a **spiral similarity** centred at Ω. If $a = 1$, then $z' = z + b$ is the translation by the vector of affix b.

Derivation. A fixed point satisfies $\omega = a\omega + b$, i.e. $(1 - a)\omega = b$. If $a \neq 1$ this gives $\omega = \dfrac{b}{1 - a}$ uniquely. Subtracting $\omega = a\omega + b$ from $z' = az + b$ yields $z' - \omega = a(z - \omega)$, which says precisely: rotate about Ω through $\arg a$ and dilate from Ω by factor $|a|$. ■

WARNING **Ratio versus coefficient**

Do not confuse the ratio of the similarity, which is $|a|$ and is always a positive real number, with the complex coefficient a itself. If $a = -2$, the ratio is 2, not -2; the negative sign contributes an angle of π. Likewise the angle is $\arg(a)$, not a.



A direct similarity $z' = az + b$ sends triangle ABC to the enlarged, rotated triangle $A'B'C'$: sides are multiplied by $|a|$ and turned through $\arg(a)$.

8.2 Case-by-Case Identification

METHOD **Identifying a transformation given by $z' = az + b$ or $z' = a\bar{z} + b$**

1. Compute $|a|$ and $\arg(a)$ (for $a \neq 0$).
2. Direct case ($z' = az + b$):
   - If $a = 1$: translation by the vector of affix b.
   - If $a \in \mathbb{R}$, $a \neq 1$: homothety of ratio a (direct if $a > 0$, ratio $|a|$ with a half-turn if $a < 0$), centre $\omega = \dfrac{b}{1-a}$.
   - If $|a| = 1$, $a \neq 1$: rotation of angle $\arg(a)$, centre $\omega = \dfrac{b}{1-a}$.
   - Otherwise ($|a| \neq 1$, $a \notin \mathbb{R}$): spiral similarity, centre $\omega = \dfrac{b}{1-a}$, ratio $|a|$, angle $\arg(a)$.
3. Indirect case ($z' = a\bar{z} + b$):
   - If $|a| = 1$: indirect isometry — reflection (if fixed points exist, i.e. line of fixed points) or glide reflection (no fixed points).
   - If $|a| \neq 1$: indirect similarity (homothety followed by reflection, or spiral similarity with reflection).

EXEMPLE **Four quick identifications — direct similarities**

- $z' = z + (2 - 3i)$: $a = 1$, translation by $\vec{u}(2, -3)$.
- $z' = 3z - 4 + 2i$: $a = 3 \in \mathbb{R}$, homothety of ratio 3, centre $\omega = \dfrac{-4 + 2i}{1 - 3} = 2 - i$.
- $z' = iz + 1$: $|a| = 1$, $\arg a = \frac{\pi}{2}$, rotation through 90° about $\omega = \dfrac{1}{1 - i} = \dfrac{1 + i}{2}$.
- $z' = (1 + i)z$: $|a| = \sqrt{2}$, $\arg a = \frac{\pi}{4}$, spiral similarity centred at O, ratio $\sqrt{2}$, angle 45°.

EXEMPLE **Indirect similarities**

- $z' = \bar{z}$: reflection in the real axis (fixed points: real axis).
- $z' = \bar{z} + 2$: glide reflection (reflection in real axis + translation by 2 parallel to axis); no fixed points.
- $z' = 2\bar{z}$: indirect similarity, ratio 2, centre O (homothety 2 + reflection in real axis).

WORKED EXAMPLE 13 **Full identification, GCE style**

A transformation S of the plane is given by $z' = (1 - i\sqrt{3})\,z + 2i\sqrt{3}$. Determine the nature of S and its characteristic elements.

SOLUTION

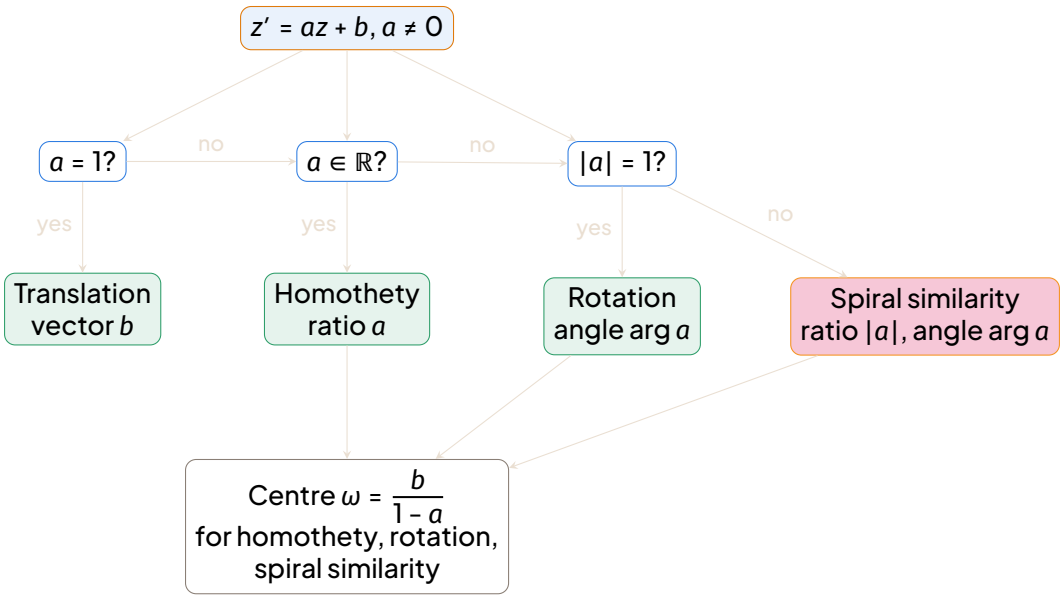
Solution. Here $a = 1 - i\sqrt{3}$ and $b = 2i\sqrt{3}$.

$$|a| = \sqrt{1 + 3} = 2, \qquad \arg(a) = -\arctan\left(\frac{\sqrt{3}}{1}\right) = -\frac{\pi}{3} \pmod{2\pi}.$$

Since $|a| = 2 \neq 1$ and $a \notin \mathbb{R}$, S is a spiral similarity (direct similarity which is neither a homothety nor an isometry). Its centre is

$$\omega = \frac{b}{1 - a} = \frac{2i\sqrt{3}}{1 - (1 - i\sqrt{3})} = \frac{2i\sqrt{3}}{i\sqrt{3}} = 2.$$

Conclusion: S is the direct similarity of centre $\Omega(2, 0)$, ratio $k = 2$ and angle $\theta = -\dfrac{\pi}{3}$; equivalently $S = h(\Omega, 2) \circ r\left(\Omega, -\frac{\pi}{3}\right)$.



Decision chart for direct similarities ($z' = az + b$). For indirect similarities ($z' = a\bar{z} + b$): if $|a| = 1$ it is a reflection or glide reflection; otherwise an indirect similarity with ratio $|a|$.

8.3 Composition of Similarities via Complex Expressions

Composing similarities is now pure algebra. If $S_1 : z' = a_1 z + b_1$ and $S_2 : z' = a_2 z + b_2$ (both direct), then

$$S_2 \circ S_1 : z \mapsto a_2(a_1 z + b_1) + b_2 = a_1 a_2\, z + (a_2 b_1 + b_2).$$

PROPERTY **Ratio and angle of a composite — direct similarities**

The composite of two direct similarities is a direct similarity with

$$\text{ratio} = k_1 k_2 \quad (|a_1 a_2| = |a_1||a_2|), \qquad \text{angle} = \theta_1 + \theta_2 \quad (\arg(a_1 a_2) = \arg a_1 + \arg a_2).$$

In particular the set of direct similarities is a group under composition; the inverse of $z' = az + b$ is $z' = \frac{1}{a}z - \frac{b}{a}$, a similarity of ratio $\frac{1}{|a|}$ and angle $-\arg a$.

PROPERTY **Composition involving indirect similarities**

- Direct $\circ$ Indirect = Indirect; Indirect $\circ$ Direct = Indirect.
- Indirect $\circ$ Indirect = Direct.
- In all cases, the ratio of the composite is the product of the ratios ($|a_1||a_2|$).
- For the angle: direct similarities add their angles; an indirect similarity contributes

$-\arg(a)$ (or equivalently, the oriented angle is reversed).

EXEMPLE **Composing a rotation and a homothety**

Let $r : z' = iz$ (rotation about O through $\frac{\pi}{2}$) and $h : z' = 2z - 3$ (homothety of ratio 2). Then

$$h \circ r : z \mapsto 2(iz) - 3 = 2iz - 3,$$

a spiral similarity of ratio 2, angle $\frac{\pi}{2}$, centre $\omega = \dfrac{-3}{1-2i} = \dfrac{-3(1+2i)}{5} = -\dfrac{3}{5} - \dfrac{6}{5}i$. Note that $r \circ h : z \mapsto i(2z-3) = 2iz - 3i$ has the same ratio and angle but a different centre: composition is not commutative.

8.4 Geometric Applications: Triangles and Circles

PROPERTY **Conservation properties of similarities**

Every similarity of ratio k:

- preserves angles (direct ones preserve oriented angles, indirect ones reverse them);
- maps three collinear points to three collinear points and preserves ratios of distances along a line;
- maps a line to a line and a circle to a circle, the radius being multiplied by k;
- maps a triangle to a similar triangle (same shape, sides multiplied by k).

Why circles map to circles. A circle of centre $\Omega(\omega)$ and radius R is the set $\{z : |z - \omega| = R\}$. Under $z' = az + b$, writing $z = \dfrac{z' - b}{a}$, the condition becomes $\left|\dfrac{z' - b}{a} - \omega\right| = R$, i.e. $|z' - (a\omega + b)| = |a|R$: a circle of centre $\Omega'(a\omega + b)$ and radius kR. The same holds for $z' = a\bar{z} + b$ because $|\bar{z} - \bar{\omega}| = |z - \omega|$. ■

EXEMPLE **Similar triangles via the complex ratio**

Triangles ABC and $A'B'C'$ (affixes $z_A, \dots, z_{C'}$) are directly similar (same orientation) if and only if

$$\frac{z_{B'} - z_{A'}}{z_{C'} - z_{A'}} = \frac{z_B - z_A}{z_C - z_A}.$$

Indeed this single complex equality encodes both the ratio of sides (moduli) and the angle at A (arguments). This is the analytic version of the classical criteria (SSS, SAS, AA) for similar triangles.

8.5 Spiral Similarities in Problem-Solving

A spiral similarity is determined by one pair of corresponding points together with its centre, or equivalently by two segments: given segments AB and $A'B'$, there is a unique direct similarity sending $A \mapsto A'$, $B \mapsto B'$, namely $z' = az + b$ with $a = \dfrac{z_{B'} - z_{A'}}{z_B - z_A}$ and $b = z_{A'} - az_A$.

METHOD **Proving collinearity, concyclicity or equal angles using similarities**

1. Identify (or construct) a spiral similarity $S : z' = az + b$ sending a known configuration to the configuration under study.
2. Collinearity: to prove P, Q, R collinear, show that P, Q are images under S of two points whose line must contain the preimage of R; or show $\dfrac{z_R - z_P}{z_Q - z_P} \in \mathbb{R}$.
3. Equal angles: use $\arg\left(\dfrac{z_C - z_A}{z_B - z_A}\right)$, which is invariant under a direct similarity.
4. Concyclicity: four points are concyclic (or collinear) iff their cross-ratio $\dfrac{(z_A - z_C)(z_B - z_D)}{(z_A - z_D)(z_B - z_C)}$ is real; similarities preserve cross-ratios, hence concyclicity.

WORKED EXAMPLE 14 **The classic two-triangle configuration**

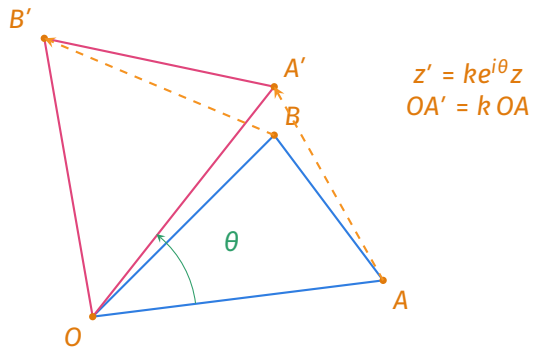
Triangles OAB and $OA'B'$ are directly similar at the common vertex O, with A', B' the images of A, B under the spiral similarity S of centre O, ratio k, angle θ. Show that the lines AA' and BB' make the same angle with OA and OB respectively, and that $\dfrac{AA'}{BB'} = \dfrac{OA}{OB}$.

SOLUTION

Solution. Place O at the origin; then $S : z' = ke^{i\theta}z$, so $z_{A'} = az_A$, $z_{B'} = az_B$ with $a = ke^{i\theta}$. Then

$$z_{A'} - z_A = (a-1)z_A, \qquad z_{B'} - z_B = (a-1)z_B.$$

Hence $\dfrac{z_{A'} - z_A}{z_{B'} - z_B} = \dfrac{z_A}{z_B}$, a constant complex number. Taking moduli: $\dfrac{AA'}{BB'} = \dfrac{OA}{OB}$. Taking arguments: the directed angle from BB' to AA' equals the angle from OB to OA, i.e. the two segments AA' and BB' are inclined to OA, OB in exactly the same way. In particular, if O, A, B are collinear then A, A', B, B' lie in a configuration where $AA' \parallel BB'$ — a standard route to proving parallelism and collinearity in GCE problems.



Two directly similar triangles sharing the vertex O: the spiral similarity $z' = ke^{i\theta}z$ sends $A \mapsto A'$ and $B \mapsto B'$. The dashed segments AA', BB' are in a constant ratio and constant relative inclination.

DID YOU KNOW? **Spiral similarities and the "pivot theorem"**

Spiral similarities are the secret engine behind many olympiad and GCE geometry facts: the centre of the spiral similarity sending segment AB to $A'B'$ lies at the intersection of the circles through A, A' and the intersection point of lines AA', BB'. Napoleon's theorem, the Miquel point of a complete quadrilateral, and the properties of the nine-point circle can all be proved elegantly this way.

8.6 Synthesis Problems, Assessment and Remediation

WORKED EXAMPLE 15 **GCE Paper 2 style — mixing homothety, rotation and complex numbers**

The plane is referred to an orthonormal frame $(O;\vec{u},\vec{v})$. Let T be the transformation with complex expression $z' = \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)z + 1 - \sqrt{3}i$.

(a) Show that T is a rotation and determine its centre and angle.

(b) Let h be the homothety of centre O and ratio 2. Write the complex expression of $S = h \circ T$.

(c) Deduce the nature and characteristic elements of S.

(d) The circle $\mathcal{C}$ of centre O and radius 1 is transformed by S into $\mathcal{C}'$. Give the centre and radius of $\mathcal{C}'$.

SOLUTION

Solution. (a) $a = \frac{1}{2} + \frac{\sqrt{3}}{2}i = e^{i\pi/3}$, so $|a| = 1$, $a \neq 1$: T is a rotation of angle $\frac{\pi}{3}$. Centre: $\omega = \frac{b}{1-a} = \frac{1-\sqrt{3}i}{1-\frac{1}{2}-\frac{\sqrt{3}}{2}i} = \frac{1-\sqrt{3}i}{\frac{1}{2}-\frac{\sqrt{3}}{2}i} = \frac{1-\sqrt{3}i}{e^{-i\pi/3}}$. Since $1-\sqrt{3}i = 2e^{-i\pi/3}$, we get $\omega = 2$. Centre $\Omega(2, 0)$.
(b) $h : z' = 2z$. $S = h \circ T : z \mapsto 2\left(e^{i\pi/3}z + 1 - \sqrt{3}i\right) = (1+\sqrt{3}i)\,z + 2 - 2\sqrt{3}i$.
(c) $|1+\sqrt{3}i| = 2$, $\arg(1+\sqrt{3}i) = \frac{\pi}{3}$. Since $|a| = 2 \neq 1$: S is a spiral similarity of ratio 2, angle $\frac{\pi}{3}$, centre $\omega_S = \frac{2-2\sqrt{3}i}{1-1-\sqrt{3}i} = \frac{2-2\sqrt{3}i}{-\sqrt{3}i} = \frac{(2-2\sqrt{3}i)\,i}{\sqrt{3}} = 2 + \frac{2i}{\sqrt{3}} = 2 + \frac{2\sqrt{3}}{3}i$.
(d) C' is a circle (similarities map circles to circles) of radius $2 \times 1 = 2$, centred at $S(O)$: $z_{O'} = 2 - 2\sqrt{3}i$, i.e. $O'(2, -2\sqrt{3})$.

WARNING **Exam tip — GCE Paper 2**

In Paper 2, part (a) usually asks for the complex form and part (b) for the geometric elements. Answer both explicitly and completely: state the nature (translation / homothety / rotation / reflection / glide reflection / spiral similarity), the centre (affix $\omega = \frac{b}{1-a}$ for direct, or fixed point analysis for indirect), the ratio $|a|$ and the angle $\arg(a)$. A common loss of marks: giving the ratio as a complex number, or forgetting that a translation ($a = 1$) has no centre.

EXERCISE 8 **Assessment**

1. Identify each transformation and give its characteristic elements:
   - $z' = -2z + 6$; $z' = e^{-i\pi/4}z + i$; $z' = (3-3i)z + 4i$; $z' = \bar{z} + 2$.
2. $S_1 : z' = iz + 2$ and $S_2 : z' = 2z - i$. Find the complex expressions of $S_2 \circ S_1$ and $S_1 \circ S_2$, and the nature and elements of each composite. Comment.
3. $A(1)$, $B(2+i)$, $C(i)$ are three points and $S : z' = (1+i)z - i$. Find the images A', B', C'; show that ABC and $A'B'C'$ are similar and give the ratio.
4. The circle $\mathcal{C} : |z - 1 + i| = 3$ is transformed by $z' = 2i\bar{z} + 1$. Find the centre and radius of the image circle. Is the similarity direct or indirect?

SOLUTION **Exercise 1**

1. (i) $a = -2 \in \mathbb{R}$: homothety of ratio -2 (i.e. ratio 2 with a half-turn), centre $\omega = \dfrac{6}{1-(-2)} = 2$.

(ii) $a = e^{-i\pi/4}$, $|a| = 1$, $a \neq 1$: rotation of angle $-\dfrac{\pi}{4}$. Centre $\omega = \dfrac{i}{1-e^{-i\pi/4}}$. Compute exactly:

$1 - e^{-i\pi/4} = 1 - \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$. $\omega = \dfrac{i\left(1-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right)}{\left(1-\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2} = \dfrac{\frac{\sqrt{2}}{2} + i\left(1-\frac{\sqrt{2}}{2}\right)}{2-\sqrt{2}} = \dfrac{1+\sqrt{2}}{2} + \frac{i}{2}$. So centre $\Omega\left(\frac{1+\sqrt{2}}{2}, \frac{1}{2}\right)$.

(iii) $a = 3 - 3i = 3\sqrt{2}e^{-i\pi/4}$: $|a| = 3\sqrt{2}$, $\arg a = -\frac{\pi}{4}$. Spiral similarity, centre $\omega = \dfrac{4i}{1-3+3i} = \dfrac{4i}{-2+3i} = \dfrac{4i(-2-3i)}{13} = \dfrac{12-8i}{13}$, ratio $3\sqrt{2}$, angle $-\frac{\pi}{4}$.

(iv) $z' = \bar{z} + 2$: indirect isometry ($|a| = 1$). Fixed points satisfy $z = \bar{z} + 2 \Rightarrow z - \bar{z} = 2 \Rightarrow 2i\,\mathrm{Im}(z) = 2$, impossible for real $\mathrm{Im}(z)$. Hence no fixed points: it is a glide reflection (reflection in the real axis followed by translation by 2 parallel to that axis).

2. $S_2 \circ S_1 : z \mapsto 2(iz+2) - i = 2iz + 4 - i$: spiral similarity, ratio 2, angle $\frac{\pi}{2}$, centre $\omega = \dfrac{4-i}{1-2i} = \dfrac{(4-i)(1+2i)}{5} = \dfrac{6+7i}{5}$.

$S_1 \circ S_2 : z \mapsto i(2z-i) + 2 = 2iz + 3$: same ratio and angle, centre $\omega = \dfrac{3}{1-2i} = \dfrac{3+6i}{5}$.

The composites differ: composition of similarities is not commutative in general.

3. $z_{A'} = (1+i)(1) - i = 1$; $z_{B'} = (1+i)(2+i) - i = 1 + 2i$; $z_{C'} = (1+i)(i) - i = -1$.

Check: $\dfrac{z_{B'} - z_{A'}}{z_{C'} - z_{A'}} = \dfrac{2i}{-2} = -i$ and $\dfrac{z_B - z_A}{z_C - z_A} = \dfrac{1+i}{-1+i} = \dfrac{(1+i)(-1-i)}{2} = -i$. Equal, so the triangles are directly similar; ratio $|1+i| = \sqrt{2}$.

4. $z' = 2i\bar{z} + 1$ is an indirect similarity (presence of $\bar{z}$) of ratio $|2i| = 2$. The centre $1 - i$ maps to $2i\overline{(1-i)} + 1 = 2i(1+i) + 1 = -1 + 2i$. Image circle: centre $(-1, 2)$, radius $2 \times 3 = 6$.

KEY POINTS Key points of the section

- Direct similarity: $z' = az + b$ ($a \neq 0$); indirect: $z' = a\bar{z} + b$. Ratio = $|a|$, angle = $\arg(a)$ (direct) or orientation-reversing (indirect).
- If $a \neq 1$, unique fixed point (centre) $\omega = \dfrac{b}{1-a}$ for direct similarities; for indirect, solve $z = a\bar{z} + b$.
- $a = 1$: translation; $a \in \mathbb{R}$: homothety; $|a| = 1$: rotation; otherwise: spiral similarity (direct). Indirect: $|a| = 1$ gives reflection/glide reflection; $|a| \neq 1$ gives indirect similarity.
- Composition multiplies ratios and adds angles (direct) or subtracts (indirect): $a_{\text{composite}} = a_1 a_2$ for direct∘direct.
- Similarities preserve angles, collinearity and concyclicity; circles map to circles with radius multiplied by k; triangles map to similar triangles.
- In GCE answers, always state nature, centre, ratio and angle explicitly.

9 INTEGRATION ACTIVITY

9.1 Aims of the activity

This integration activity mobilises, in a single realistic project, all the skills developed in this chapter on **plane transformations**. By the end of the activity, you should be able to:

- **Compose two reflections** across intersecting axes and identify the resulting rotation, giving its centre and angle;
- **Construct and compute** the images of a figure under homotheties of given ratios, analytically and geometrically;
- **Write the complex expression** $z' = az + b$ of a direct plane similarity (spiral similarity) and determine its characteristic elements: centre, ratio and angle;
- **Use the area-scaling property** $A' = k^2 A$ of a similarity of ratio k to solve a concrete estimation problem;
- **Justify** each transformation rigorously by its characteristic elements, in the style required at the GCE Advanced Level examination;
- Work collaboratively, present constructions on grid paper, and defend your reasoning orally.

KEY POINTS **Skills mobilised**

Composition of isometries (reflections, rotations, translations), glide reflections, homotheties and their composition, complex representation of direct similarities $z' = az + b$, ratio and angle of a similarity, effect of a similarity on lengths ($\times|a|$) and on areas ($\times|a|^2$).

9.2 Situation

Designing a Logo for a Yaoundé Printing Press.

Mvogo & Sons Press, a small printing company located near the Mokolo market in Yaoundé, wants a new logo. The designer proposes a pattern built from a single basic triangular motif, repeated by rotation around a central point O, and then produced in three sizes: a small version for **business cards**, a medium version for **posters**, and a large version for **billboards**.

The press works with a graphic designer who uses a coordinate plane. The unit on both axes is 1 cm on the business-card artwork. The basic motif is the triangle $T = ABC$ with

$$A(1\,;\,0), \qquad B(3\,;\,0), \qquad C(2\,;\,1.5),$$

and the centre of the pattern is the origin $O(0\,;\,0)$.

To generate the repeated pattern, the designer reflects the motif successively across two axes passing through O: the x-axis (Δ_1) and the line (Δ_2) of equation $y = x$. The motif is first reflected across (Δ_1), and the image is then reflected across (Δ_2). The press owner wants to understand

which single transformation achieves the same result, so that the pattern can be repeated around O.

For the three sizes, the artwork is enlarged from O using homotheties of centre O and ratios

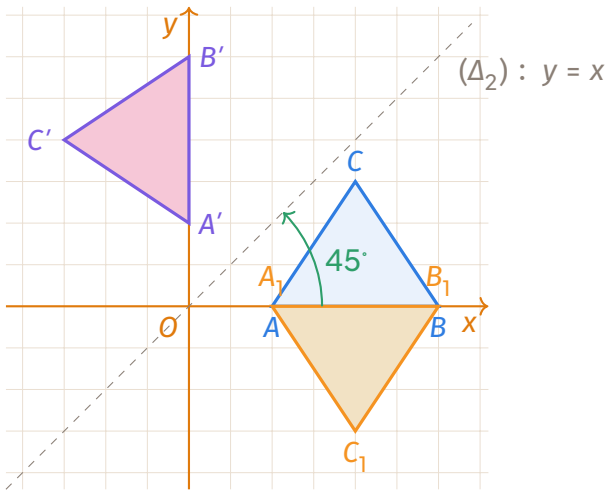
$$k_1 = 1 \quad \text{(business card)}, \qquad k_2 = 2.5 \quad \text{(poster)}, \qquad k_3 = 6 \quad \text{(billboard)}.$$

Finally, for an advertising animation, the designer wants to map the small motif ABC directly onto a rotated and enlarged copy $A''B''C''$, where A'' and B'' are the images of A and B under the direct similarity s of complex expression

$$z' = az + b, \qquad a = 1 + i\sqrt{3}, \qquad b = 2.$$

The complex plane is referred to the orthonormal frame $(O\,;\,\vec{u},\vec{v})$.

The printing cost of the coloured motif is estimated at 2500 FCFA per square decimetre of inked surface. The owner asks: what will the ink for one motif cost on each of the three supports?



The motif $T = ABC$ (blue), its image under r_1 (orange) and its image under $f = r_2 \circ r_1$ (purple): a quarter-turn about O.

9.3 Tasks

Work in groups of three or four. Present all constructions on grid paper and justify every transformation by its **characteristic elements**.

1. **Placing the motif.** Draw the coordinate plane (unit 1 cm), plot the triangle $T = ABC$, and compute its area A in cm^2.

2. **Composing two reflections.** Let r_1 be the reflection across (Δ_1) (the x-axis) and r_2 the reflection across (Δ_2) (the line $y = x$).

   - Determine the coordinates of $A_1 = r_1(A)$, $B_1 = r_1(B)$, $C_1 = r_1(C)$, then those of $A' = r_2(A_1)$, $B' = r_2(B_1)$, $C' = r_2(C_1)$.

- State the theorem on the composition of two reflections across **intersecting axes**. Identify the single transformation $f = r_2 \circ r_1$, giving its centre and its angle (measure the angle from (Δ_1) to (Δ_2)).
- Verify numerically that A' is indeed the image of A under this rotation.
- Deduce the transformation that generates the whole repeated pattern around O, and state how many copies of the motif complete one full turn.

3. **The three sizes.** Let h_k denote the homothety of centre O and ratio k.
   - Give the analytical expression of h_k, i.e. the coordinates of $M'(x' ; y')$, image of $M(x ; y)$.
   - Compute the coordinates of the vertices of the poster motif ($k_2 = 2.5$) and of the billboard motif ($k_3 = 6$).
   - State the property giving the image of a triangle under a homothety, and the effect on side lengths.

4. **The spiral similarity.** Consider $s : z' = (1 + i\sqrt{3})\,z + 2$.
   - Compute the modulus and an argument of $a = 1 + i\sqrt{3}$. Deduce the ratio and the angle of s.
   - Show that s has a unique fixed point Ω, and compute its affix ω. Conclude on the nature and the characteristic elements of s.
   - Compute the affixes of $A'' = s(A)$ and $B'' = s(B)$, then the distance $A''B''$. Verify that $A''B'' = |a| \cdot AB$.

5. **Ink cost estimation.**
   - State the area-scaling property of a similarity of ratio k.
   - Compute the area of one motif on each support, in cm^2 then in dm^2 ($1\ \text{dm}^2 = 100\ \text{cm}^2$).
   - Estimate the ink cost of one motif on each support at 2500 FCFA per dm^2.

6. **Synthesis.** Each group presents its grid-paper constructions and completes a summary table: transformation, definition used, characteristic elements, image of the motif, effect on lengths and areas.

9.4 Model answer

1. Placing the motif.

The triangle ABC has base AB on the x-axis with $AB = 3 - 1 = 2$ cm, and the height from C to the line (AB) is the ordinate of C, namely 1.5 cm. Hence

$$A = \frac{1}{2} \times AB \times 1.5 = \frac{1}{2} \times 2 \times 1.5 = 1.5\ \text{cm}^2.$$

2. Composing two reflections.

The reflection r_1 across the x-axis maps $(x ; y) \mapsto (x ; -y)$:

$$A_1(1 ; 0), \qquad B_1(3 ; 0), \qquad C_1(2 ; -1.5).$$

The reflection r_2 across the line $y = x$ maps $(x\,;\,y) \mapsto (y\,;\,x)$:

$$A'(0\,;\,1), \qquad B'(0\,;\,3), \qquad C'(-1.5\,;\,2).$$

Theorem (composition of two reflections, intersecting axes): if (Δ_1) and (Δ_2) intersect at O, then $r_2 \circ r_1$ is the **rotation** of centre O whose angle is twice the directed angle from the first axis to the second:

$$r_2 \circ r_1 = \mathrm{Rot}\big(O\,;\,2\,(\Delta_1,\hat{\Delta}_2)\big).$$

Here $(\Delta_1,\hat{\Delta}_2) = 45^\circ$, so $f = r_2 \circ r_1$ is the rotation of centre O and angle $2 \times 45^\circ = 90^\circ$.

Verification: the rotation of centre O and angle 90° maps $(x\,;\,y) \mapsto (-y\,;\,x)$. Applied to $A(1\,;\,0)$ it gives $(0\,;\,1) = A'$; applied to $B(3\,;\,0)$ it gives $(0\,;\,3) = B'$; applied to $C(2\,;\,1.5)$ it gives $(-1.5\,;\,2) = C'$. ■

The pattern is therefore generated by successive rotations of angle 90° about O; since $\dfrac{360^\circ}{90^\circ} = 4$, exactly **four copies** of the motif complete one full turn.

3. The three sizes.

The homothety h_k of centre O and ratio k is defined by $\vec{OM'} = k\,\vec{OM}$; its analytical expression is therefore

$$M(x\,;\,y) \longmapsto M'(kx\,;\,ky).$$

For the **poster** ($k_2 = 2.5$): $A_2(2.5\,;\,0)$, $B_2(7.5\,;\,0)$, $C_2(5\,;\,3.75)$.

For the **billboard** ($k_3 = 6$): $A_3(6\,;\,0)$, $B_3(18\,;\,0)$, $C_3(12\,;\,9)$.

Property: a homothety maps a triangle onto a triangle whose corresponding sides are parallel, and it multiplies all lengths by $|k|$ (here $k > 0$, so lengths are multiplied by k).

4. The spiral similarity.

For $a = 1 + i\sqrt{3}$: $|a| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$, and writing $a = 2\left(\dfrac{1}{2} + i\dfrac{\sqrt{3}}{2}\right)$ gives $\arg(a) = \dfrac{\pi}{3}\,[2\pi]$. The ratio of s is therefore $k = |a| = 2$ and its angle is $\theta = \arg(a) = \dfrac{\pi}{3}$.

Fixed point: a point of affix z is fixed by s when $z = az + b$, i.e. $(1 - a)z = b$. Since $a = 1 + i\sqrt{3} \neq 1$, there is a **unique** fixed point Ω, of affix

$$\omega = \frac{b}{1-a} = \frac{2}{1-(1+i\sqrt{3})} = \frac{2}{-i\sqrt{3}} = \frac{2i}{\sqrt{3}} = \frac{2i\sqrt{3}}{3}.$$

Thus s is the **direct similarity** (spiral similarity) of centre $\Omega\left(0\,;\,\dfrac{2\sqrt{3}}{3}\right)$, ratio 2 and angle $\dfrac{\pi}{3}$: it is the composition, in either order, of the homothety $h(\Omega\,;\,2)$ and the rotation $\mathrm{Rot}\left(\Omega\,;\,\dfrac{\pi}{3}\right)$.

Images: $z_A = 1$, so $z_{A''} = (1 + i\sqrt{3})(1) + 2 = 3 + i\sqrt{3}$; $z_B = 3$, so $z_{B''} = 3(1 + i\sqrt{3}) + 2 = 5 + 3i\sqrt{3}$. Then

$$A''B'' = |z_{B''} - z_{A''}| = |2 + 2i\sqrt{3}| = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = 4 = 2 \times AB,$$

which confirms that a direct similarity of ratio $|a|$ multiplies all distances by $|a|$. ■

5. Ink cost estimation.

Property: a similarity of ratio k multiplies areas by k^2. With $A = 1.5\ \mathrm{cm}^2$:

Support	Ratio k	Area $= 1.5k^2$ cm^2	Cost (FCFA)
Business card	1	1.5 cm^2 = 0.015 dm^2	0.015 × 2500 = 37.5
Poster	2.5	9.375 cm^2 = 0.09375 dm^2	≈ 234.4
Billboard	6	54 cm^2 = 0.54 dm^2	0.54 × 2500 = 1350

6. Synthesis.

Transformation	Characteristic elements	Effect
$r_2 \circ r_1$	Rotation, centre O, angle $2 \times 45^\circ = 90^\circ$	Preserves lengths and areas
$h(O\,;\,k)$	Centre O, ratio $k \in \{1\,;\,2.5\,;\,6\}$	Lengths $\times k$, areas $\times k^2$
$s : z' = az + b$	Centre Ω, ratio $\lvert a\rvert = 2$, angle $\arg a = \frac{\pi}{3}$	Lengths $\times 2$, areas $\times 4$

WARNING **Common mistakes to avoid**

- The angle of the rotation $r_2 \circ r_1$ is twice the angle between the axes, measured from the first axis to the second: the order of composition matters ($r_2 \circ r_1 \neq r_1 \circ r_2$ in general).
- The centre of $z' = az + b$ is not the point of affix b: it is the fixed point of affix $\omega = \dfrac{b}{1-a}$.
- Areas scale by k^2, not by k: the billboard motif is 36 times, not 6 times, the area of the card motif.
- Do not forget unit conversions (cm^2 → dm^2) before computing the cost.

EXERCISE 9 **Extension: a glide reflection for the logo**

The press owner now wants a motif mapped onto a shifted copy by a glide reflection. Let t be the translation of vector $\vec{u}(4\,;\,0)$ and r the reflection across the x-axis.

1. Compute the coordinates of the images of A, B, C under $g = r \circ t$, then under $g' = t \circ r$.
2. What do you observe? Explain, using the fact that $\vec{u}$ is parallel to the axis of r.
3. Justify that g is an indirect isometry and prove that it has no fixed point.

SOLUTION **Extension: a glide reflection for the logo**

1. Under t: $(x\,;\,y) \mapsto (x+4\,;\,y)$; under r: $(x\,;\,y) \mapsto (x\,;\,-y)$.

For $g = r \circ t$ (translate first, then reflect): $(x\,;\,y) \mapsto (x+4\,;\,-y)$, giving

$$A \mapsto (5\,;\,0), \qquad B \mapsto (7\,;\,0), \qquad C \mapsto (6\,;\,-1.5).$$

For $g' = t \circ r$ (reflect first, then translate): $(x\,;\,y) \mapsto (x+4\,;\,-y)$ as well, giving the same images.
2. We observe that $g = g'$: the translation and the reflection commute. This is because the vector $\vec{u}$ is parallel to the axis of reflection: reflecting across the x-axis does not change a horizontal displacement. The composite $g = r \circ t$ with $\vec{u}$ parallel to the axis is precisely a **glide reflection**.
3. The reflection r is an indirect isometry and t is a direct isometry, so their composition is an indirect isometry (it reverses orientation). A fixed point would satisfy $(x+4\,;\,-y) = (x\,;\,y)$, i.e. $x + 4 = x$, which is impossible. Hence g has no fixed point, as expected for a glide reflection with a non-zero translation vector.

10 METHODS AND WORKED EXAMPLES

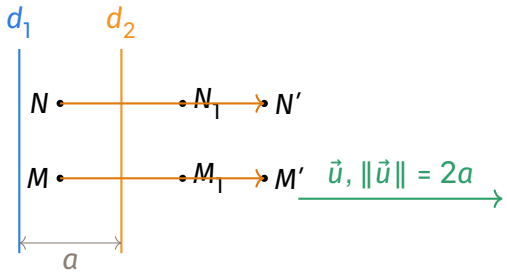
10.1 Composing two reflections (orthogonal symmetries)

Recall that the **orthogonal symmetry** (reflection) S_d in a line d sends a point M to the point M' such that d is the perpendicular bisector of $[MM']$ (and fixes every point of d). What happens when we apply two reflections one after the other? The answer depends only on the relative position of the two axes, and it produces the two fundamental direct isometries: translations and rotations.

PROPERTY **Composite of two reflections — proof not examinable**

Let d_1 and d_2 be two lines of the plane.

- If $d_1 \parallel d_2$, then $S_{d_2} \circ S_{d_1}$ is the **translation** of vector $\vec{u} = 2\vec{H_1H_2}$, where $H_1 \in d_1$ and $H_2 \in d_2$ are the feet of a common perpendicular. In particular $\|\vec{u}\| = 2\,\mathrm{dist}(d_1, d_2)$.
- If d_1 and d_2 intersect at O, then $S_{d_2} \circ S_{d_1}$ is the **rotation** of centre O and angle 2θ, where $\theta = (d_1, d_2)$ is the oriented angle from d_1 to d_2.



Two reflections in parallel axes d_1, d_2 give a translation of vector twice the gap between the axes.

METHOD **Identifying the composite of two reflections**

To determine the nature of $S_{d_2} \circ S_{d_1}$ where d_1, d_2 are lines in the plane:

1. Draw or visualise the two axes d_1 and d_2. Decide whether they are **parallel** or **intersecting**.
2. Parallel axes (distance a apart): the composite is a **translation** of vector $\vec{u} = 2\vec{H_1H_2}$, where $H_1 \in d_1$, $H_2 \in d_2$ are feet of a common perpendicular. Hence $\|\vec{u}\| = 2a$, directed from d_1 towards d_2.
3. Intersecting axes at O, with angle θ from d_1 to d_2: the composite is the **rotation** of centre O and angle 2θ.

4. Remember the order: $S_{d_2} \circ S_{d_1}$ means S_{d_1} is applied first. Reversing the order reverses the translation vector or the rotation angle.

5. To verify, test the composite on one or two well-chosen points (e.g. a point on d_2 is fixed by S_{d_2}, so only S_{d_1} moves it).

WORKED EXAMPLE 16 **Composite of reflections in two parallel lines**

In the plane with an orthonormal frame $(O;\vec{i},\vec{j})$, let d_1 be the line of equation $x = 1$ and d_2 the line of equation $x = 4$. Determine the nature and characteristic elements of $f = S_{d_2} \circ S_{d_1}$, and find the image of $A(2, 3)$ under f.

SOLUTION

Solution.
Step 1 — Nature of the axes. Both d_1 and d_2 are vertical lines, hence parallel. The distance between them is $a = 4 - 1 = 3$.
Step 2 — Apply the theorem. The composite of two reflections in parallel axes is a translation of vector $\vec{u} = 2\overrightarrow{H_1H_2}$, where H_1, H_2 are the intersections of a common perpendicular with d_1, d_2. Take the horizontal perpendicular: $H_1(1, 0) \in d_1$, $H_2(4, 0) \in d_2$. Then

$$\vec{u} = 2\overrightarrow{H_1H_2} = 2(4 - 1)\,\vec{i} = 6\vec{i}.$$

So f is the translation of vector $\vec{u} = 6\vec{i}$, i.e. $M(x, y) \mapsto M'(x + 6, y)$.
Step 3 — Direct verification. For $M(x, y)$: S_{d_1} sends M to M_1 with the same ordinate and abscissa x_1 satisfying $\frac{x+x_1}{2} = 1$, so $x_1 = 2 - x$. Then S_{d_2} sends M_1 to M' with $\frac{x_1+x'}{2} = 4$, giving $x' = 8 - x_1 = 8 - (2 - x) = x + 6$. Confirmed: f is the translation by $6\vec{i}$.
Step 4 — Image of $A(2, 3)$. $A' = f(A) = (2 + 6,\ 3) = (8, 3)$.

WORKED EXAMPLE 17 **Composite of reflections in two intersecting lines**

Let d_1 and d_2 be two lines through the point O, such that the oriented angle from d_1 to d_2 is $\frac{\pi}{6}$. Show that $f = S_{d_2} \circ S_{d_1}$ is a rotation, give its elements, and deduce the image of a point $M \neq O$ with $OM = 4$ cm when $M \in d_1$.

SOLUTION

Solution.
Step 1 — Apply the theorem on intersecting axes. The composite of two reflections in axes intersecting at O is the rotation of centre O whose angle is twice the angle from the first

axis to the second:

$$f = r\left(O,\ 2 \times \tfrac{\pi}{6}\right) = r\left(O,\ \tfrac{\pi}{3}\right).$$

Step 2 — Justification. The point O lies on both axes, so $S_{d_1}(O) = O$ and $S_{d_2}(O) = O$: O is fixed by f. Each reflection preserves distances, so f is an isometry fixing O; being a composite of two indirect isometries, f is direct, hence a rotation about O. For the angle, let α and β be the angles that d_1 and d_2 make with a fixed reference direction, so $\beta - \alpha = \theta = \frac{\pi}{6}$. A reflection in a line through O making angle α sends a direction of angle φ to a direction of angle $2\alpha - \varphi$. Hence, if $\vec{OM}$ has angle φ, then $\vec{OM_1}$ (with $M_1 = S_{d_1}(M)$) has angle $2\alpha - \varphi$, and $\vec{OM'}$ has angle

$$2\beta - (2\alpha - \varphi) = \varphi + 2(\beta - \alpha) = \varphi + 2\theta.$$

Therefore $\left(\vec{OM}, \vec{OM'}\right) = 2\theta = \dfrac{\pi}{3}$, as claimed.

Step 3 — Image of $M \in d_1$ with $OM = 4$ cm. Since $M \in d_1$, $S_{d_1}(M) = M$, so $M' = S_{d_2}(M)$: M' is the mirror image of M in d_2. Equivalently, $M' = r(O, \frac{\pi}{3})(M)$, so $OM' = OM = 4$ cm and $\left(\vec{OM}, \vec{OM'}\right) = \frac{\pi}{3}$. The triangle OMM' is isosceles at O with vertex angle $\frac{\pi}{3}$, hence equilateral: $MM' = 4$ cm.

10.2 Composing translations, rotations and glide reflections

Translations and rotations are the direct isometries; composing them can never produce a reflection. The results below are obtained either by Chasles' relation (for translations) or by decomposing each rotation into two reflections with a cleverly chosen common axis.

METHOD **Composing translations and rotations**

1. Two translations: $t_{\vec{u}} \circ t_{\vec{v}} = t_{\vec{u}+\vec{v}}$ (Chasles). Always a translation, commutative.
2. Two rotations of the same centre O: $r(O, \alpha) \circ r(O, \beta) = r(O, \alpha + \beta)$. If $\alpha + \beta \equiv 0 \pmod{2\pi}$, the composite is the identity.
3. Two rotations of distinct centres A, B: the composite $r(B, \beta) \circ r(A, \alpha)$ is a rotation of angle $\alpha+\beta$ if $\alpha+\beta \not\equiv 0 \pmod{2\pi}$, and a translation if $\alpha+\beta \equiv 0 \pmod{2\pi}$. To find the new centre, decompose each rotation into two reflections sharing a common axis (usually the line (AB)).
4. Translation $\circ$ reflection (or reflection $\circ$ translation): in general a **glide reflection**; it reduces to a pure reflection exactly when the translation vector is perpendicular to the axis.
5. Reflex: to locate the centre of a composite rotation, look for a fixed point: solve $f(M) = M$.

WORKED EXAMPLE 18 **Composite of two rotations with distinct centres**

Let A and B be two distinct points. Let $r_A = r\left(A, \frac{\pi}{2}\right)$ and $r_B = r\left(B, \frac{\pi}{2}\right)$. Determine the nature of $f = r_B \circ r_A$ and construct its centre when $AB = 2$ cm.

SOLUTION

Solution.
Step 1 — Nature. The sum of the angles is $\frac{\pi}{2} + \frac{\pi}{2} = \pi \not\equiv 0 \pmod{2\pi}$. Therefore f is a rotation of angle π, i.e. a **central symmetry** (half-turn). Its centre Ω remains to be found.
Step 2 — Decomposition into reflections. Write each rotation as a composite of two reflections whose axes pass through its centre, choosing the common axis (AB):

$$r_A = S_{(AB)} \circ S_{d_1}, \qquad r_B = S_{d_2} \circ S_{(AB)},$$

where d_1 is the line through A making angle $-\frac{\pi}{4}$ with (AB), and d_2 the line through B making angle $+\frac{\pi}{4}$ with (AB) (each rotation angle is twice the angle between its axes). Then

$$f = r_B \circ r_A = S_{d_2} \circ S_{(AB)} \circ S_{(AB)} \circ S_{d_1} = S_{d_2} \circ S_{d_1}.$$

Step 3 — The centre. The lines d_1 and d_2 make angles $\mp\frac{\pi}{4}$ with (AB); they intersect at a point Ω such that the triangle $A\Omega B$ has angles $\frac{\pi}{4}, \frac{\pi}{4}, \frac{\pi}{2}$: it is right-angled isosceles at Ω. Hence Ω is the point with $\Omega A = \Omega B$ and $(\vec{\Omega A}, \vec{\Omega B}) = \frac{\pi}{2}$ — the apex of the right isosceles triangle built on $[AB]$. With $AB = 2$ cm: $\Omega A = \Omega B = \frac{AB}{\sqrt{2}} = \sqrt{2}$ cm.
Conclusion. f is the half-turn about the point Ω such that $A\Omega B$ is right-angled isosceles at Ω.

WORKED EXAMPLE 19 **Recognising a glide reflection**

In an orthonormal frame, let d be the x-axis and $\vec{u} = 4\vec{i}$. Let $f = t_{\vec{u}} \circ S_d$. Find the image of $M(x, y)$, show that f has no fixed point, and express f as a glide reflection (axis and vector).

SOLUTION

Solution.
Step 1 — Image of $M(x, y)$. First $S_d(M) = M_1(x, -y)$. Then $t_{\vec{u}}(M_1) = M'(x + 4, -y)$. Hence

$$f : (x, y) \longmapsto (x + 4,\ -y).$$

Step 2 — Fixed points. M is fixed iff $x + 4 = x$ and $-y = y$. The first equation gives $4 = 0$: impossible. So f has no fixed point.
Step 3 — Nature. Since $\vec{u} = 4\vec{i}$ is parallel to the axis d (the x-axis), the decomposition cannot be simplified to a pure reflection: f is a glide reflection with axis d (the line $y = 0$) and vector $\vec{u} = 4\vec{i}$.
Step 4 — Consistency check. $f \circ f = t_{\vec{u}} \circ S_d \circ t_{\vec{u}} \circ S_d = t_{2\vec{u}}$ (since $S_d \circ t_{\vec{u}} \circ S_d = t_{\vec{u}}$ when $\vec{u} \parallel d$).

Indeed $f(f(x,y)) = f(x+4,-y) = (x+8,y)$, a translation by $8\vec{i}$: the square of a glide reflection is always a translation. ■

10.3 Classifying a plane isometry from given data

An **isometry** of the plane is a transformation that preserves distances: $M'N' = MN$ for all points M, N. Isometries are of two kinds: **direct** isometries, which preserve the orientation of angles (translations and rotations), and **indirect** isometries, which reverse it (reflections and glide reflections). The classification theorem states that these four types are the only plane isometries; the fixed points tell us which type we are facing.

METHOD **Recognition table for isometries**

An isometry is **direct** (preserves orientation: translations, rotations) or **indirect** (reflections, glide reflections). Use fixed points:

1. Compute (or test) the fixed points of f.
2. Three non-collinear fixed points ⇒ identity. Exactly a line of fixed points ⇒ reflection in that line.
3. Exactly one fixed point ⇒ rotation about that point (find the angle from one point and its image).
4. No fixed point and direct ⇒ translation (vector = $\vec{MM'}$ for any M).
5. No fixed point and indirect ⇒ glide reflection: find the axis as the line through the midpoints of the segments $[MM']$, and the vector from the displacement along the axis.
6. Orientation test: a direct isometry preserves the sign of angles; analytically, the matrix $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ (with $a^2+b^2=1$) is direct, while $\begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ is indirect.

WORKED EXAMPLE 20 **Identifying an isometry given analytically**

The plane is referred to an orthonormal frame $(O;\vec{i},\vec{j})$. A transformation f maps $M(x,y)$ to $M'(x',y')$ with

$$x' = \frac{1}{2}x - \frac{\sqrt{3}}{2}y + 1, \qquad y' = \frac{\sqrt{3}}{2}x + \frac{1}{2}y - 2.$$

Show that f is an isometry, determine its nature and its characteristic elements.

SOLUTION

Solution.

Step 1 — Isometry and orientation. The linear part has matrix $\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$. Its columns are orthonormal: $\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 = 1$ and the determinant is $\frac{1}{4} + \frac{3}{4} = 1 > 0$. Hence f is a direct isometry: a translation or a rotation.
Step 2 — Fixed points. Solve $f(M) = M$:

$$\begin{cases} \frac{1}{2}x - \frac{\sqrt{3}}{2}y + 1 = x \\ \frac{\sqrt{3}}{2}x + \frac{1}{2}y - 2 = y \end{cases} \iff \begin{cases} -\frac{1}{2}x - \frac{\sqrt{3}}{2}y = -1 \\ \frac{\sqrt{3}}{2}x - \frac{1}{2}y = 2 \end{cases} \iff \begin{cases} x + \sqrt{3}\,y = 2 \\ \sqrt{3}\,x - y = 4. \end{cases}$$

Multiply the second equation by $\sqrt{3}$: $3x - \sqrt{3}y = 4\sqrt{3}$. Adding to the first: $4x = 2 + 4\sqrt{3}$, so $x = \frac{1+2\sqrt{3}}{2}$. Then $y = \sqrt{3}x - 4 = \frac{\sqrt{3}+6}{2} - 4 = \frac{\sqrt{3}-2}{2}$.
There is exactly one fixed point $\Omega\left(\frac{1+2\sqrt{3}}{2}, \frac{\sqrt{3}-2}{2}\right)$, so f is a rotation of centre Ω.
Step 3 — Angle. The linear part is the matrix of the rotation of angle θ with $\cos\theta = \frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$, i.e. $\theta = \frac{\pi}{3}$.
Conclusion. $f = r\left(\Omega, \frac{\pi}{3}\right)$ with Ω as above.

10.4 Constructing and using a homothety

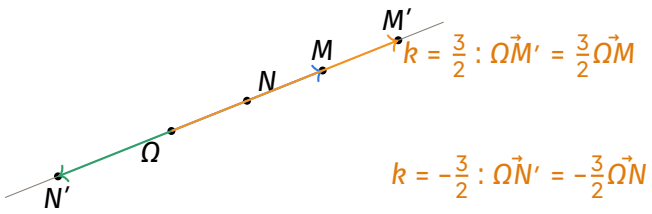
Enlarging a photograph, drawing a plan of a school campus to scale, projecting a shadow: all these situations suggest a transformation that multiplies all distances by the same factor from a fixed point. This is the idea of a homothety.

DEFINITION **Homothety**

Let Ω be a point and k a non-zero real number. The **homothety** of centre Ω and ratio k, denoted $h(\Omega, k)$, is the transformation sending every point M to the point M' defined by

$$\vec{\Omega M'} = k\,\vec{\Omega M}.$$

If $k > 0$ the homothety is called **direct** (positive); if $k < 0$ it is called **indirect** (negative), and M' lies on the ray opposite to $[\Omega M)$.



Homothety of centre Ω: positive ratio keeps the direction, negative ratio reverses it.

METHOD **Homothety $h(\Omega, k)$: construction and properties**

1. Image of a point: $M' = h(M) \iff \vec{\Omega M'} = k\,\vec{\Omega M}$. If $k > 0$, M' is on the ray $[\Omega M)$; if $k < 0$, on the opposite ray (indirect/negative homothety).
2. Image of a line d: a line d' parallel to d (if $\Omega \notin d$); d is globally invariant if $\Omega \in d$.
3. Image of a circle $C(O, R)$: the circle of centre $O' = h(O)$ and radius $|k|R$.
4. Distances are multiplied by $|k|$; areas by k^2; angles and alignment are preserved.
5. To find the centre of a homothety mapping $A \mapsto A', B \mapsto B'$: the centre lies at the intersection of the lines (AA') and (BB'); the ratio is $k = \dfrac{\vec{\Omega A'}}{\vec{\Omega A}}$.
6. $k = 1$ gives the identity; $k = -1$ gives the central symmetry about Ω.

WORKED EXAMPLE 21 **Determining a homothety from two pairs of points**

In an orthonormal frame, $A(1, 2)$, $B(4, 2)$ have images $A'(3, 6)$, $B'(9, 6)$ under a homothety h. Find the centre and ratio of h, then the image of the circle of centre $C(2, 0)$ and radius 1.

SOLUTION

Solution.
Step 1 — The centre. The centre Ω lies on (AA') and on (BB'). Line (AA'): through $A(1, 2)$ and $A'(3, 6)$, direction $(2, 4) \parallel (1, 2)$, equation $y = 2x$. Line (BB'): through $B(4, 2)$ and $B'(9, 6)$, direction $(5, 4)$, equation $y - 2 = \frac{4}{5}(x - 4)$.
Intersecting: $2x - 2 = \frac{4}{5}(x - 4) \Rightarrow 10x - 10 = 4x - 16 \Rightarrow 6x = -6 \Rightarrow x = -1, y = -2$. Hence $\Omega(-1, -2)$.
Step 2 — The ratio. $\vec{\Omega A} = (1 - (-1), 2 - (-2)) = (2, 4)$ and $\vec{\Omega A'} = (3 - (-1), 6 - (-2)) = (4, 8) = 2(2, 4)$.
So $k = 2$. Check with B: $\vec{\Omega B} = (5, 4)$, $2\vec{\Omega B} = (10, 8) = \vec{\Omega B'}$. ✓
Thus $h = h(\Omega, 2)$ with $\Omega(-1, -2)$.

Step 3 — Image of the circle. The image is the circle of centre $C' = h(C)$ and radius $|k| \times 1 = 2$.

$$\vec{\Omega C'} = 2\vec{\Omega C} = 2(2 - (-1), 0 - (-2)) = 2(3, 2) = (6, 4),$$

so $C' = \Omega + (6, 4) = (5, 2)$. The image circle has centre $(5, 2)$ and radius 2: equation $(x - 5)^2 + (y - 2)^2 = 4$.

10.5 Analytical and complex representation of homothety

METHOD **Formulas for $h(\Omega, k)$**

1. Vector form: $\vec{\Omega M'} = k\,\vec{\Omega M}$.
2. Analytical form (frame $(O; \vec{i}, \vec{j})$, $\Omega(a, b)$):

$$\begin{cases} x' = kx + (1 - k)a \\ y' = ky + (1 - k)b. \end{cases}$$

3. Complex form (Ω of affix ω, M of affix z): $z' = kz + (1 - k)\omega$ with $k \in \mathbb{R}^*$.
4. Conversely, any map $z \mapsto kz + p$ with $k \in \mathbb{R} \setminus \{0, 1\}$ is a homothety of ratio k; its centre has affix $\omega = \dfrac{p}{1 - k}$ (the unique fixed point).
5. Reflex: to identify the transformation $z \mapsto az + b$, first check whether $a \in \mathbb{R}$ (homothety/translation) or $|a| = 1$ (rotation/translation).

WORKED EXAMPLE 22 **Identifying a homothety from its complex expression**

The plane is equipped with an orthonormal frame; to each point M we associate its affix z. Let f be the map defined by

$$z' = -3z + 8 - 4i.$$

Determine the nature of f, its centre and its ratio. Find the image of the point E of affix $1 + i$.

SOLUTION

Solution.
Step 1 — Nature. The map has the form $z' = az + b$ with $a = -3 \in \mathbb{R}^* \setminus \{1\}$. Hence f is a homothety of ratio $k = -3$ (an indirect, i.e. negative, homothety).
Step 2 — Centre. The centre Ω is the unique fixed point: $\omega = -3\omega + 8 - 4i$, so $4\omega = 8 - 4i$ and

$$\omega = 2 - i.$$

Check with the formula $\omega = \dfrac{b}{1-a} = \dfrac{8-4i}{1-(-3)} = \dfrac{8-4i}{4} = 2-i$. ✓
Step 3 — Standard form. $f = h(\Omega, -3)$, i.e. $z' - \omega = -3(z-\omega)$. Indeed $-3(z-2+i)+2-i = -3z+6-3i+2-i = -3z+8-4i$. ✓
Step 4 — Image of $E(1+i)$. $z'_E = -3(1+i)+8-4i = -3-3i+8-4i = 5-7i$. So E' has affix $5-7i$.
Verification: $\vec{\Omega E}$ has affix $(1+i)-(2-i) = -1+2i$; $-3(-1+2i) = 3-6i$; $\Omega + (3-6i) = 5-7i$. ✓

10.6 Composing two homotheties

METHOD **Composite of two homotheties**

Let $h_1 = h(\Omega_1, k_1)$ and $h_2 = h(\Omega_2, k_2)$.

1. Same centre Ω: $h(\Omega, k_2) \circ h(\Omega, k_1) = h(\Omega, k_1 k_2)$. The homotheties of centre Ω commute and form a group.
2. Distinct centres: the composite $h_2 \circ h_1$ multiplies distances by $|k_1 k_2|$ and sends every line to a parallel line.
   - If $k_1 k_2 \neq 1$: it is a homothety of ratio $k_1 k_2$ whose centre Ω is aligned with Ω_1 and Ω_2; more precisely $\vec{\Omega_2\Omega} = \dfrac{k_2(1-k_1)}{1-k_1k_2}\vec{\Omega_2\Omega_1}$. In practice: solve $f(M) = M$.
   - If $k_1 k_2 = 1$: it is a translation of vector $(1-k_2)\vec{\Omega_1\Omega_2}$. In practice, compute the image of one point: $\vec{u} = \vec{M f(M)}$.
3. Strategy: write the complex or analytical expressions and compose them; then identify.

WORKED EXAMPLE 23 **Composite of two homotheties with distinct centres**

Let $h_1 = h(A, 2)$ and $h_2 = h(B, \frac{1}{2})$ where A and B have affixes $a = 0$ and $b = 3$ in an orthonormal frame. Determine the nature and elements of $f = h_2 \circ h_1$, then of $g = h_1 \circ h_2$. Comment.

SOLUTION

Solution.
Step 1 — Complex expressions. $h_1 : z \mapsto 2z$ (centre 0, ratio 2). $h_2 : z \mapsto \frac{1}{2}z + (1-\frac{1}{2})\cdot 3 = \frac{1}{2}z + \frac{3}{2}$.
Step 2 — Composite $f = h_2 \circ h_1$.

$$f(z) = h_2(2z) = \tfrac{1}{2}(2z) + \tfrac{3}{2} = z + \tfrac{3}{2}.$$

Since $k_1 k_2 = 2 \times \frac{1}{2} = 1$, f is a translation of vector affix $\frac{3}{2}$, i.e. $\vec{u} = \frac{3}{2}\vec{i}$. Note $\vec{u} = (1-k_2)\vec{AB} = \frac{1}{2}\vec{AB}$: consistent, since $\vec{AB}$ has affix 3.

Step 3 — Composite $g = h_1 \circ h_2$.

$$g(z) = h_1\left(\tfrac{1}{2}z + \tfrac{3}{2}\right) = 2\left(\tfrac{1}{2}z + \tfrac{3}{2}\right) = z + 3.$$

So g is the translation of vector affix 3, i.e. $\vec{v} = \vec{AB}$.
Step 4 — Comment. $f \neq g$: composition of homotheties with distinct centres is not commutative. Both are translations (because the product of ratios is 1), but with different vectors: $\vec{v} = 2\vec{u}$. This is a classic GCE trap: always compute the composite in the order requested.

10.7 Plane similarities: recognition and applications

Homotheties multiply distances by $|k|$; rotations and translations preserve them. Composing such maps yields transformations that multiply all distances by a fixed positive factor while preserving angles: these are the **plane similarities**, the most general "shape-preserving" transformations of the plane.

METHOD **Studying a plane similarity**

1. A **similarity** of ratio $k > 0$ multiplies all distances by k: $M'N' = k \cdot MN$. It preserves angles, alignment, parallelism, midpoints, and sends circles to circles.
2. Complex test: $z' = az + b$ ($a \neq 0$) is a **direct similarity**: ratio $|a|$, angle $\arg(a)$. $z' = a\bar{z} + b$ is an **indirect similarity**.
3. Identification of $z' = az + b$:
   - $a = 1$: translation of vector affix b.
   - $|a| = 1$, $a \neq 1$: rotation of angle $\arg a$ about the fixed point $\omega = \frac{b}{1-a}$.
   - $a \in \mathbb{R} \setminus \{0, 1\}$: homothety of ratio a, centre $\omega = \frac{b}{1-a}$.
   - General case ($|a| \neq 1$, $a \notin \mathbb{R}$): direct similarity $= h(\Omega, |a|) \circ r(\Omega, \arg a)$, centre Ω of affix $\frac{b}{1-a}$.
4. A **spiral similarity** is exactly the composite $h \circ r$ of a homothety and a rotation of the same centre: it is the general direct similarity with a fixed point.
5. Applications: two circles are always homothetic (the two homothety centres lie on the line of centres); similar triangles are linked by one similarity.

WORKED EXAMPLE 24 **Identifying and decomposing a direct similarity**

Let s be the plane transformation given in complex form by

$$z' = (1+i)\,z + 2 - i.$$

1. Show that s is a direct similarity; give its ratio and angle.
2. Determine its fixed point Ω.
3. Write s as the composite of a homothety and a rotation of centre Ω.
4. Find the image of the circle C of centre O (affix 0) and radius 2.

SOLUTION

Solution.
Step 1 — Nature, ratio, angle. s has the form $z' = az+b$ with $a = 1+i \neq 0$: it is a direct similarity. Ratio: $k = |a| = |1+i| = \sqrt{2}$. Angle: $\arg(a) = \arg(1+i) = \frac{\pi}{4}$. Since $|a| \neq 1$ and $a \notin \mathbb{R}$, s is neither a rotation nor a homothety: it is a genuine spiral similarity.
Step 2 — Fixed point. Solve $\omega = (1+i)\omega + 2 - i$:

$$\omega - (1+i)\omega = 2 - i \iff -i\omega = 2 - i \iff \omega = \frac{2-i}{-i} = \frac{(2-i)\,i}{(-i)(i)} = \frac{2i+1}{1} = 1 + 2i.$$

Hence Ω has affix $\omega = 1 + 2i$.
Step 3 — Decomposition. Any direct similarity with fixed point Ω writes $z' - \omega = a(z - \omega)$, i.e.

$$s = h(\Omega, \sqrt{2}) \circ r\left(\Omega, \tfrac{\pi}{4}\right) = r\left(\Omega, \tfrac{\pi}{4}\right) \circ h(\Omega, \sqrt{2}),$$

the two maps commuting since they share the same centre. Check: $z' = \omega + (1+i)(z-\omega) = (1+i)z + \omega(1 - 1 - i) = (1+i)z - i(1+2i) = (1+i)z + 2 - i$. ✓
Step 4 — Image of the circle $C(O, 2)$. A similarity of ratio $\sqrt{2}$ sends a circle to a circle whose radius is multiplied by $\sqrt{2}$ and whose centre is the image of the centre. Image of O (affix 0): $z'_O = (1+i)\cdot 0 + 2 - i = 2 - i$. Hence $s(C)$ is the circle of centre $O'(2-i)$ and radius $2\sqrt{2}$.

WARNING **Frequent errors to avoid**

- Confusing the order in $f \circ g$: g acts first. For reflections, $S_{d_2} \circ S_{d_1}$ has vector/angle directed from d_1 to d_2.
- Forgetting the factor 2: the translation vector is twice the gap between parallel axes; the rotation angle is twice the angle between intersecting axes.
- Claiming two rotations of distinct centres always give a rotation: when the angles sum to $0 \pmod{2\pi}$ the composite is a translation.
- Writing the homothety ratio as k in the radius: the radius is multiplied by $|k|$, never by

a negative number.

- For homotheties with $k_1 k_2 = 1$ and distinct centres, concluding "homothety of ratio 1": it is a translation, not the identity in general.
- In $z' = az + b$, concluding "rotation" without checking $|a| = 1$, or "homothety" without checking $a \in \mathbb{R}$.

11 EXERCISES

11.1 I check my knowledge

EXERCISE 10 **MCQ: nature of a composite**

For each question, choose the single correct answer and justify briefly.

1. The composite $s_2 \circ s_1$ of the reflections in two parallel lines d_1, d_2 with $d_1 \neq d_2$ is:
   (a) a rotation
   (b) a translation
   (c) a reflection
   (d) a glide reflection
2. The composite $r_2 \circ r_1$ of two rotations of angles θ_1 and θ_2 with $\theta_1 + \theta_2 \not\equiv 0 \pmod{2\pi}$ is:
   (a) always a translation
   (b) always a rotation
   (c) always a homothety
   (d) a reflection
3. The transformation of complex expression $z' = 3z - 4 + 2i$ is:
   (a) a translation
   (b) a rotation
   (c) a homothety of ratio 3
   (d) an indirect similarity
4. A homothety of ratio $k = -2$ multiplies areas by:
   (a) -2
   (b) 2
   (c) 4
   (d) -4

EXERCISE 11 **True or false**

Say whether each statement is true or false, and justify your answer.

1. Every isometry of the plane is a translation, a rotation or a reflection.
2. The composite of two translations is a translation whose vector is the sum of the two vectors.
3. Two homotheties of distinct centres $O_1 \neq O_2$ and ratios k_1, k_2 with $k_1 k_2 = 1$ compose to give a translation.
4. A direct similarity of ratio $k = 1$ is an isometry.
5. A homothety of ratio -1 is a reflection in a line.

EXERCISE 12 **Definitions**

Give a precise definition of each of the following, stating its characteristic elements:

1. a glide reflection;
2. a homothety of centre Ω and ratio k;
3. a direct plane similarity;
4. an indirect plane similarity.

EXERCISE 13 **Direct calculations**

1. Let $t_{\vec{u}}$ and $t_{\vec{v}}$ be the translations of vectors $\vec{u}(2, -1)$ and $\vec{v}(3, 4)$. Determine the vector of $t_{\vec{v}} \circ t_{\vec{u}}$ and the image of $M(1, 5)$.
2. Find the complex expression of the homothety of centre Ω of affix $1 - i$ and ratio $k = -3$, then the image of the point of affix $2 + i$.
3. Find the complex expression of the rotation of centre $A(1, 0)$ and angle $\frac{\pi}{3}$, and the image of $B(4, 0)$.

11.2 I practise

EXERCISE 14 **Composing reflections in parallel lines**

d_1 and d_2 are two parallel lines 3 cm apart, and s_1, s_2 are the reflections in d_1, d_2.

1. Describe precisely the transformation $s_2 \circ s_1$ (nature and characteristic elements).
2. Construct the image of a given triangle ABC by $s_2 \circ s_1$.
3. Find two lines whose reflections compose to give the translation of vector $\vec{u}(4, 0)$.

EXERCISE 15 **Composing two rotations**

Let $r_1 = r(A, 60^\circ)$ and $r_2 = r(B, -60^\circ)$ with $A \neq B$.

1. Prove that $r_2 \circ r_1$ is a translation.
2. Determine its vector in terms of A and B.
3. What is the nature of $r_2 \circ r_1$ if instead $r_2 = r(B, 120^\circ)$? Give its characteristic elements.

EXERCISE 16 **Recognising an isometry from images**

An isometry f maps $A(0, 0)$ to $A'(2, 1)$, $B(3, 0)$ to $B'(5, 1)$ and $C(0, 4)$ to $C'(-2, 5)$.

1. Show that f is a translation and give its vector.
2. What is the nature of f if instead $C' = (6, 1)$? Determine its characteristic elements.

EXERCISE 17 **Homothety from its complex expression**

Let f be the transformation of the complex plane given by $z' = 3z - 4 + 2i$.

1. Prove that f is a homothety, and find its centre and ratio.
2. Compute the image by f of the circle of centre $1 + i$ and radius 2.
3. Determine the image of the line of equation $z + \bar{z} = 4$.

11.3 I prepare for the GCE

EXERCISE 18 **Composing homotheties**

(8 marks) Let $h_1 = h(O_1, 2)$ and $h_2 = h\left(O_2, \frac{1}{2}\right)$ with $O_1 \neq O_2$.

1. Prove that $h_2 \circ h_1$ is a translation and express its vector in terms of $\vec{O_1O_2}$. (3 marks)

2. Construct this vector on a figure. (1 mark)
3. Repeat the study with $h_2 = h(O_2, 3)$: prove that $h_2 \circ h_1$ is a homothety, give its ratio, and determine the position of its centre on the line (O_1O_2). (4 marks)

EXERCISE 19 **Centres of similitude of two circles**

(10 marks) Two circles C and C' have centres A, B with $AB = 10$ cm and radii 2 cm and 6 cm respectively.

1. Prove that there exist exactly two homotheties mapping C onto C', and give their ratios. (3 marks)
2. Locate the two centres of similitude on the line (AB) by computing their distances from A and B. (3 marks)
3. Show that a common external tangent to the two circles passes through one of the centres of similitude. (2 marks)
4. Construct the two centres of similitude and the common tangents on a figure. (2 marks)

EXERCISE 20 **Euler line and direct similarity**

(12 marks) Part A. In a triangle ABC, let G be the centroid, O the circumcentre and H the orthocentre.

1. Show that the homothety $h(G, -2)$ maps the midpoint of $[BC]$ onto A, and deduce that it maps O onto H. (3 marks)
2. Deduce that O, G, H are collinear and that $GH = 2\,GO$. (2 marks)

Part B. A direct similarity S has complex expression $z' = (1 + i\sqrt{3})z + 2 - i$.

1. Determine the ratio and the angle of S, and classify S. (3 marks)
2. Determine the affix of the fixed point (centre) of S. (2 marks)
3. Find the image of the point of affix 1. (1 mark)
4. Prove that the composite of a glide reflection and a homothety of ratio $k \neq 1$ is an indirect similarity with exactly one fixed point, and illustrate with $z' = 2\bar{z} + 3i$. (1 mark)

12 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — MCQ: nature of a composite**

1. Answer: (b) a translation. The composite of the reflections in two distinct parallel lines $d_1 \parallel d_2$ is the translation of vector $2\vec{u}$, where $\vec{u}$ is the vector from d_1 to d_2 perpendicular to both lines. It cannot be a rotation (no fixed point, and the angle of rotation would be 0 with distinct axes).

2. Answer: (b) always a rotation. The composite of two rotations of angles θ_1, θ_2 is a rotation of angle $\theta_1 + \theta_2$ when $\theta_1 + \theta_2 \not\equiv 0 \pmod{2\pi}$ (and a translation when the sum is 0 modulo 2π). Since the sum is non-zero here, the composite is a rotation.

3. Answer: (c) a homothety of ratio 3. The expression has the form $z' = kz + b$ with $k = 3 \in \mathbb{R}$, $k \neq 1$: this is a homothety of ratio 3 (its centre is the fixed point $\omega = \dfrac{b}{1-k} = \dfrac{-4+2i}{-2} = 2 - i$).

4. Answer: (c) 4. A homothety of ratio k multiplies lengths by $|k|$ and areas by k^2. Here $k^2 = (-2)^2 = 4$.

SOLUTION **Exercise 2 — True or false**

1. False. The classification theorem states that every plane isometry is a translation, a rotation, a reflection or a glide reflection. The glide reflection (composite of a reflection and a translation parallel to its axis) has been forgotten. Counter-example: $z' = \bar{z} + 1$ is a glide reflection, which is none of the three listed types.

2. True. For every point M, $t_{\vec{v}} \circ t_{\vec{u}}(M) = M + \vec{u} + \vec{v} = M + (\vec{u} + \vec{v})$, so the composite is the translation of vector $\vec{u} + \vec{v}$. (Translations commute: $t_{\vec{v}} \circ t_{\vec{u}} = t_{\vec{u}} \circ t_{\vec{v}}$.)

3. True. $h(O_2, k_2) \circ h(O_1, k_1)$ has ratio $k_1 k_2 = 1$. A homothety-type composite of ratio 1 with distinct centres is a translation; its vector is $(1-k_2)\vec{O_1O_2}$ (equivalently $(k_1-1)\vec{O_2O_1}$), which is non-zero since $k_2 \neq 1$ (otherwise $k_1 k_2 = 1$ would force $k_1 = 1$ and the centres could coincide).

4. True. A direct similarity of ratio k multiplies all distances by k. If $k = 1$, distances are preserved: the similarity is a (direct) isometry, i.e. a translation or a rotation.

5. False. A homothety of ratio -1 is the central symmetry (reflection in a point, the centre), i.e. the rotation of angle π about its centre. It is a direct isometry, whereas a reflection in a line is an indirect isometry. They are never equal.

SOLUTION **Exercise 3 — Definitions**

1. Glide reflection. A glide reflection is the composite $t_{\vec{u}} \circ s_d = s_d \circ t_{\vec{u}}$ of the reflection s_d in a line d and a translation $t_{\vec{u}}$ whose vector $\vec{u} \neq \vec{0}$ is parallel to d. Characteristic elements: the axis d and the vector $\vec{u}$ (parallel to d). It is an indirect isometry with no fixed point.

2. Homothety. The homothety of centre Ω and ratio $k \in \mathbb{R}^*$ is the transformation $h(\Omega, k)$ mapping every point M to the point M' defined by $\vec{\Omega M'} = k\,\vec{\Omega M}$. Characteristic elements: the centre Ω (the unique fixed point when $k \neq 1$) and the ratio k.

3. Direct plane similarity. A direct similarity is a transformation of the plane that multiplies all distances by a fixed positive number k and preserves oriented angles. Equivalently it is the composite of a homothety of ratio k with a direct isometry (translation or rotation). Characteristic elements: the ratio $k > 0$, the angle θ, and (when $k \neq 1$) the centre Ω, its unique fixed point. Complex form: $z' = az + b$ with $a \in \mathbb{C}^*$, $|a| = k$, $\arg a = \theta$.

4. Indirect plane similarity. An indirect similarity multiplies all distances by a fixed $k > 0$ and reverses oriented angles; equivalently it is the composite of a homothety with an indirect isometry (reflection or glide reflection). Complex form: $z' = a\bar{z} + b$, $a \in \mathbb{C}^*$, with ratio $|a|$.

SOLUTION **Exercise 4 — Direct calculations**

1. The composite of two translations is the translation of the sum vector:

$$\vec{u} + \vec{v} = (2 + 3,\ -1 + 4) = (5, 3).$$

   Hence $t_{\vec{v}} \circ t_{\vec{u}} = t_{(5,3)}$ and the image of $M(1, 5)$ is

$$M' = (1 + 5,\ 5 + 3) = (6, 8).$$

2. A homothety of centre ω and ratio k has complex expression $z' - \omega = k(z - \omega)$. With $\omega = 1 - i$ and $k = -3$:

$$z' = -3(z - (1 - i)) + (1 - i) = -3z + 4(1 - i) = -3z + 4 - 4i.$$

   Image of $z = 2 + i$: $z' = -3(2 + i) + 4 - 4i = -6 - 3i + 4 - 4i = -2 - 7i$.

3. The rotation of centre A (affix $\omega = 1$) and angle $\theta = \dfrac{\pi}{3}$ has expression $z' - \omega = e^{i\pi/3}(z - \omega)$:

$$z' = e^{i\pi/3}(z - 1) + 1 = \left(\tfrac{1}{2} + i\tfrac{\sqrt{3}}{2}\right)(z - 1) + 1.$$

   For B of affix 4: $z' = \left(\tfrac{1}{2} + i\tfrac{\sqrt{3}}{2}\right) \cdot 3 + 1 = \dfrac{5}{2} + \dfrac{3\sqrt{3}}{2}i$, so $B' = \left(\dfrac{5}{2},\ \dfrac{3\sqrt{3}}{2}\right)$.

SOLUTION **Exercise 5 — Composing reflections in parallel lines**

1. Since $d_1 \parallel d_2$ with $d_1 \neq d_2$, the composite $s_2 \circ s_1$ is a translation. Its vector is $\vec{w} = 2\vec{u}$, where $\vec{u}$ is the vector going perpendicularly from d_1 to d_2. As the lines are 3 cm apart, $\|\vec{u}\| = 3$ cm, so $\|\vec{w}\| = 6$ cm, directed from d_1 towards d_2, perpendicular to both lines.

2. Construction. For each vertex, say A: drop the perpendicular from A to d_1 and d_2; translate A by the vector $\vec{w} = 2\vec{u}$ (i.e. move 6 cm along the perpendicular direction from d_1 to d_2) to obtain A'. Repeat for B and C, then join $A'B'C'$. Alternatively, reflect ABC in d_1 to get $A_1B_1C_1$, then reflect $A_1B_1C_1$ in d_2.

3. We need $2\vec{u} = \vec{u}_{\text{target}} = (4, 0)$, so $\vec{u} = (2, 0)$: the two lines must be vertical lines 2 units apart, the second being to the right of the first. For example $d_1 : x = 0$ and $d_2 : x = 2$; then $s_{d_2} \circ s_{d_1}$ is the translation of vector $(4, 0)$. (Infinitely many pairs work: $x = a$ and $x = a + 2$.)

SOLUTION **Exercise 6 — Composing two rotations**

1. The angles sum to $60^\circ + (-60^\circ) = 0$. By the theorem on the composition of two rotations, when the sum of the angles is $0 \pmod{360^\circ}$ and the centres are distinct, the composite is a translation. Hence $r_2 \circ r_1$ is a translation.

2. Write the rotations as composites of reflections through the line (AB): $r_1 = s_{(AB)} \circ s_{d_1}$ where d_1 passes through A with $(d_1, (AB)) = 30^\circ$, and $r_2 = s_{d_2} \circ s_{(AB)}$ where d_2 passes through B with $((AB), d_2) = -30^\circ$. Then

$$r_2 \circ r_1 = s_{d_2} \circ s_{(AB)} \circ s_{(AB)} \circ s_{d_1} = s_{d_2} \circ s_{d_1}.$$

   Since $(d_1, d_2) = 30^\circ + (-30^\circ) \cdot$ (sign check): the lines d_1 and d_2 are parallel (both make an angle of 30° with (AB) on appropriate sides), so $s_{d_2} \circ s_{d_1}$ is the translation of vector $2\vec{u}$ where $\vec{u}$ goes from d_1 to d_2. A direct computation with complex numbers, $z' = e^{-i\pi/3}\left(e^{i\pi/3}(z - a) + a - b\right) + b = z + (a - b)\left(1 - e^{-i\pi/3}\right)$, gives the vector of affix $(a - b)\left(1 - e^{-i\pi/3}\right)$, i.e.

$$\vec{w} = \left(1 - e^{-i\pi/3}\right)\vec{BA}.$$

   Since $1 - e^{-i\pi/3} = 1 - \frac{1}{2} + i\frac{\sqrt{3}}{2} = e^{i\pi/3}$, the vector is $\vec{w} = e^{i\pi/3}\vec{BA}$: it has length AB and makes an angle of 60° with $\vec{BA}$.

3. Now the angles sum to $60^\circ + 120^\circ = 180^\circ \neq 0$, so $r_2 \circ r_1$ is a rotation of angle 180°, i.e. a central symmetry. Its centre Ω is found from the fixed-point condition. Using complex expressions with $e^{i\pi/3}$ and $e^{2i\pi/3}$:

$$z' = e^{2i\pi/3}\left(e^{i\pi/3}(z - a) + a - b\right) + b = -z + a\left(e^{2i\pi/3} - e^{i\pi}\right) + b\left(1 - e^{2i\pi/3}\right),$$

and the centre is $\omega = \left.\frac{z' + z}{2}\right|_{z=\omega}$, giving $\omega = \frac{a\,e^{2i\pi/3} + a + b\left(1 - e^{2i\pi/3}\right)}{2}$. Geometrically, Ω is the point such that the triangle built on A, B, Ω encodes the half-angles: Ω is the vertex of the triangle with $\Omega\hat{A}B = 30^\circ$ and $A\hat{B}\Omega = 60^\circ$ (half-angles at A and B), hence a right angle at Ω.

SOLUTION **Exercise 7 — Recognising an isometry from images**

1. Compute the displacement vectors:

$$\vec{AA'} = (2,1), \qquad \vec{BB'} = (5-3, 1-0) = (2,1), \qquad \vec{CC'} = (-2-0, 5-4) = (2,1).$$

All three displacement vectors are equal to $(2,1)$. Since an isometry mapping three non-collinear points is uniquely determined, f coincides with the translation of vector $\vec{u}(2,1)$ on the whole plane. Check distances: $AB = 3 = A'B'$, $AC = 4 = A'C'$, consistent. Hence $f = t_{\vec{u}}$ with $\vec{u}(2,1)$.

2. Now $C' = (6,1)$. We still have $A'(2,1)$, $B'(5,1)$. Note $A'B' = 3 = AB$ and $A'C' = \sqrt{36+1} = \sqrt{37}$, while $AC = 4$: distances are not preserved, so no isometry maps A, B, C to A', B', C' in this case. (Indeed $B'C' = \sqrt{1+0} = 1$ while $BC = 5$.) Therefore such an isometry f does not exist; the data are inconsistent with an isometry.

Remark. The examiner's trap: always verify that all pairwise distances are preserved before concluding. Three image points that fail the distance test cannot come from an isometry.

SOLUTION **Exercise 8 — Homothety from its complex expression**

1. The expression $z' = 3z-4+2i$ has the form $z' = kz+b$ with $k = 3 \in \mathbb{R}\setminus\{1\}$: this is a homothety of ratio 3. Its centre is the unique fixed point: $\omega = 3\omega - 4 + 2i \Rightarrow 2\omega = 4 - 2i \Rightarrow \omega = 2 - i$. Indeed $z' - (2-i) = 3\big(z - (2-i)\big)$. Centre $\Omega(2,-1)$, ratio $k = 3$.

2. A homothety of ratio k maps a circle of centre C and radius R to the circle of centre $C' = f(C)$ and radius $|k|R$. Centre: C has affix $1+i$, so $c' = 3(1+i) - 4 + 2i = -1 + 5i$. Radius: $3 \times 2 = 6$. Image: circle of centre $(-1,5)$ and radius 6.

3. The line $d : z + \bar{z} = 4$ is the vertical line $x = 2$ (since $z + \bar{z} = 2\,\mathrm{Re}(z)$). A homothety of ratio 3 maps a line not through the centre to a parallel line. Write $z = 2 + iy$; then $z' = 3(2+iy) - 4 + 2i = 2 + (3y+2)i$, whose real part is 2. Wait — check: the centre $\Omega = 2 - i$ has real part 2, so the line $x = 2$ passes through the centre! A line through the centre of a homothety is globally invariant: its image is the line itself, $x = 2$, i.e. $z + \bar{z} = 4$. (Each point is mapped to another point of the same line.)

SOLUTION **Exercise 9 — Composing homotheties (8 marks)**

1. (3 marks) For any point M, set $M_1 = h_1(M)$ and $M' = h_2(M_1)$. Then

$$\vec{O_1M_1} = 2\vec{O_1M}, \qquad \vec{O_2M'} = \tfrac{1}{2}\vec{O_2M_1}.$$

Compute $\vec{MM'} = \vec{MO_1} + \vec{O_1O_2} + \vec{O_2M'}$. Alternatively use affixes with o_1, o_2: $z_1 - o_1 = 2(z - o_1)$ so $z_1 = 2z - o_1$; then $z' - o_2 = \tfrac{1}{2}(z_1 - o_2)$ gives

$$z' = \tfrac{1}{2}(2z - o_1 - o_2) + o_2 = z + \tfrac{1}{2}(o_2 - o_1).$$

This is the translation of vector $\frac{1}{2}\vec{O_1O_2}$. Since $O_1 \neq O_2$ the vector is non-zero. Mark scheme: 1 mark for writing both homothety relations; 1 mark for correct composition to $z' = z + b$ (or vector form); 1 mark for identifying the translation and its vector $\tfrac{1}{2}\vec{O_1O_2}$.

2. (1 mark) Draw O_1 and O_2; the vector of the translation is the vector from O_1 to the mid-point of $[O_1O_2]$ (half of $\vec{O_1O_2}$, same direction). 1 mark for a correct arrow of length $\frac{O_1O_2}{2}$ parallel to (O_1O_2).

3. (4 marks) Now $h_2 = h(O_2, 3)$: $z_1 = 2z - o_1$ and $z' - o_2 = 3(z_1 - o_2)$, so

$$z' = 3(2z - o_1 - o_2) + o_2 = 6z - 3o_1 - 2o_2.$$

This has the form $z' = Kz + b$ with $K = 6 \in \mathbb{R} \setminus \{1\}$: a homothety of ratio $k_1k_2 = 6$. Its centre Ω is the fixed point: $\omega = 6\omega - 3o_1 - 2o_2 \Rightarrow 5\omega = 3o_1 + 2o_2$, i.e.

$$\omega = \frac{3o_1 + 2o_2}{5}, \qquad \vec{O_1\Omega} = \frac{2}{5}\vec{O_1O_2}.$$

So Ω lies on the segment $[O_1O_2]$, at $\frac{2}{5}$ of the way from O_1 to O_2 (equivalently $\vec{\Omega O_2} = \tfrac{3}{5}\vec{O_1O_2}$… precisely $O_1\Omega = \tfrac{2}{5}O_1O_2$). Mark scheme: 1 mark for the composite expression; 1 mark for ratio 6; 1 mark for the fixed-point equation; 1 mark for the position $\vec{O_1\Omega} = \tfrac{2}{5}\vec{O_1O_2}$ on line (O_1O_2).

SOLUTION **Exercise 10 — Centres of similitude of two circles (10 marks)**

1. (3 marks) A homothety mapping $C(A, 2)$ onto $C'(B, 6)$ must send the centre A to the centre B (centres correspond) and multiply radii by $|k|$: $|k| = \frac{6}{2} = 3$, so $k = 3$ or $k = -3$. For each value, the centre Ω is determined by $\vec{\Omega B} = k\vec{\Omega A}$, which has exactly one solution since $k \neq 1$. Hence exactly two homotheties, of ratios $\boxed{3}$ and $\boxed{-3}$.

2. (3 marks) Let Ω on line (AB) with $\vec{\Omega B} = k\vec{\Omega A}$. For $k = 3$: $B - \Omega = 3(A - \Omega)$ gives $2\Omega = 3A - B$, i.e.

$\vec{A\Omega} = -\frac{1}{2}\vec{AB}$: Ω_1 is on line (AB), on the side of A opposite to B, with $\Omega_1 A = \frac{AB}{2} = 5$ cm and $\Omega_1 B = 15$ cm. (External centre: $\frac{\Omega_1 A}{\Omega_1 B} = \frac{2}{6}$.) For $k = -3$: $B - \Omega = -3(A - \Omega)$ gives $4\Omega = 3A + B$, i.e. $\vec{A\Omega_2} = \frac{1}{4}\vec{AB}$: Ω_2 lies between A and B with $\Omega_2 A = \frac{AB}{4} = 2.5$ cm and $\Omega_2 B = 7.5$ cm. (Internal centre, dividing $[AB]$ in the ratio of the radii $2 : 6$.)

3. (2 marks) Let t be a common external tangent touching C at T and C' at T'. The radii AT and BT' are both perpendicular to t, hence parallel, pointing to the same side. The homothety $h(\Omega_1, 3)$ sends A to B and, since $\vec{BT'} = 3\vec{AT}$ (same direction, lengths $6 = 3 \times 2$), sends T to T'; therefore $T' = h(T)$ lies on the line $(\Omega_1 T)$: the points Ω_1, T, T' are collinear, i.e. the tangent line at T and T' passes through Ω_1. (Similarly the common internal tangents pass through Ω_2.)

4. (2 marks) Construction: draw (AB), mark Ω_1 with A between Ω_1 and B, $\Omega_1 A = 5$ cm; mark Ω_2 between A and B with $A\Omega_2 = 2.5$ cm. For the external tangents from Ω_1: draw the circle of diameter $[\Omega_1 A]$ meeting C at the tangent points T_1, T_2; the lines $(\Omega_1 T_i)$ are the external tangents (also tangent to C'). Mark scheme: 1 mark for each centre correctly placed, tangents drawn with tangent points constructed via a right angle.

SOLUTION **Exercise 11 — Euler line and direct similarity (12 marks)**

Part A.

1. (3 marks) Let M be the midpoint of $[BC]$. Since G is the centroid, $\vec{GA} = -2\vec{GM}$ (the centroid divides the median in ratio $2 : 1$ from the vertex). Hence $h(G, -2)$ maps M onto A. The perpendicular bisector of $[BC]$ passes through O and is perpendicular to $[BC]$ at M; its image under $h(G, -2)$ is the line through A parallel to it (a homothety maps a line to a parallel line), i.e. the line through A perpendicular to $[BC]$ — the altitude from A. Doing the same with the other sides, O (intersection of the perpendicular bisectors) is mapped to the intersection of the three altitudes, which is H. Hence $h(G, -2)(O) = H$.

2. (2 marks) Since $h(G, -2)$ maps O to H, we have $\vec{GH} = -2\vec{GO}$; the vectors are collinear, so O, G, H lie on one line (the Euler line), and $GH = 2\,GO$ with G between O and H.

Part B.

1. (3 marks) $z' = az + b$ with $a = 1 + i\sqrt{3}$: this is a direct similarity of ratio $k = |a| = \sqrt{1+3} = 2$ and angle $\theta = \arg a = \frac{\pi}{3}$. Since $k = 2 \neq 1$, S is the composite of the homothety of ratio 2 and the rotation of angle $\frac{\pi}{3}$ about the same centre (a direct similarity with centre, ratio 2, angle $\frac{\pi}{3}$ — sometimes called a spiral similarity).

2. (2 marks) The centre Ω is the fixed point: $\omega = a\omega + b$ with $b = 2 - i$:

$$\omega = \frac{b}{1-a} = \frac{2-i}{1-(1+i\sqrt{3})} = \frac{2-i}{-i\sqrt{3}} = \frac{(2-i)\,i}{\sqrt{3}} = \frac{1+2i}{\sqrt{3}} = \frac{\sqrt{3}}{3} + \frac{2\sqrt{3}}{3}i.$$

3. (1 mark) Image of $z = 1$: $z' = (1 + i\sqrt{3})(1) + 2 - i = 3 + i(\sqrt{3} - 1)$.

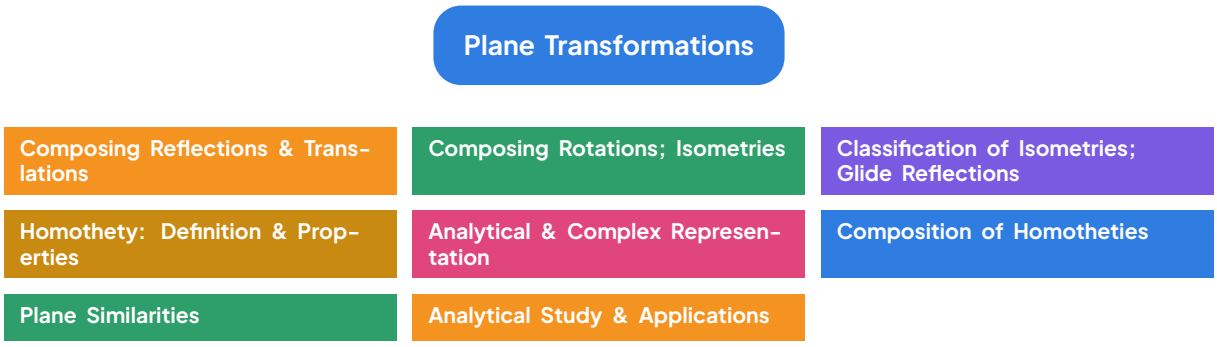
4. (1 mark) A glide reflection has complex form $z' = e^{i\alpha}\bar{z} + c$ (indirect isometry); composing with a homothety of ratio $k \neq 1$ gives $z' = a\bar{z} + b$ with $|a| = k \neq 1$: an indirect similarity. Its fixed points satisfy $z = a\bar{z} + b$. Conjugating gives $\bar{z} = \bar{a}z + \bar{b}$. Substituting: $z = a(\bar{a}z + \bar{b}) + b = |a|^2 z + a\bar{b} + b$, hence $(1 - |a|^2)z = b + a\bar{b}$. Since $|a| = k \neq 1$, this linear equation has a unique solution $z = \dfrac{b + a\bar{b}}{1 - k^2}$. Illustration: $z' = 2\bar{z} + 3i$ is an indirect similarity of ratio 2; fixed point: $z = 2\bar{z} + 3i$; conjugate $\bar{z} = 2z - 3i$; substitute: $z = 2(2z - 3i) + 3i = 4z - 3i \Rightarrow z = i$: unique fixed point i.

Examiner expectations. Part A rewards the median relation $\vec{GA} = -2\vec{GM}$ (1 mark), the image of the perpendicular bisector being the altitude (1 mark), the conclusion $O \mapsto H$ (1 mark), then collinearity and the ratio (2 marks). Part B: correct modulus/argument (2 marks) and classification (1 mark); fixed-point computation with correct conjugate manipulation (2 marks); image computation (1 mark); the final proof requires the general form $z' = a\bar{z} + b$ and the conjugate substitution argument leading to a unique solution (1 mark).

13 SUMMARY SHEET

13.1 Chapter at a Glance

This chapter completes your study of plane transformations begun in Lower Sixth. You already know the four basic **isometries** (translation, reflection, rotation, central symmetry); here you will learn what happens when they are composed, how every plane isometry can be classified, and you will meet two new families of transformations: **homotheties** and, more generally, **plane similarities**, studied both geometrically and analytically (with coordinates and with complex numbers). The mind map below shows the route we shall follow.



Mind map of the chapter Plane Transformations: from composing isometries to homotheties and similarities.

13.2 Summary Sheet

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- A **reflection** (orthogonal symmetry) S_Δ maps M to M' where Δ is the perpendicular bisector of $[MM']$; points of Δ are fixed.
- A **translation** $t_{\vec{u}}$ maps M to M' with $M\vec{M'} = \vec{u}$; it has no fixed point unless $\vec{u} = \vec{0}$ (the identity).
- A **rotation** $r(\Omega, \theta)$ maps M to M' with $\Omega M = \Omega M'$ and $\text{mes}(\vec{\Omega M}, \vec{\Omega M'}) = \theta$; its only fixed point is Ω (unless $\theta = 0\ [2\pi]$).
- A **central symmetry** is the rotation of angle π, i.e. the homothety of ratio -1.
- A **glide reflection** is the composite $S_\Delta \circ t_{\vec{u}}$ with $\vec{u}$ parallel to Δ; it has no fixed point and reverses orientation.
- A **homothety** $h(\Omega, k)$, $k \neq 0$, maps M to M' with $\vec{\Omega M'} = k\,\vec{\Omega M}$; it is **direct** if $k > 0$, **indirect** if

$k < 0$.

- A **plane similarity** of ratio $k > 0$ multiplies all distances by k; it is **direct** (preserves oriented angles) or **indirect** (reverses oriented angles).

Laws, formulas and theorems

- $S_{\Delta_2} \circ S_{\Delta_1}$ with $\Delta_1 \parallel \Delta_2$: translation by $2\vec{u}$, where $\vec{u}$ is the vector sending Δ_1 to Δ_2 perpendicularly.
- $S_{\Delta_2} \circ S_{\Delta_1}$ with $\Delta_1 \cap \Delta_2 = \{\Omega\}$: rotation of centre Ω and angle $2(\Delta_1, \Delta_2)$.
- $t_{\vec{v}} \circ t_{\vec{u}} = t_{\vec{u}+\vec{v}}$: composition of translations is commutative.
- $r(\Omega, \theta') \circ r(\Omega, \theta) = r(\Omega, \theta + \theta')$; for distinct centres, the composite is a rotation of angle $\theta + \theta'$ unless $\theta + \theta' = 0\ [2\pi]$, in which case it is a translation.
- $S_\Delta \circ t_{\vec{u}}$ with $\vec{u}$ not parallel to Δ is still a glide reflection: decompose $\vec{u} = \vec{u}_1 + \vec{u}_2$ with $\vec{u}_1 \parallel \Delta$, $\vec{u}_2 \perp \Delta$, and absorb $t_{\vec{u}_2}$ by shifting the axis.
- Classification theorem: every plane isometry is a translation, a rotation, a reflection or a glide reflection; the first two are **direct** (preserve orientation), the last two are **indirect** (reverse orientation).
- Homothety $h(\Omega, k)$: $M'N' = |k|\,MN$; areas are multiplied by k^2; the image of a line is a parallel line; the image of a circle of radius R is a circle of radius $|k|R$, the centre being the image of the centre.
- $h(\Omega, k_2) \circ h(\Omega, k_1) = h(\Omega, k_1 k_2)$; with distinct centres and $k_1 k_2 \neq 1$, the composite is a homothety of ratio $k_1 k_2$ whose centre lies on the line of the two centres; if $k_1 k_2 = 1$, it is a translation.
- Complex form: homothety of centre $\Omega(a)$ and ratio $k \in \mathbb{R}^*$: $z' = kz + a(1 - k)$; direct similarity: $z' = az + b$ ($|a|$ is the ratio, $\arg a$ the angle); indirect similarity: $z' = a\bar{z} + b$.
- A direct similarity with ratio $k \neq 1$ has a unique fixed point Ω (its centre) and equals $r(\Omega, \theta) \circ h(\Omega, k) = h(\Omega, k) \circ r(\Omega, \theta)$; it is called a **spiral similarity** when $k \neq 1$ and $\theta \neq 0\ [2\pi]$.

Methods

- To identify a composite: count the fixed points, check whether orientation is preserved or reversed, then test candidate transformations on two well-chosen points.
- To construct $M' = h(\Omega, k)(M)$: draw the line (ΩM) and mark M' with $\Omega\vec{M}' = k\Omega\vec{M}$ (same side as M if $k > 0$, opposite side if $k < 0$).
- To find the centre of a homothety given $A \mapsto A'$ and $B \mapsto B'$: the centre is the intersection of (AA') and (BB'); the ratio is $k = \dfrac{\Omega\vec{A}'}{\Omega\vec{A}}$.
- To find the centre of a spiral similarity sending $A \mapsto A'$, $B \mapsto B'$: the centre lies at the intersection of the circles through (A, B, P) and (A', B', P), where $P = (AB) \cap (A'B')$.

- In analytical work: write $z' = az + b$, compute a and b from two point-image pairs, then read off the nature from a ($a \in \mathbb{R}$: homothety or translation; $|a| = 1$: isometry).

Common mistakes

- Forgetting the factor 2 in the angle $2(\Delta_1, \Delta_2)$ of the composite of two reflections.
- Believing composition is always commutative: in general $g \circ f \neq f \circ g$ (order matters; $g \circ f$ means f first, then g).
- Confusing the ratio k with $|k|$: distances are multiplied by $|k|$, areas by k^2.
- Forgetting the special cases: $k = 1$ gives the identity, $k = -1$ gives the central symmetry about Ω.
- Forgetting that when $\theta + \theta' = 0\,[2\pi]$ the composite of two rotations is a translation, not a rotation; similarly $k_1 k_2 = 1$ for homotheties with distinct centres gives a translation.
- Claiming a glide reflection has a fixed point: it has none, although its axis is globally invariant.

GCE tips

- Always state the **characteristic elements** of a transformation: centre, ratio, angle, axis or vector, as appropriate.
- In complex-number questions, write the form $z' = az + b$ first, then read off the nature of the transformation from a; this earns method marks.
- Show constructions with dashed lines, mark equal lengths and angles, and justify each step; method marks are awarded for correct reasoning even if the final identification slips.
- Check your answer on a simple figure (a triangle or a square) before generalising; a quick sketch often reveals a wrong ratio or a wrong angle.

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